

Deciding Conjugacy of a Rational Relation

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Abstract

A rational relation is conjugate if every pair of words in the relation are conjugates, i.e., cyclic shifts of each other. We show that checking whether a rational relation is conjugate is decidable.

We assume that the rational relation is given as a rational expression over pairs of words. Every rational expression is effectively equivalent to a sum of sumfree expressions, possibly with an exponential size blow-up. The general problem reduces to determining the conjugacy of sumfree rational expressions. To solve this specific case, we generalise the classical Lyndon-Schützenberger theorem from word combinatorics that equates conjugacy of a pair of words (u, v) and the existence of a word z (called a *witness*) such that $uz = zv$ (or $zu = vz$). We give two generalisations of this result. We say that a set of conjugate pairs has a *common witness* if there is a word that is a witness for every pair in the set. The generalisations are as follows:

1. If G is an arbitrary set of conjugate pairs, then G^* is conjugate if and only if there is a common witness for G . Moreover, a word is a common witness for G^* if and only if it is a common witness for G .
2. If $G_1^*, \dots, G_k^*$, $k > 0$ are arbitrary sets of conjugate pairs and $(\alpha_0, \beta_0), \dots, (\alpha_k, \beta_k)$ are arbitrary pairs of words, then the set of words

$$G = (\alpha_0, \beta_0)G_1^*(\alpha_1, \beta_1) \cdots (\alpha_{k-1}, \beta_{k-1})G_k^*(\alpha_k, \beta_k)$$

is conjugate if, and only if,

- either it has a common witness,
- or there exists words $w \sim \alpha_0 \dots \alpha_k$ and ρ such that each pair generated by M is conjugate to a word in $w\rho^*$.

Moreover, the common witnesses of G are computable in polynomial time from the common witnesses of $G_1^*, \dots, G_k^*$.

Further, the set of common witnesses, if it exists, is computable inductively. Using the above results, we give a polynomial-time algorithm for checking the conjugacy of a sumfree expression and an exponential-time procedure for the general problem.

Keywords: Rational relations, Finite state transducers, Conjugacy of words, Combinatorics of words

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1. Introduction

Conjugacy of two elements u and v in a group can be defined as any of the following equivalent cases:

1. $uz = zv$ for some z ,
2. $u = xy$ and $v = yx$ for some x, y .

The conjugacy problem asks if a given pair of elements in a finitely presented group (typically infinite) is conjugate. It along with the word and isomorphism problems constitute the classical triad of decision problems on groups identified by Dehn in 1912 [1]. Dehn's prescient choice turned out to be instrumental not only in mathematics, but also to the theory of semigroups/monoids and automata in computer science. It turns out that the above conditions are equivalent for free monoids (i.e., when u, v, z, x, y are taken to be words over some finite alphabet). This is the well-known second theorem of Lyndon-Schützenberger. But unlike in the case of groups where condition (1) is taken to be the definition of conjugacy, in the case of monoids condition (2) is taken as the definition of conjugacy. Hence the statement reads the following way.

Theorem 1 (Proposition 1.3.4 of [2]). *A pair of nonempty words (u, v) is conjugate if and only if there exists a word z such that $uz = zv$. Moreover, $z \in (xy)^*x$ where x and y are such that $u = xy$ and $v = yx$.*

Conjugacy problem is solvable in polynomial time over free monoids and free groups. We consider a generalisation of the problem to a finitely-presented possibly infinite set of pairs. Let A be a finite alphabet. A relation $R \subseteq A^* \times A^*$ over the free monoid A^* is conjugate if each pair $(u, v) \in R$ is conjugate. Consider the following decision version:

1. Given a relation R over A^* , is it conjugate?

Of particular interest is when R is automata-definable because of motivations detailed later. First we recall the class of rational relations. The family of *rational subsets* of a monoid M is the smallest class containing $\emptyset$, all singleton subsets of M and closed under union, product and Kleene closure. A natural way to present a rational subsets of M is as a rational expression: $\emptyset, m \in M$ are rational expressions, and if E_1, E_2 are rational expressions then $E_1 \cdot E_2, E_1 + E_2$, and E_1^* are also rational expressions. A rational relation over A^* is a rational subset of the product monoid $A^* \times A^*$. Coincidentally, rational relations are precisely those that are defined by nondeterministic finite state transducers. Checking conjugacy of rational relation is equivalent to checking if the input output pairs of a relation defined by a nondeterministic transducer are conjugates. A rational expression over $A^* \times A^*$ is conjugate if the rational relation it represents is conjugate.

Example 1. The rational expression $E_1 = (\epsilon, a)(ab, ba)^*(a, \epsilon)$ denotes the relation $\{((ab)^n a, a(ba)^n) \mid n \geq 0\}$. The expression $E_2 = ((a, aa) + (b, \epsilon))^*$ represents $\{(u, v) \mid v \text{ is obtained from } u \text{ by duplicating } a\text{'s and discarding } b\text{'s}\}$. The expression E_1 is conjugate, but E_2 is not.

A strong justification for the above problem comes from the theory of word transducers. Checking a number of properties of word transducers, for instance sequentiality (can the given transducer be input-determinised?) or finite sequentiality (is the given transducer equivalent to a disjoint union of input-deterministic transducers?), bounded edit distance [3] (is the edit distance between the respective outputs of the given transducers bounded?) etc. amounts to checking conjugacy of the rational relations defined by the strongly connected components of the transducer and certain specific properties of the underlying acyclic graph of strongly connected components. Loosely speaking, conjugacy of the relations defined by the strongly connected components imply that the loops of the transducer are pumpable. Likewise, given two functional transducers with identical domain, checking if the output pairs of these transducers are conjugates on all input, reduces to checking conjugacy of the rational relation of output pairs defined by their product automata. A converse result also holds: checking conjugacy of rational relation is equivalent to checking conjugacy of output pairs of two sequential transducer with identical domain by virtue of Nivat's theorem [4, 3].

Our main result is summarised by the following theorem.

Theorem 2. *Conjugacy of rational relations is decidable.*

The union operation of rational relations preserves conjugacy, i.e., if R_1 and R_2 are conjugate, then $R_1 \cup R_2$ is also conjugate. However, conjugacy is not preserved under the product and Kleene closure.

Example 2. Consider $E_1 = (ab, ba)$, $E_2 = (ca, ac)$ and $E_3 = (ac, ca)$, all of which are conjugates. $E_1 \cdot E_3$ is conjugate but $E_1 \cdot E_2$ is not. Also $(E_1 + E_3)^*$ is conjugate, but $(E_1 + E_2)^*$ is not.

Assume that the rational relation is given by a rational expression. Every rational expression is effectively equivalent to a sum of sumfree expressions (called sumfree normal form (SNF)), by inductively rewriting the expression using the identities $(a + b)^* = (a^*b^*)^*$ and $(a + b) \cdot (c + d) = ac + ad + bc + bd$ (see Lemma 20). This fact is the rational-expression analogue of the factorisation forest theorem of Simon, a deep result from the theory of finite semigroups [5]. Rewriting a rational expression in SNF may result in an exponential blow-up, both in the number of summands and the size of each summand.

Example 3. For the expression $E = (a + b)^n$ for some $n > 0$, it can be shown that any equivalent expression in SNF will have at least 2^n summands. For $E' = \$(E\#)^*$, any equivalent SNF expression will have at least one summand of exponential size, and the expression $E \cdot E'$ in SNF will have exponentially many summands of exponential size.

For proving Theorem 2 we only need to decide the conjugacy of a rational relation given by a sumfree expression. The decidability of conjugacy of a sumfree expression hinges on the notion of a common witness of a relation, inherited from Lyndon-Schützenberger's theorem. A *witness* of a conjugate pair (u, v) is a word z such that either $uz = zv$ (*inner witness*) or $zu = vz$ (*outer witness*). A word z is a *common inner (resp. outer) witness* of a relation, if for every pair (u, v) in the relation, z is an inner witness (resp. outer witness) of (u, v) . By Theorem 1, if a relation has a common inner witness or common outer witness then it is conjugate. However, the converse is easily shown to be false.

We show that a sumfree rational relation is conjugate if and only if either it has a common witness — either a common inner witness or a common outer witness, but not necessarily both — or there exist words u and v such that every pair generated by the relation is a cyclic shift of some word in uv^* . This characterisation of conjugacy is a main contribution of our paper. It is in fact a generalisation of Lyndon-Schützenberger theorem characterising conjugacy of two words.

There are two interesting questions regarding common witnesses:

I. *Is there a common witness for the relation R ?*

II. *Given a word z , is it a common witness of R ?*

Question II proves to be comparatively more tractable, as it can be reduced to verifying whether the rational relation $R' = \{(uz, zv) \mid (u, v) \in R\}$ (or, $R' = \{(zu, vz) \mid (u, v) \in R\}$) consists of only identical pairs [6]. In fact, the decidability of the twinning property of a transducer is connected to Question II. It is further elaborated in Section 1.2.

Question I, on the other hand, is more difficult *a priori* as we do not have a bound on the size of a possible common witness. We provide a decision procedure for Question I. This is another main contribution of the article. Our characterisation of conjugacy via common witness, together with this procedure, yields an algorithm for deciding conjugacy.

In the rest of this section, we give an overview of determining the conjugacy of sumfree expressions. Subsequently, it is argued that decidability follows from two specific questions (Question 1 and Question 2). Finally, we discuss related works.

1.1. Conjugacy of a Sumfree Expression

A rational expression is *sumfree* if it does not use the union (i.e., $+$). The set of sumfree expressions is formally defined as a hierarchy. Fix a monoid $\mathbf{M} = (M, \cdot, 1)$. Given a class C of expressions over $\mathbf{M}$, the *Kleene closure* of C , denoted as $\mathcal{KC}$, is the class of expressions

$$\mathcal{KC} = C \cup \{E^* \mid E \in C\}.$$

The *monoid closure* of C , denoted as $\mathcal{MC}$, is the class of expressions

$$\mathcal{MC} = C \cup \{E_1 \cdots E_k \mid E_i \in C \text{ for each } 1 \leq i \leq k \text{ and } k \in \mathbb{N}\}.$$

Definition 1 (Sumfree Expression). *The family $\mathcal{F}$ of sumfree expressions is defined inductively. Let $\mathcal{F}_0 = M \cup \{\emptyset\}$ and $\mathcal{F}_{i+1} = \mathcal{MKF}_i$ for each $i \geq 0$. We define*

$$\mathcal{F} = \bigcup_{i \geq 0} \mathcal{F}_i.$$

The star height of an expression E is the smallest $k \in \mathbb{N}$ such that E belongs to $\mathcal{F}_k$.

Over the free monoid A^* , the set of expressions $\mathcal{F}_0$ is $A^* \cup \{\emptyset\}$ and $\mathcal{KF}_0$ is the set of expressions $\mathcal{F}_0 \cup \{w^* \mid w \in A^*\}$ (for convenience we assume that $\emptyset$ is not used in any other expression other than $\emptyset$ itself). It is not difficult to see that $\mathcal{MKF}_0$ is the set of expressions $\mathcal{KF}_0 \cup \{u_1 v_1^* u_2 v_2^* \cdots u_k v_k^* u_{k+1} \mid u_i, v_i \in A^*, k \in \mathbb{N}\}$.

We now proceed to solve the conjugacy of sumfree expressions. We use pairs of lowercase Greek letters (α, β) with suitable modifications to denote pairs of words over $A^* \times A^*$. Clearly $\emptyset$ and (ϵ, ϵ) are conjugates. For an expression of the form (α, β) , it is straightforward to check conjugacy (see Proposition 5). Thus, the conjugacy problem is decidable for the class of expressions $\mathcal{F}_0$.

To show the decidability of the conjugacy problem for the whole family $\mathcal{F}$, it suffices to show that if the problem is decidable for $\mathcal{F}_i$, $i \geq 0$, then it is also decidable for $\mathcal{KF}_i$ and $\mathcal{F}_{i+1} = \mathcal{MKF}_i$. Then by induction on i the decidability extends to the whole family $\mathcal{F}$.

Assume that conjugacy is decidable for $\mathcal{F}_i$. Let E be an expression in $\mathcal{F}_i$ and hence $E^* \in \mathcal{KF}_i$. Since $L(E) \subseteq L(E^*)$,

Proposition 1. *If the expression E^* is conjugate, then E is conjugate.*

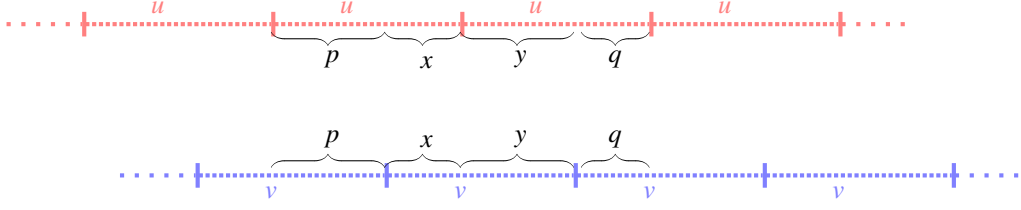


Figure 1: v as infix of uu .

Because conjugacy is decidable for $\mathcal{F}_i$, we can check whether E is conjugate. Therefore, to show the decidability of conjugacy for $\mathcal{KF}_i$, it suffices to show the decidability of the following question.

Question 1 (Conjugacy of Kleene Closures). *Given a conjugate sumfree expression E , is E^* conjugate?*

Next, assume that conjugacy is decidable for $\mathcal{KF}_i$. Let $E = (\alpha_0, \beta_0)E_1^*(\alpha_1, \beta_1) \cdots E_k^*(\alpha_k, \beta_k)$ be an expression in $\mathcal{MKF}_i$ where $E_1^*, \dots, E_k^*$ are from $\mathcal{KF}_i$. Analogous to the case of Kleene closures, E is conjugate only if $E_1^*, \dots, E_k^*$ are conjugate, as the next lemma shows.

Lemma 1. *If the expression $E = (\alpha_0, \beta_0)F^*(\alpha_1, \beta_1)$ is conjugate, then F^* is conjugate.*

Proof. If F^* is an empty set, then it is conjugate. Otherwise, assume that (u, v) is a nonempty pair that belongs to F^* . Therefore, (u^ℓ, v^ℓ) for each $\ell \geq 0$ also belongs to F^* . We can safely assume that $|u| = |v|$, otherwise each iteration will increase the difference in length between u^ℓ and v^ℓ , leading to nonconjugacy of E .

Let $k = |\alpha_0| + |\beta_0| + |\alpha_1| + |\beta_1|$. Consider the pair $(\alpha_0, \beta_0)(u^\ell, v^\ell)(\alpha_1, \beta_1)$ where ℓ is some value much larger than k , say 2^k . Since ℓ is much larger than k and $(\alpha_0 u^\ell \alpha_1, \beta_0 v^\ell \beta_1)$ is conjugate, there exist large factors of u^ℓ and v^ℓ that match as shown in Figure 1. Since $|u| = |v|$, we can infer that u is a factor of vv , and v is a factor of uu .

Since v is an infix of uu , the following holds as shown in Figure 1. There exist words x, y, p , and q such that $v = xy$ and $u = px = yq$. Since $|u| = |v|$, length of p and length of y are the same, that implies $p = y$ (since $u = px = yq$). Therefore, $u = yx$. Hence u and v are conjugate words. Since the pair (u, v) was arbitrary, F^* is conjugate. $\square$

We can generalize the above lemma to the general form of sumfree expressions.

Corollary 1. *If the expression $E = (\alpha_0, \beta_0)E_1^*(\alpha_1, \beta_1) \cdots (\alpha_{k-1}, \beta_{k-1})E_k^*(\alpha_k, \beta_k)$, for $k > 0$, is conjugate, then each of $E_1^*, \dots, E_k^*$ is conjugate.*

Proof. If E is conjugate, then for each $i \in \{1, \dots, k\}$,

$$(\alpha_0 \cdots \alpha_{i-1}, \beta_0 \cdots \beta_{i-1})E_i^*(\alpha_i \cdots \alpha_k, \beta_i \cdots \beta_k) \subseteq E$$

is conjugate. Therefore, from Lemma 1 we get that each of $E_1^*, \dots, E_k^*$ is conjugate. $\square$

Since the conjugacy of $\mathcal{KF}_i$ is decidable, we can check whether $E_1^*, \dots, E_k^*$ are conjugate expressions. Thus, to show the decidability of $\mathcal{MKF}_i$, it suffices to show the decidability of the following question.

Question 2 (Conjugacy of Monoid Closures). *Given conjugate sumfree expressions $E_1^*, \dots, E_k^*$, is the expression $E = (\alpha_0, \beta_0)E_1^*(\alpha_1, \beta_1) \cdots (\alpha_{k-1}, \beta_{k-1})E_k^*(\alpha_k, \beta_k)$ conjugate?*

We show that Question 1 and Question 2 can be effectively answered. The idea is to use the notion of common witness that we mentioned in the beginning (further elaborated in Definition 5).

We present two common witness theorems that address the above questions:

1. Let G be an arbitrary set of conjugate pairs. The set G^* is conjugate if and only if G has a common witness (Theorem 6).

2. Let $G_1^*, \dots, G_k^*, k > 0$, be arbitrary sets of conjugate pairs. The set

$$G = (\alpha_0, \beta_0)G_1^*(\alpha_1, \beta_1) \cdots (\alpha_{k-1}, \beta_{k-1})G_k^*(\alpha_k, \beta_k),$$

called a *sumfree set*, is conjugate if and only if either it has a common witness, or each G_i^* has a common witness and there exist words u, v such that each pair of words in G is a cyclic shift of a word in uv^* . (Theorem 7).

Remark 1. Note that the assumption of conjugacy of the sets $G, G_1^*, \dots, G_k^*$ is not necessary. However, if they are not conjugate then the corresponding sets will neither have a common witness nor be conjugate, and the statements will be vacuously true (Since Proposition 1 and Corollary 1 also hold for arbitrary sets).

Item 2 is a generalisation of Item 1, and its proof relies on Item 1. Both theorems are generalisations of the Lyndon-Schützenberger theorem.

In the case of rational sumfree expressions of pairs the above theorems are *effective*. In Section 7 we show that a common witness of a rational sumfree expression, if exists, is computable in polynomial time in the length of the expression. Finally, we give a decision procedure proving our main theorem (Theorem 2) in Section 8.

1.2. Related Work

Conjugate Post Correspondence Problem: A problem much related to Theorem 6 is the *Conjugate Post Correspondence problem*: given a finite set of pairs G , does there exist of a pair $(u, v) \in G^*$ such that u and v are conjugate? This problem is shown to be undecidable by reduction to the word problem of a special type of semi-Thue systems [7]. In Section 3, we show that the universal version of this problem — checking if all the pairs in G^* are conjugate — is decidable.

Edit-Distance between Transducers: Distance between two functional transducers are studied in [3]. For a metric d on words, the distance between two functional transducers $\mathcal{T}_1, \mathcal{T}_2$ from A^* to B^* is given by

$$d(\mathcal{T}_1, \mathcal{T}_2) = \begin{cases} \sup \{ d(\mathcal{T}_1(w), \mathcal{T}_2(w)) \mid w \in \text{dom}(\mathcal{T}_1) \} & \text{if } \text{dom}(\mathcal{T}_1) = \text{dom}(\mathcal{T}_2) \\ \infty & \text{otherwise} \end{cases}$$

Two transducers are said to be *close* if the distance between them is finite ($< \infty$). This allows to compare two transducers beyond their equivalence. One important class of metrics over words is edit distance, in particular Levenshtein edit distance. Levenshtein edit distance between two words is the minimum number of insertions, deletions, and substitutions required to convert one word to another. In the case of edit distances, closeness guarantees that the output of one transducer can be converted to the output of another by doing a bounded number of edits. It is shown that two functional transducers with identical domain are close with respect to Levenshtein edit distance if and only if all the loops in the cartesian product of the transducers are conjugates [3]. Our result gives an algorithm for checking the latter, and hence decidability of closeness of transducers follows.

Twinning and Sequentiality: Rational functions are functions defined by a finite state transducer. A rational function is *sequential* if it can be realised by a sequential transducer, i.e., those that are deterministic in the input. These were originally called subsequential in the literature by Schützenberger [8]. Sequentiality of rational functions is a decidable property due to a topological characterisation called the *twinning property* by Choffrut [9].

A transducer $\mathcal{T}$ from A^* to B^* is an automaton over $A^* \times B^*$. A transition of $\mathcal{T}$ from state p to state q is of the form $(p, (u, v), q)$ where the word $u \in A^*$ is called the input and the word $v \in B^*$ is called the output. A path from state p to q on an input word w producing an output word x is represented as $p \xrightarrow{w|x} q$. The transducer $\mathcal{T}$ realises the rational relation $\{(w, x) \mid q_0 \xrightarrow{w|x} q_f\}$ over $A^* \times B^*$ where q_0, q_f is an initial and a final state respectively.

The *delay* between two words u and v , denoted by $\text{delay}(u, v)$, is the pair (u', v') such that $u = \ell u'$ and $v = \ell v'$ where ℓ is the longest common prefix of u and v .

A transducer with the initial state q_0 is *twinning* if for all states p, q and for all words $w_1, w_2 \in A^*$ and $x, y, u, v \in B^*$, if $q_0 \xrightarrow{w_1|x} p \xrightarrow{w_2|u} q$ and $q_0 \xrightarrow{w_1|y} q \xrightarrow{w_2|v} q$, then $\text{delay}(x, y) = \text{delay}(xu, yv)$. This is equivalent to either $u = v = \epsilon$, or $u \neq \epsilon \neq v$ and u and v are conjugates with $\text{delay}(x, y)$ being a witness of (u, v) (Proposition 6.2 of [10]).

Since the twinning property compares paths with the same input label, an equivalent definition for twinning can be defined on the square of the transducer [11, 12]. The *square* of a transducer $\mathcal{T}$, denoted by $\mathcal{T}^2$, is a cartesian product of $\mathcal{T}$ by itself, equivalent to the transducer from A^* into $B^* \times B^*$. The original definition of twinning has the following equivalent form. Let $\mathcal{T}$ be a trim transducer. Two states p and q of $\mathcal{T}$ are *twinning* if whenever (u, v) is a nonempty output pair of a loop in $\mathcal{T}^2$ rooted at a reachable state (p, q) , and for any path from initial state to (p, q) in $\mathcal{T}^2$ with output (x, y) , the following holds: $\text{delay}(x, y) = \text{delay}(xu, yv)$, or equivalently, $\text{delay}(x, y)$ is a witness of (u, v) . A transducer $\mathcal{T}$ is twinning if, for any pair of states p and q that are reachable in $\mathcal{T}^2$, the states p and q are twinning.

Thus checking twinning property reduces to computing a number of instances of the following problem: decide if a given rational relation R is conjugate with a given witness z . This is straightforward to check: verify whether the rational relation $R' = \{(uz, zv) \mid (u, v) \in R\}$ (or, $R' = \{(zu, vz) \mid (u, v) \in R\}$) consists of only identical pairs.

Generalisation of the twinning property called *weak twinning* is used to characterise *multi-sequential* (also called *plurisubsequential* or *finitely sequential*) functions [13] and relations [14]. This property also can be stated as conjugacy of suitably defined rational relations with respect to certain predetermined witnesses, and hence decidable albeit higher complexity.

The results in this paper holds the potential to define more general properties of transducers where the witnesses are not known a priori, for instance determinising approximately.

Others: A generalisation of Lyndon-Schützenberger’s theorem to infinite sets, though with no comparison to ours, is considered in [15, 16], where solutions to the language equation $XZ = ZY$, where X, Y, Z are sets of words, are given for special cases. The general solution is still open.

1.3. Organisation of the Paper

In Section 2, we revisit the standard tools from combinatorics of words required to state and prove our main theorems. We present the common witness theorems for addressing Question 1 and Question 2, along with the proofs of the easier directions in Section 3. However, the difficult directions require a detailed case analysis, and we complete the proof of common witness theorem for Kleene closure in Section 4. We prove the common witness and conjugacy theorem of monoid closures in Section 5 and Section 6 respectively. We outline the decision procedure for computing the witness in Section 7, and deciding conjugacy in Section 8. In Section 9, we state some future directions and conclude.

2. Combinatorial Tools on Words

In this section, we introduce some preliminary notions and results that we need for the rest of the paper.

Let $\mathbb{N}$ denote the set $\{1, 2, \dots\}$. For $i, j \in \mathbb{N} \cup \{0\}$, define $[i, j]$ as the set $\{i, i+1, \dots, j\}$ if $i \leq j$, and as $\{i, i-1, \dots, j\}$ otherwise. Given a word u and $k \in \mathbb{N} \cup \{\omega\}$, we denote by u^k the concatenation of u with itself k times. A word u is called a *factor* (respectively *prefix*, *suffix*) of a word v , if there exist words $x, y \in A^*$ such that $v = xuy$ (respectively $v = uy$, $v = xu$). Let $u[i \cdot j]$ denote the factor of u from index i to j where $1 \leq i \leq j \leq |u|$. A *cyclic shift* of a word $w = uv$ is a word vu for some words $u, v \in A^*$. Let $\text{lshift}_i(u)$, denote the word obtained after i left cyclic shifts of a word u for $i \geq 0$. If u and v are words such that u is a prefix of v , the *left quotient* of v by u , denoted by $u^{-1}v$, is the word x such that $v = ux$. Similarly, the *right quotient* of $v = xu$ by u , denoted as vu^{-1} , is the word x .

Given a set of pairs of words G , let $\text{fst}(G)$ denote the projection of G onto the first component, i.e., $\text{fst}(G) = \{u \mid (u, v) \in G\}$. Similarly, $\text{snd}(G)$ denote the projection of G onto the second component, i.e., $\text{snd}(G) = \{v \mid (u, v) \in G\}$.

Two words u and v are conjugate if they are cyclic shifts of each other, i.e., there exist words x, y such that $u = xy$ and $v = yx$. We use $u \sim v$ to denote that u and v are conjugate words. Let $X, Y \subseteq A^*$. We say X is a *conjugate subset* of Y , denoted by $X \subseteq_c Y$, if for every word $x \in X$ there exists a word $y \in Y$ such that $x \sim y$. Also, X and Y are *conjugate*, denoted by $X \sim Y$, if $X \subseteq_c Y$ and $Y \subseteq_c X$.

2.1. Primitive words, Roots and Conjugacy

A word is *primitive* if it cannot be expressed as a power of any strictly smaller word. For example, *aba* is primitive but *abab* is not.

A word ρ is called a *primitive root* of a word u if $u = \rho^n$ for $n \geq 1$ and ρ is a primitive word. Every word u has a unique primitive root, denoted by ρ_u ([2], Proposition 1.3.1).

Proposition 2. *The primitive root of a word can be computed in time linear in the length of the word.*

Proof. For a word w , find the first nontrivial occurrence of w in ww , and let the offset be the prefix of ww before this occurrence. This can be done in linear time using the classical pattern matching algorithm by Knuth-Morris-Pratt [17]. If the length of the offset divides the length of w , the offset is the primitive root of the word w . $\square$

Proposition 3. *Let u, v be nonempty words such that their primitive roots are distinct, i.e., $\rho_u \neq \rho_v$. Let $w = uv$. Then $\rho_w \neq \rho_u$ and $\rho_w \neq \rho_v$.*

Proof. Consider the word $w = uv$. We show that $\rho_w \neq \rho_u$ and $\rho_w \neq \rho_v$. Assume for contradiction that $\rho_w = \rho_u$. Then $uv \in (\rho_u)^*$. Since $u \in (\rho_u)^*$, it follows that $v \in (\rho_u)^*$ as well. Since primitive root of a word is unique, we get $\rho_v = \rho_u$, which contradicts the assumption. By a similar argument, we get $\rho_w \neq \rho_v$. Therefore, $\rho_w \neq \rho_u$ and $\rho_w \neq \rho_v$, which completes the proof. $\square$

Proposition 4. *Let $u, v \in A^*$ be words such that $\rho_u \sim \rho_v$ and $\rho_u \neq \rho_v$. Then any word*

$$w \in (u + v)^* \tag{1}$$

in which both u and v appear at least once, satisfies $\rho_w \not\sim \rho_u$.

Proof. Suppose, for contradiction, that $\rho_w \sim \rho_u$. Then, we have $|\rho_w| = |\rho_u| = |\rho_v|$. Since $w \in (\rho_w)^*$ and both u and v are factors of w with lengths divisible by $|\rho_w|$, it follows that $u, v \in (\rho_w)^*$. By uniqueness of primitive roots, this implies $\rho_u = \rho_w = \rho_v$ contradicting $\rho_u \neq \rho_v$. $\square$

First Theorem of Lyndon-Schützenberger relates primitivity and commutativity.

Theorem 3 ([18], Lemma 3). *Two words $u, v \in A^*$ commute, i.e., $uv = vu$, if and only if they have the same primitive root.*

We lift the notion of primitive root to a pair and a set of pairs as follows.

Definition 2 (Primitive Root of a Set of Pairs). *The primitive root of a pair (u, v) is the pair (ρ_u, ρ_v) . The primitive root of a set of pairs P , denoted by $R(P)$, is the set of all primitive roots of each pair in P , i.e., $R(P) = \{(\rho_u, \rho_v) \mid (u, v) \in P\}$.*

For example, $\{(ab, ba), (bab, abb)\}$ is the primitive root of the set $\{(abab, baba), (bab, abb)\}$.

Recall that a pair of words (u, v) is conjugate if v can be obtained from u by cyclic shifts, i.e., there exist words x, y such that $u = xy$ and $v = yx$. For example, $(aaab, aaba)$ is a conjugate pair with $x = a$ and $y = aab$. It is not difficult to see that conjugacy relation is an equivalence relation on the set of words.

Proposition 5. *Deciding if a pair of words is conjugate can be done in linear time.*

Proof. Two words u and v are conjugate if and only if either $u = v$ or $|u| = |v|$ and there is exactly one nontrivial occurrence of v in uu .

If u and v are distinct conjugates, then $u = xy$ and $v = yx$ for some nonempty words x, y , implying $uu = xyxy = xvy$, where v occurs nontrivially in uu exactly once. Conversely, if v occurs nontrivially in uu exactly once and $|u| = |v|$, then there exist words x, y, p and q such that $v = xy$ and $u = px = yq$. Since $|u| = |v|$, we get $|p| = |y|$. This implies $p = y$ since $u = px = yq$. Therefore, $u = yx$. Hence u and v are conjugate words.

Therefore, deciding whether a pair of words (u, v) is conjugate can be done by checking if $u = v$. If not, verify $|u| = |v|$ and whether v occurs in uu nontrivially exactly once. This can be done in linear time using the Knuth-Morris-Pratt algorithm [17]. $\square$

A conjugate pair of words and its primitive root are connected.

Proposition 6 ([19], Lemma 1). *If a pair (u, v) is conjugate, then its primitive root (ρ_u, ρ_v) is also conjugate. Moreover, $(u, v) = (\rho_u, \rho_v)^n$ for some $n \geq 1$.*

Below is a fundamental result on words by Fine and Wilf.

Theorem 4 (Theorem 5 of [19]). *Two nonempty words u and v have the same primitive root if and only if the words u^ω and v^ω have a common prefix of length $|u| + |v| - \gcd(|u|, |v|)$.*

The above theorem can be adapted to yield conjugate primitive roots.

Theorem 5 ([19]). *For any two words $u, v \in A^+$, if u^ω and v^ω have a common factor of length at least $|u| + |v| - \gcd(|u|, |v|)$, then the primitive roots of u and v are conjugates.*

The bound $|u| + |v| - \gcd(|u|, |v|)$ is known as the *Fine and Wilf index* of u and v .

2.2. Cuts and Uniqueness of Cuts of Primitive Pairs

Definition 3 (Cut). *A cut of a conjugate pair (u, v) is a pair of words (x, y) such that $u = xy$ and $v = yx$. Alternatively, we say that u has a cut at position $|x|$, or equivalently, v has a cut at position $|y|$.*

If either x or y is the empty word, then we say the cut is empty. Otherwise the cut is nonempty.

For example, the pair $(aabb, bbaa)$ has a cut (aa, bb) . There can be several cuts for a conjugate pair. For instance, the pair $(abab, baba)$ has cuts (a, bab) and (aba, b) . Empty cuts are possible only for pair of identical words.

Definition 4 (Matching of a Cut). *Let u and v be conjugate words, and let (x, y) be a cut of (u, v) with $|x| = k$. It corresponds to a matching between positions of u and v , namely the circular permutation given by:*

$$\begin{array}{ccc} |x| + 1 & \mapsto & 1 \\ |x| + 2 & \mapsto & 2 \\ & \vdots & \\ |x| + |y| & \mapsto & |y| \end{array} \quad \begin{array}{ccc} 1 & \mapsto & |y| + 1 \\ 2 & \mapsto & |y| + 2 \\ & \vdots & \\ |x| & \mapsto & |x| + |y| \end{array}$$

This terminology is extended to factors of u and v , namely given a cut (x, y) of u and v , and factors u' and v' of u and v respectively, we say u' and v' match if the positions of u' and v' match according to the above bijection.

If u and v are conjugates and one of them is primitive, by Proposition 6, the other is also primitive. A pair (u, v) is primitive if both u and v are primitive words. For such pairs, their cuts are also special.

Proposition 7 (Uniqueness of Cuts of Primitive Pairs). *Let (u, v) be a conjugate primitive pair. If (u, v) is distinct, then (u, v) has a unique cut (x, y) . If (u, v) is not distinct (i.e., $u = v$), the only two possible cuts of (u, v) are (u, ϵ) and (ϵ, v) .*

Proof. By definition, if pair (u, v) is conjugate, then there exist a cut (x, y) such that $u = xy$ and $v = yx$. Assume u and v are distinct, and hence x and y have to be nonempty. It suffices to show that x and y are unique if u and v are primitive.

For the sake of contradiction, assume that (x, y) is not unique, i.e., there exists a different cut (x', y') for (u, v) , i.e., $u = x'y'$, $v = y'x'$ and $x' \neq x, y' \neq y$. WLOG, assume that $|x| > |x'|$. Therefore there exists a nonempty word p such that $x = x'p$ and $y' = py$. Substituting for x in v , we get

$$v = yx = yx'p$$

and substituting for y' in v , we obtain

$$v = y'x' = pyx'.$$

Therefore yx' and p commutes. By Theorem 3, they have the same primitive root. Since p and yx' are nonempty words, v is a power of some smaller word. Hence v is not primitive and it is a contradiction.

In the case where $u = v$ and u being primitive, two possible cuts are (u, ϵ) and (ϵ, v) , i.e., the empty cuts. Imagine there is a nonempty cut (x, y) . Since $u = v$, it follows that $xy = yx$. Using Theorem 3, u and v are powers of a smaller word and hence not primitive. Therefore, when u is primitive and $u = v$, the only possible cuts are the empty cuts. $\square$

It is known that a primitive word cannot be equal to any of its nontrivial cyclic shifts, see for instance [20, 21]. We give a reformulation of this fact below. The statement of the lemma is given in a fashion that is suitable for some of the proofs in the rest of the paper.

Lemma 2 (Cut Lemma). *Assume (u, v) is a conjugate primitive pair.*

I. *If (u, v) is a distinct pair with the unique cut (x, y) , then the following equalities cannot hold for any nonempty words x', x'', y', y'' such that $x = x'x''$ and $y = y'y''$.*

- (a) $xy = x''yx'$
- (b) $xy = y''xy'$
- (c) $yx = y''xy'$
- (d) $yx = x''yx'$
- (e) $xy = yx$

II. *In the special case when $u = v$, there are two empty cuts (u, ϵ) and (ϵ, u) . In both cases, the equality $u = u''u'$ cannot hold for any nonempty words u', u'' such that $u = u'u''$.*

Proof. Consider the case when (u, v) is a distinct pair with the unique nonempty cut (x, y) . It suffices to show that if any of the equalities hold, there exists a different nonempty cut of the primitive pair (u, v) contradicting Proposition 7.

1. In the case of I. (a), the other nonempty cut is (x'', yx') since $(xy, yx) = (x''yx', yx'x'')$.
2. In the case of I. (b), the other nonempty cut is $(y''x, y')$ since $(xy, yx) = (y''xy', y'y''x)$.
3. When I. (c) is true, we obtain a different nonempty cut (xy', y'') because $(xy, yx) = (xy'y'', y''xy')$.
4. If I. (d) holds, the other nonempty cut is $(x', x''y)$ since $(xy, yx) = (x'x''y, x''yx')$.
5. If I. (e) holds, the other nonempty cut is (y, x) since $(xy, yx) = (yx, xy)$ and $x \neq y$ (since $u \neq v$).

Consider the special case when $u = v$. If the equality $u = u''u'$ holds, then we obtain $u = u'u'' = u''u'$. Therefore, u' and u'' commutes. Since u' and u'' are nonempty words, u is a power of some smaller word using Theorem 3. Hence u is not primitive and it is a contradiction. $\square$

A consequences of the above lemma is that the cut of the primitive root decides the cuts of its powers.

Proposition 8. *Let (u, v) is a distinct conjugate primitive pair with the unique cut (x, y) . Any cut of the pair (u^n, v^n) for $n \geq 1$ is of the form $((xy)^*x, (yx)^*y)$.*

Proof. Let $(u', v') = (u^n, v^n)$ for some $n \geq 1$. The lemma is trivially true for $n = 1$ by the uniqueness of cut of primitive pairs by Proposition 7.

Consider the case when $n \geq 2$. Substituting for $u = xy$ and $v = yx$ in u' and v' ,

$$\begin{aligned} u' &= \overbrace{u \cdots u}^{n \text{ times}} = xy \cdots xy \\ v' &= v \cdots v = yx \cdots yx \end{aligned}$$

We show that cut in u' will always be at the end of some x and all other cases leads to one of Cases I. (a) to I. (e) of Cut Lemma.

Case 1: When the cut is at the end of y. I.e., there exists a cut (p, q) for (u', v') such that $p \in (xy)^+$. Then

$$u' = \overbrace{xy \cdots xy}^p \overbrace{xy \cdots xy}^q \tag{2}$$

$$v' = yx \cdots yx = qp = \overbrace{yx \cdots yx}^q \overbrace{yx \cdots yx}^p \tag{3}$$

Equating the suffixes of v' of length $|xy|$ in both side of the Equation (3), we deduce $xy = yx$, i.e., $u = v$. It satisfies Case I. (e) of Cut Lemma. Hence a contradiction.

Case 2: When the cut is strictly within some x or y . We will make a further case analysis: when there is an xy present before the cut, and when there is an xy present after the cut (since $n \geq 2$).

Suppose the cut in u' is in the i^{th} xy for $i > 1$, i.e., there is an xy present before the cut.

1. When the cut is within x , i.e., there exists a cut (p, q) of (u', v') such that $p \in (xy)^+ x'$ where x' is a nonempty proper prefix of x and $x = x'x''$ for some word x'' . Now,

$$u' = \overbrace{\dots x'x''yx'}^p \overbrace{x''y\dots}^q \quad (4)$$

$$v' = yx'x''\dots yx'x'' = qp = \overbrace{x''y\dots}^q \overbrace{\dots x'x''yx'}^p \quad (5)$$

As before, equating the suffixes of v' of length $|xy|$ on both sides of Equation (5), we obtain

$$yx = yx'x'' = x''yx'$$

Here x' and x'' satisfies Case I. (d) of Cut Lemma. Hence a contradiction.

2. When the cut is within y , i.e., there exists a cut (p, q) of (u', v') such that $p \in (xy)^+ xy'$ where y' is a nonempty prefix of y and $y = y'y''$ for some word y'' . Then,

$$u' = \overbrace{\dots xy'y''xy'}^p \overbrace{y''\dots}^q \quad (6)$$

$$v' = y'y''x\dots y'y''x = qp = \overbrace{y''\dots}^q \overbrace{\dots xy'y''xy'}^p \quad (7)$$

On both sides of the Equation (7), equating the suffixes of v' of length $|xy|$, we get

$$yx = y'y''x = y''xy'$$

that is Case I. (c) of Cut Lemma. Hence a contradiction.

The case when there is an xy after the cut is symmetric and leads to Cases I. (a) and I. (b) of Cut Lemma.

Since we have eliminated all of other cases, the only possible cuts of the pair (u', v') are of the form $((xy)^*x, (yx)^*y)$. $\square$

The below lemma gives the relationship between the cuts of two conjugate primitive pairs, particularly when there is a specific combination of these pairs which is conjugate.

Lemma 3 (Equal Length Lemma). *Let $(u_1, v_1), (u_2, v_2)$ be two conjugate primitive pairs of equal length (i.e., $|u_1| = |u_2|$). If pair $(u_1, v_1)^{\ell_1}(u_2, v_2)^{\ell_2}$ where $\ell_1 > 2, \ell_2 > \ell_1 + 2$ is conjugate, then there exist cuts (x_1, y_1) of (u_1, v_1) and (x_2, y_2) of (u_2, v_2) such that either $x_1 = x_2$ or $y_1 = y_2$.*

Proof. Let $(u, v) = (u_1, v_1)^{\ell_1}(u_2, v_2)^{\ell_2}$ such that $\ell_1 > 2$ and $\ell_2 > \ell_1 + 2$ suffices.

$$\begin{aligned} u &= \overbrace{u_1 \dots u_1}^{\ell_1 \text{ times}} \overbrace{u_2 u_2 \dots u_2 u_2}^{\ell_2 \text{ times}} \\ v &= v_1 \dots v_1 v_2 v_2 \dots v_2 v_2 \end{aligned}$$

If (u, v) is conjugate, then they have a cut say (p, q) . If either p or q is empty then both the pairs are identical. In this case, $u_1 = x_1 y_1 = y_1 x_1 = v_1$ and $u_2 = x_2 y_2 = y_2 x_2 = v_2$. By Proposition 7, either $x_1 = x_2 = \varepsilon$ or $y_1 = y_2 = \varepsilon$. Thus, assume that both p and q are nonempty. There are two possibilities for a cut of (u, v) : when the cut in u is within $u_1^{\ell_1} u_2$ or it is after $u_1^{\ell_1} u_2$.

In both the cases we show that either $x_1 = x_2$, or $y_1 = y_2$ or both.

Case 1: When the cut in u is within $u_1^{\ell_1}u_2$. In this case, the cut in v is within the suffix $v_2^{\ell_1+1}$ since the $|u_1| = |u_2| = |v_2|$ and $\ell_2 > \ell_1 + 2$. Substituting (u_1, v_1) and (u_2, v_2) with (x_1y_1, y_1x_1) and (x_2y_2, y_2x_2) ,

$$\begin{aligned} u &= x_1y_1 \cdots x_1y_1 x_2y_2 \cdots x_2y_2 = pq \\ v &= y_1x_1 \cdots y_1x_1 \cdots y_2x_2 \underbrace{y_2x_2}_{\text{cut region}} \cdots = qp \end{aligned}$$

Since $\ell_2 > \ell_1 + 2$, there exist at least one y_2x_2 before the cut in v . We compare the suffixes of q in both u and v . Since q ends with x_2y_2 in u , the cut in v should be at the end of a y_2 by Case I. of Cut Lemma. Note that by Case II. of Cut Lemma, the cut can end after x_2 but then $u_2 = x_2y_2 = y_2x_2 = v_2$ and hence either $x_2 = \varepsilon$ or $y_2 = \varepsilon$. In both case, we can say that the cut is at the end of a y_2 .

Hence p can be of the form x_2 or $(x_2y_2)^+x_2$ depending upon if the cut in v is within the last y_2x_2 or not.

$$\begin{aligned} u &= x_1y_1 \cdots x_1y_1 x_2y_2 \cdots x_2y_2 = pq \\ v &= \underbrace{y_1x_1 \cdots y_1x_1 \cdots y_2x_2y_2}_q \underbrace{x_2}_{p} \cdots \end{aligned}$$

Suppose $p \in (x_2y_2)^+x_2$, then equating the prefixes of p in u and v of length $|x_2y_2| = |x_1y_1|$ (Since $|u_1| = |u_2|$), we obtain $x_2y_2 = x_1y_1$. Substituting this in u ,

$$\begin{aligned} u &= x_1y_1 \cdots x_1y_1 x_1y_1 \cdots x_1y_1 = pq \\ v &= \underbrace{y_1x_1 \cdots y_1x_1 \cdots y_2x_2y_2}_q \underbrace{x_2y_2 \cdots x_2}_p \end{aligned}$$

Now we compare the prefixes of q in u and v . Since q starts with y_1x_1 in v , from Cut Lemma, the cut in u should be at the end of x_1 . Therefore, $p \in (x_1y_1)^+x_1$ in u . Also, $p \in (x_2y_2)^+x_2$ in v . Since $|x_1y_1| = |x_2y_2|$ and $|x_1|, |x_2| < |x_1y_1|$, we can deduce $p = (x_1y_1)^i x_1 = (x_2y_2)^i x_2$ for some i . Hence, $x_1 = x_2$. Therefore, it implies $y_1 = y_2$ since $x_1y_1 = x_2y_2$ and hence, (u_1, v_1) and (u_2, v_2) are identical.

Suppose $p = x_2$ in v . Here, p in u is within the first x_1y_1 since $|x_1y_1| = |x_2y_2|$. Moreover $p = x_1$ since the only possible cut in u will be at the end of x_1 by Cut Lemma (comparing the prefixes of q in u and v). Hence $x_1 = x_2$.

Case 2: Cut in u is after $u_1^{\ell_1}u_2$. This case is symmetric. For the sake of completeness we prove it.

Substituting (u_1, v_1) and (u_2, v_2) with (x_1y_1, y_1x_1) and (x_2y_2, y_2x_2) ,

$$\begin{aligned} u &= x_1y_1 \cdots x_1y_1 \cdots x_2y_2 \underbrace{x_2y_2}_{\text{cut region}} \cdots = pq \\ v &= y_1x_1 \cdots y_1x_1 y_2x_2 \cdots y_2x_2 = qp \end{aligned}$$

Note that there is at least one x_2y_2 before the cut. We compare the suffixes of p in u and v . Since p ends with y_2x_2 in v , the cut in u should be at the end of x_2 by Case I. of Cut Lemma. Note that by Case II. of Cut Lemma, the cut can end after y_2 but then $u_2 = x_2y_2 = y_2x_2 = v_2$ and hence either $x_2 = \varepsilon$ or $y_2 = \varepsilon$. Hence q is of the form y_2 or $(y_2x_2)^+y_2$ depending upon if the cut in u is within the last x_2y_2 or not.

$$\begin{aligned} u &= \underbrace{x_1y_1 \cdots x_1y_1 \cdots x_2y_2x_2}_p \underbrace{\cdots y_2}_q \\ v &= y_1x_1 \cdots y_1x_1 y_2x_2 \cdots y_2x_2 = qp \end{aligned}$$

If $q \in (y_2x_2)^+y_2$, then comparing the prefixes of q of length $|y_2x_2| = |y_1x_1|$ (Since $|v_1| = |v_2|$) in u and v , we obtain $y_2x_2 = y_1x_1$. Substituting this in v ,

$$\begin{aligned} u &= \underbrace{x_1y_1 \cdots x_1y_1 \cdots x_2y_2x_2}_p \underbrace{\cdots y_2}_q \\ v &= y_1x_1 \cdots y_1x_1 y_1x_1 \cdots y_1x_1 = qp \end{aligned}$$

We compare the prefixes of p in u and v . Since p starts with x_1y_1 in u , the cut in v should be at the end of y_1 using Cut Lemma. Therefore, $q \in (y_1x_1)^+y_1$ in v and $q \in (y_2x_2)^+y_2$ in u . Hence, as before we can deduce that $y_1 = y_2$. It also implies $x_1 = x_2$ since $y_1x_1 = y_2x_2$ and thus (u_1, v_1) and (u_2, v_2) are identical.

If $q = y_2$. The cut in v is within first y_1x_1 . In fact, $q = y_1$ since the only possible cut in u will be at the end of y_1 by Cut Lemma (comparing the prefixes of p in u and v). Hence $y_1 = y_2$. $\square$

3. Common Witness Theorems

In this section, it is shown that an infinite set of pairs that is generated by a sumfree set is conjugate if and only if there is a word witnessing its conjugacy.

3.1. Common Witness and its Characterisations

Recall from Theorem 1 that a pair of words (u, v) is conjugate, then there exists a word z such that $uz = zv$ where $u = xy, v = yx$ and $z \in (xy)^*x$. By symmetry of conjugacy, there also exists a word z' such that $z'u = vz'$ where $z' \in (yx)^*y$. We call z (resp. z') in the above characterisation as an *inner witness* (resp. *outer witness*) of the pair (u, v) (since z is appended to the inner ends).

Given a conjugate pair (u, v) , the set of all inner witnesses of (u, v) is $\{z \mid uz = zv\} = \cup_{\{(x,y) \mid u=xy, v=yx\}} (xy)^*x$. Similarly, the set of all outer witnesses of (u, v) is $\{z \mid zu = vz\} = \cup_{\{(x,y) \mid u=xy, v=yx\}} (yx)^*y$.

For example, the pair (aba, baa) has inner witnesses $(aba)^*a$ and outer witnesses $(baa)^*ba$.

An inner witness of a pair (u, v) is an outer witness of the pair (v, u) . We say that a pair of words has a *witness* if it has either an inner witness or an outer witness.

The following proposition shows that a pair and its primitive root shares the same set of witnesses.

Proposition 9. *Let (u, v) be a conjugate pair with the primitive root (ρ_u, ρ_v) . The following are equivalent for a word z .*

1. z is an inner (resp. outer) witness of (u, v) .
2. z is an inner (resp. outer) witness of (ρ_u, ρ_v) .

Proof. We prove (1) $\iff$ (2) for inner witness. The outer witness case is symmetric.

(2) $\Rightarrow$ (1): Assume that z is an inner witness of (ρ_u, ρ_v) . By induction on n , we prove that z is also an inner witness of $(\rho_u, \rho_v)^n$ for all $n \geq 1$, i.e., $\rho_u^n z = z \rho_v^n$. It is true when $n = 1$. For all $n > 1$,

$$\begin{aligned} \rho_u^n z &= \rho_u^{n-1} \rho_u z \\ &= \rho_u^{n-1} z \rho_v && \text{(Since } \rho_u z = z \rho_v \text{)} \\ &= z \rho_v^{n-1} \rho_v && \text{(Inductive Hypothesis)} \\ &= z \rho_v^n \end{aligned}$$

Since (u, v) is a conjugate pair, there exists an integer $n \geq 1$ such that $(u, v) = (\rho_u, \rho_v)^n$ by Proposition 6. Hence, z is also an inner witness for (u, v) .

(1) $\Rightarrow$ (2): Since (u, v) is conjugate, its primitive root (ρ_u, ρ_v) is conjugate as well by Proposition 6. Consider the case when $u \neq v$. It follows that $\rho_u \neq \rho_v$. According to Proposition 7, the pair (ρ_u, ρ_v) has a unique cut, denoted as (x, y) . From Proposition 8, all cuts of (u, v) are of the form $((xy)^*x, (yx)^*y)$. From Theorem 1, an inner witness of (u, v) belongs to

$$((xy)^*x(yx)^*y)^*(xy)^*x = (xy)^*x$$

and hence is an inner witness of (ρ_u, ρ_v) .

In the case where $u = v$, it follows that $\rho_u = \rho_v$. Consequently, any witness z for the pair (u, u) belongs to the set u^* that is a subset of ρ_u^* . Thus, z is also a witness for the pair (ρ_u, ρ_v) , since ρ_u^* consists of witnesses of (ρ_u, ρ_v) . $\square$

Proposition 10. Let ρ, ρ_1, ρ_2 be nonempty primitive words that are pair-wise conjugate, and let α be an arbitrary word. If $\rho_1\alpha\rho_2$ is an infix of a word in ρ^* , then α is an inner witness of (ρ_1, ρ_2) , i.e. $\rho_1\alpha = \alpha\rho_2$.

Proof. Since (ρ_1, ρ_2) are conjugate, there exists a cut (x, y) such that $\rho_1 = xy$ and $\rho_2 = yx$. Moreover, since $\rho \sim \rho_1 \sim \rho_2$, all three words have the same length, say $p > 0$. Further, since $\rho \sim \rho_1 = xy$, we have $\rho^+ = s\rho_1^*t = s(xy)^*t$ for some words s, t such that $|st| = p$.

Assume that $\rho_1\alpha\rho_2$ is an infix of a word $w \in \rho^+$. By the previous observation $w = s(xy)^it$ for some sufficiently large $i > 0$. Firstly, assume $\rho_1 \neq \rho_2$. Then, the cut (x, y) is non-empty and unique by Proposition 7. Since ρ_1 and ρ_2 are primitive, by Item II. of Cut Lemma, neither xy nor yx is equal to any of its non-trivial cyclic shifts. Given that $\rho_1\alpha\rho_2 = xy\alpha yx$ occurs as an infix of w . We do a case analysis. If $xy\alpha yx$ lies entirely within the factor $(xy)^i$ of w , then by Cut Lemma, the factor xy of $xy\alpha yx$ must align exactly with a factor xy of w , and likewise yx should match exactly with a factor yx of w . As a consequence, $\alpha \in (xy)^*x$ which is an inner witness of (ρ_1, ρ_2) .

Next, assume for the sake of contradiction that $xy\alpha yx$ is a factor of $s(xy)^i$, but not a factor of $(xy)^i$. Hence a suffix of s is a prefix of xy (recall that $|s| \leq p = |xy|$). Since $\rho \sim xy$ and $|s| \leq p$, s is an infix of xy . Hence, the prefix xy of $\rho_1\alpha\rho_2$ matches with a nontrivial cyclic shift of xy in w . This contradicts Cut Lemma. A symmetric argument rules out the case when $xy\alpha yx$ is a factor of $(xy)^it$, but not a factor of $(xy)^i$.

When $\rho_1 = \rho_2$, the only cuts possible are empty cuts (i.e., either x or y is empty) by Proposition 7. Thus $\rho_1 = \rho_2 = x$, and the previous argument applies directly, and we get $\alpha \in x^*$, and hence is an inner witness of (ρ_1, ρ_2) . $\square$

We generalise the notion of a witness of a pair to a set of pairs.

Definition 5 (Common Witness). A word is a common inner witness of a set of pairs P if it is an inner witness of each pair in P . Similarly, a word is a common outer witness of P if it is an outer witness of each pair in P .

A set of pairs has a common witness if it has either a common inner witness or a common outer witness. We denote by $W(P)$ the set of all common witnesses of P .

The structure of a common witness of a set of pairs is obtained from Theorem 1.

Proposition 11. Let P be a set of pairs of words. The following are equivalent.

1. z is a common inner witness of P .
2. There exists a cut (x, y) of each pair $(u, v) \in P$ such that $z \in \bigcap_{(u,v) \in P} (xy)^*x$.
3. $z \in \bigcap_{(u,v) \in P} \bigcup_{\{(x,y) | u=xy, v=yx\}} (xy)^*x$

The statement for common outer witness is analogous.

Proof. Follows from the definition of common witness and Theorem 1. $\square$

Example 4. Consider the set $P = \{(ab, ba), (abab, baba)\}$. The pair (ab, ba) has a unique cut (a, b) , and the pair $(abab, baba)$ has two cuts: (a, bab) and (aba, b) . The word a is a common inner witness of P since a belongs to both $(ab)^*a$ and $(abab)^*a$ (using the first cut). Similarly, aba is also a common inner witness of P since aba belongs to both $(ab)^*a$ and $(abab)^*aba$ (using the second cut). Notice that aba is not in the intersection of $(ab)^*a$ and $(abab)^*a$.

Proposition 9 connecting witness of a conjugate pair and its root can be lifted to a set of conjugate pairs and its root as follows.

Proposition 12. The common witnesses of a set of conjugate pairs G and its root $R(G)$ are the same, i.e., a word z is a common inner (resp. outer) witness of G iff z is a common inner (resp. outer) witness of $R(G)$.

Proof. We prove for common inner witness. The common outer witness case is symmetric.

z is a common inner witness of G

$$\iff z \text{ is an inner witness of } (u, v) \text{ for each } (u, v) \in G$$

$$\iff z \text{ is an inner witness of } (\rho_u, \rho_v) \text{ for each } (u, v) \in G \quad (\text{By Proposition 9})$$

$$\iff z \text{ is a common inner witness of } R(G)$$

$\square$

When a set is not conjugate, clearly it has no common witness. However, even when a set is conjugate, it may have both common inner and outer witnesses, or only common inner witness, or only common outer witness, or neither of them as shown below.

Example 5. Consider the set $P = \{(ab, ba), (ac, ca)\}$. The pair (ab, ba) has inner witnesses $(ab)^*a$ and outer witnesses $(ba)^*b$. Similarly, the pair (ac, ca) has inner witnesses $(ac)^*a$ and outer witnesses $(ca)^*c$. According to Proposition 11, the set P has a unique common inner witness $a = (ab)^*a \cap (ac)^*a$, but it does not have any common outer witness since $(ba)^*b \cap (ca)^*c = \emptyset$.

The set $\{(ab, ba), (abab, baba)\}$ has both common inner witnesses

$$(ab)^*a = (ab)^*a \cap ((abab)^*aba \cup (abab)^*a)$$

and common outer witnesses

$$(ba)^*b = (ba)^*b \cap ((baba)^*b \cup (baba)^*bab).$$

However, the set $\{(ab, ba), (ba, ab)\}$ has no common witnesses since $(ab)^*a \cap (ba)^*b = \emptyset$.

Next we analyse the number of common witnesses a set of pairs can have.

Proposition 13. Let G be a set of conjugate pairs of words. The following are equivalent.

1. G has more than one common witness.
2. G has infinitely many common witnesses.
3. G has infinitely many common inner witnesses.
4. G has infinitely many common outer witnesses.
5. All the pairs in G have the same primitive root.

Proof. We prove $(1) \Rightarrow (2) \Rightarrow (1)$ and $(1) \Rightarrow (5) \Rightarrow (3), (4) \Rightarrow (1)$.

$(2) \Rightarrow (1)$ and $(3), (4) \Rightarrow (1)$ is obvious. We first show $(5) \Rightarrow (3), (4)$. Since all the pairs in G have the same primitive root, $R(G)$ is singleton. Thus, $R(G)$ has infinitely many common inner as well as common outer witnesses by Theorem 1. Since G and $R(G)$ have the same witnesses (Proposition 12), G also has infinitely many common inner and common outer witnesses.

Now it remains to show that $(1) \Rightarrow (2)$ and $(1) \Rightarrow (5)$. If the set G has two common witnesses, namely z_1 and z_2 , then according to Proposition 12, z_1 and z_2 are also common witnesses of $R(G)$. Let $R(G) = \{(u_i, v_i) \mid i \in I\}$ where I is an Index set.

Suppose z_1 and z_2 are two common *inner* witnesses of $R(G)$. We choose cuts (x_i, y_i) for each $(u_i, v_i) \in R(G)$ such that both z_1 and z_2 belongs to $\bigcap_{i \in I} (x_i y_i)^* x_i$. Since $R(G)$ is a set of primitive pairs, (x_i, y_i) is the unique cut of (u_i, v_i) when $u_i \neq v_i$ by Proposition 7. When $u_i = v_i$, it has only two empty cuts, namely (ϵ, u_i) and (u_i, ϵ) (if $u_i = v_i$). The inner witnesses obtained using cut (ϵ, u_i) is a superset of inner witnesses obtained using (u_i, ϵ) . Hence, we can choose cut $(x_i, y_i) = (\epsilon, u_i)$ for pair (u_i, v_i) when $u_i = v_i$. Therefore, as stated in Proposition 11, both z_1 and z_2 belongs to $\bigcap_{i \in I} (x_i y_i)^* x_i$.

Without loss of generality, assume that $|z_1| < |z_2|$. As depicted in Figure 2, a common factor $w \in \bigcap_{i \in I} (y_i x_i)^{\geq 1}$ exists for each u_i^ω that can be repeated one after another in u_i^ω to get longer and longer common inner witnesses. By symmetry, when z_1 and z_2 are both common *outer* witnesses of $R(G)$, we get infinitely many common outer witnesses.

Now, WLOG assume that z_1 is a common *inner* witness and z_2 is a common *outer* witness of $R(G)$, where $z_1 \neq z_2$. Here also we choose cuts (x_i, y_i) for each $(u_i, v_i) \in R(G)$ such that z_1 belongs to $\bigcap_{i \in I} (x_i y_i)^* x_i$ and z_2 belongs to $\bigcap_{i \in I} (y_i x_i)^* y_i$. For each distinct primitive pair in $R(G)$, there exists a unique cut. However, for identical primitive pairs, we fix a cut based on the values of z_1 and z_2 . We consider two cases: either $z_1, z_2 \neq \epsilon$, or exactly one of z_1 and z_2 is equal to ϵ . In the case where $z_1, z_2 \neq \epsilon$, for primitive pairs (u_i, v_i) such that $u_i = v_i$, we can choose either of the two empty cuts as (x_i, y_i) , resulting in $z_1 \in (x_i y_i)^* x_i$ and $z_2 \in (y_i x_i)^* y_i$. In the second case, if $z_1 = \epsilon$, we select the cut $(x_i, y_i) = (\epsilon, u_i)$. This choice ensures that $z_1 \in (x_i y_i)^* x_i$ and $z_2 \in (y_i x_i)^* y_i$ (since $z_2 \neq \epsilon$). Similarly, if $z_2 = \epsilon$,

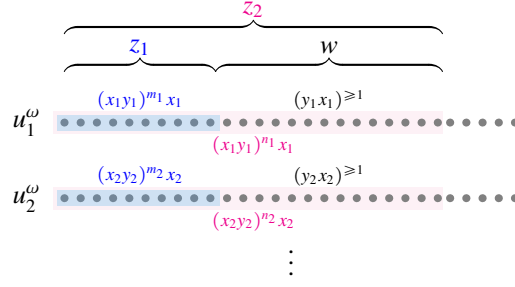


Figure 2: When there are at least two common inner witnesses z_1, z_2 .

we choose the cut $(x_i, y_i) = (u_i, \epsilon)$. Consequently, we can conclude that z_1 belongs to $\bigcap_{i \in I} (x_i y_i)^* x_i$ and z_2 belongs to $\bigcap_{i \in I} (y_i x_i)^* y_i$.

As shown in Figure 3, concatenating $z_1 \cdot z_2 \cdot z_1$ in u_i^ω , we get one more common inner witness z_3 for $R(G)$. By the above argument, $R(G)$ has infinitely many common witnesses. Since witnesses of $R(G)$ are also witnesses of G (Proposition 12), it implies that G itself has infinitely many common witnesses. This completes the proof of (1) $\Rightarrow$ (2).

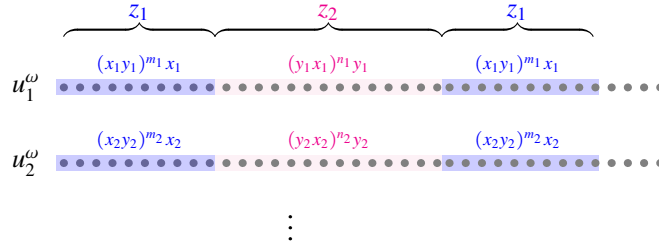


Figure 3: When there are 1 common inner witness z_1 and 1 common outer witness z_2 .

In both the cases, we get that $(x_1 y_1)^\omega = (x_2 y_2)^\omega = \dots$ and $(y_1 x_1)^\omega = (y_2 x_2)^\omega = \dots$. Hence from Fine and Wilf Theorem 4, all u_i 's has the same primitive root. Similarly, all v_i 's has the same primitive root. This proves (1) $\Rightarrow$ (5). $\square$

Therefore, a set of pairs can have no common witness, a unique common witness, or infinitely many common witnesses.

Corollary 2. *For all set of pairs G , if $W(G) \neq \emptyset$, then one of the following is true.*

1. $|W(G)| = \infty$ and $R(G)$ is singleton.
2. $|W(G)| = 1$ and $R(G)$ is not singleton.

3.2. Common Witness Theorem for Kleene Closure

Lyndon-Schützenberger theorem characterises conjugacy of a pair of words. We generalise the notion in Theorem 1 to an infinite set of pairs closed under concatenation. The question we ask is:

“Given an arbitrary set of pairs G , is G^* conjugate?”

If G^* has a common witness, then each pair in G^* has a witness and is conjugate. Hence G^* is conjugate. We prove the converse, namely, if G^* is conjugate, then it has a common witness. The below theorem characterises conjugacy of a freely generated set of pairs of words.

Theorem 6 (Common Witness Theorem for Kleene Closure). *Let G be an arbitrary conjugate set of pairs of words. The following are equivalent.*

1. G^* is conjugate.
2. G^* has a common witness z .
3. G has a common witness z .
4. $R(G)$ has a common witness z .

Proof. We prove $(4) \Rightarrow (3) \Rightarrow (2) \Rightarrow (1) \Rightarrow (4)$. The only nontrivial direction is $(1) \Rightarrow (4)$ that is proved in Section 4.

$(4) \Rightarrow (3)$ Follows from Proposition 12.

$(3) \Rightarrow (2)$ Suppose there exists a common inner witness z of the set G . Hence $\forall (u, v) \in G, uz = zv$. Let (u', v') be any arbitrary element from G^* , i.e., $(u', v') = (u_1 \cdots u_n, v_1 \cdots v_n)$ for some $n \geq 1$ and $(u_i, v_i) \in G$ for $1 \leq i \leq n$. By induction on n , we equate $u'z = zv'$ as follows. Thus, z is a common inner witness of G^* . The proof for common outer witness is symmetric.

$$\begin{aligned}
u'z &= u_1 \cdots u_{n-1} u_n z \\
&= u_1 \cdots u_{n-1} z v_n && \text{(Since } u_n z = z v_n \text{)} \\
&= z v_1 \cdots v_{n-1} v_n && \text{(Inductive Hypothesis)} \\
&= z v'
\end{aligned}$$

$(2) \Rightarrow (1)$ Follows from Theorem 1.

□

Remark 2. Note that whenever G^* is conjugate even though G and $R(G)$ have the same common witnesses, G^* need not be equal to $R(G)^*$. Indeed, $G^* \subseteq R(G)^*$.

As a corollary, we get that for any rational expression E , the expression E^* is conjugate if, and only if, E has a common witness. Below is an illustration of the common witness theorem for a set of pairs that is not rational.

Example 6. Let $G = \{(ab^p, b^p a) \mid p \text{ is a prime number}\}$. The word $a \in \bigcap_{p \in \mathbb{N}, p \text{ is a prime}} (ab^p)^* a$ is a common inner witness of the set G . It is also easy to verify that G^* is conjugate and a is a common inner witness of G^* .

3.3. Conjugacy and Common Witness Theorem for Monoid Closure

Next we state the conjugacy theorem for monoid closures, i.e., sumfree sets of the form

$$M = (\alpha_0, \beta_0) G_1^* (\alpha_1, \beta_1) G_2^* \cdots (\alpha_{k-1}, \beta_{k-1}) G_k^* (\alpha_k, \beta_k), k > 0.$$

where $G_1^*, G_2^*, \dots, G_k^*$ are arbitrary nonempty sets of conjugate pairs.

Theorem 7. ²[Conjugacy Theorem for Monoid Closure] Let $M = (\alpha_0, \beta_0) (\prod_{i=1}^k G_i^* (\alpha_i, \beta_i))$, $k \geq 1$, be a sumfree set. M is conjugate if, and only if, one of the following holds.

1. M has a common witness.
2. For each $i \in [1, k]$, G_i^* is conjugate and there exist a primitive word ρ such that $\text{fst}(G_i^*) \sim \text{snd}(G_i^*) \subseteq \rho^*$, and there exists a word $v \sim \alpha_0 \cdots \alpha_k \sim \beta_0 \cdots \beta_k$ such that $\text{fst}(M) \sim \text{snd}(M) \subseteq v\rho^*$.

We give the proof of the above theorem in Section 6. Note that the above result does not generalise to arbitrary sets of pairs, in particular, rational sets using sum.

²We thank the participants of Autoboz 2025 (Grenaa, Denmark) for identifying a missing case in the conjugacy theorem for monoid closure (Item 2 of Theorem 7).

Example 7. $(ab, ba)^* + (baa, aab)^*$ is an infinite conjugate set with no common witness. Also, it does not satisfy Item (2) of Theorem 7.

Definition 6 (Redux, Singleton Redux). Let M be the sumfree set

$$(\alpha_0, \beta_0)G_1^*(\alpha_1, \beta_1)G_2^* \cdots (\alpha_{k-1}, \beta_{k-1})G_k^*(\alpha_k, \beta_k).$$

The redux of M is the pair $(\alpha_0\alpha_1 \cdots \alpha_k, \beta_0\beta_1 \cdots \beta_k)$ obtained by substituting each G_i^* by the empty pair (ϵ, ϵ) . A singleton redux of M is a set obtained by substituting all but one of the G_i^* 's by the empty pair (ϵ, ϵ) . They are of the form $(\alpha_0 \cdots \alpha_{i-1}, \beta_0 \cdots \beta_{i-1})G_i^*(\alpha_i \cdots \alpha_k, \beta_i \cdots \beta_k)$ where $1 \leq i \leq k$.

Example 8. Consider the set $M = (a, a)(baa, aba)^*(b, a)(aab, baa)^*(a, b)$. The redux of M is (aba, aab) , and its singleton reduxes are $(a, a)(baa, aba)^*(ba, ab)$ and $(ab, aa)(aab, baa)^*(a, b)$.

If a sumfree set has a common witness, it is conjugate. The converse need not be true as illustrated in the below example.

Example 9. Consider the sumfree set

$$M = \begin{pmatrix} \epsilon \\ c \end{pmatrix} \begin{pmatrix} ab \\ ba \end{pmatrix}^* \begin{pmatrix} c \\ b \end{pmatrix} \begin{pmatrix} ba \\ ab \end{pmatrix}^* \begin{pmatrix} b \\ \epsilon \end{pmatrix}$$

The set $M = \{((ab)^n c (ba)^m b, c (ba)^n b (ab)^m) \mid n, m \geq 0\}$ is conjugate and has no common witness. However, $G_1^* = (ab, ba)^*$ and $G_2^* = (ba, ab)^*$ is conjugate and there exist a primitive word $\rho = ab$ such that $\text{fst}(G_i^*) \sim \text{snd}(G_i^*) \subseteq \rho^* = (ab)^*$, and there exists a word $v = cb$ (which is a cyclic shift of the redux of M) such that $\text{fst}(M) \sim \text{snd}(M) \subseteq v\rho^* = (cb)(ab)^*$.

Note that the v and ρ need not be unique. For $\rho = ba$ and $v = bc$, it is also true that $\text{fst}(M) \sim \text{snd}(M) \subseteq v\rho^* = (bc)(ba)^*$.

For sumfree sets of the form $M = (\alpha_0, \beta_0)G^*(\alpha_1, \beta_1)$ with only one Kleene closure, we show that M is conjugate iff M has a common witness. Later, we show that if a general sumfree set has a common witness then that is in the intersection of the common witnesses of the singleton reduxes of the set. Towards this, we need the following definition.

Definition 7 (Prefix Delay, Suffix Delay). If u and v are words such that one of them is a prefix of another, we define the prefix delay between u and v as

$$[u, v]_L = \begin{cases} u^{-1}v & \text{if } u \text{ is a prefix of } v \\ v^{-1}u & \text{if } v \text{ is a prefix of } u \end{cases}$$

Similarly, the suffix delay of two words u and v such that one of them is suffix of another, denoted by $[u, v]_R$, is vu^{-1} if u is a suffix of v and uv^{-1} if v is a suffix of u .

For example, $[abaa, ab]_L = aa = [ab, abaa]_L$.

Following is the common witness theorem for a sumfree set with only one Kleene star, i.e., sets of the form $(\alpha_0, \beta_0)G^*(\alpha_1, \beta_1)$. In short it states that such a set is conjugate if and only if it has a common witness that is determined by the common witnesses of $G \cup \{(\alpha_1\alpha_0, \beta_1\beta_0)\}$.

Theorem 8. Let $M = (\alpha_0, \beta_0)G^*(\alpha_1, \beta_1)$ be a sumfree set with nonempty redux. The following are equivalent.

1. M is conjugate.
2. There exist a common witness of $G \cup \{(\alpha_1\alpha_0, \beta_1\beta_0)\}$.
3. M has a common witness.

Furthermore,

- (a) If the set $G \cup \{(\alpha_1\alpha_0, \beta_1\beta_0)\}$ has a unique common inner witness, say z' , then M has a unique common witness $z = [\alpha_0z', \beta_0]_R = [\alpha_1, z'\beta_1]_L$. Moreover, if $|\alpha_0z'| \geq |\beta_0|$ or equivalently $|\alpha_1| \leq |z'\beta_1|$, then z is a common inner witness, otherwise it is a common outer witness.
- (b) If the set $G \cup \{(\alpha_1\alpha_0, \beta_1\beta_0)\}$ has a unique common outer witness, say z' , then M has a unique common witness $z = [\alpha_0, \beta_0z']_R = [z'\alpha_1, \beta_1]_L$. Moreover, if $|z'\alpha_1| \geq |\beta_1|$ or equivalently $|\alpha_0| \leq |\beta_0z'|$, then z is a common outer witness, otherwise it is a common inner witness.
- (c) If $G \cup \{(\alpha_1\alpha_0, \beta_1\beta_0)\}$ has infinitely many common witnesses, then M is a subset of powers of the primitive root of its redux. Thus, M has infinitely many common witnesses.

Example 10. Let $M = (\alpha_0, \beta_0)G^*(\alpha_1, \beta_1)$ be a sumfree set with one Kleene star where

$$\begin{pmatrix} \alpha_0 \\ \beta_0 \end{pmatrix} = \begin{pmatrix} ab \\ b \end{pmatrix}, G = \left\{ \begin{pmatrix} bab \\ abb \end{pmatrix} \right\}, \begin{pmatrix} \alpha_1 \\ \beta_1 \end{pmatrix} = \begin{pmatrix} b \\ ab \end{pmatrix}.$$

The redux of M is $(\alpha_0\alpha_1, \beta_0\beta_1) = (abb, bab)$. The set $G \cup \{(\alpha_1\alpha_0, \beta_1\beta_0)\} = \{(bab, abb)\} \cup \{(bab, abb)\} = \{(bab, abb)\}$ and, hence it has infinitely many common witnesses. By Theorem 8 (c), M is a subset of powers of the primitive root of the redux, i.e., $M = (abb, bab)^+$. Therefore, M has infinitely many witnesses same as those of (abb, bab) .

A singleton redux of a sumfree set is nothing but a sumfree set with only one Kleene star. Given any sumfree set M , if M is conjugate, each of its singleton reduxes are conjugate. From Theorem 8, a singleton redux of M has a common witness. Further, we prove that M has a common witness z iff z is the common witness of each of its singleton reduxes. The below theorem characterises the common witness of a general sumfree set.

Theorem 9 (Common Witness Theorem for Monoid Closure). *Let M be a sumfree set. The following are equivalent.*

1. z is a common witness of each singleton redux of M .
2. z is a common witness of M .

Example 11. Let $M = (\alpha_0, \beta_0)G_1^*(\alpha_1, \beta_1)G_2^*(\alpha_2, \beta_2)$ be a sumfree set with two Kleene star where

$$\begin{pmatrix} \alpha_0 \\ \beta_0 \end{pmatrix} = \begin{pmatrix} b \\ a \end{pmatrix}, G_1 = \left\{ \begin{pmatrix} ac \\ ca \end{pmatrix} \right\}, \begin{pmatrix} \alpha_1 \\ \beta_1 \end{pmatrix} = \begin{pmatrix} ab \\ b \end{pmatrix}, G_2 = \left\{ \begin{pmatrix} bab \\ bab \end{pmatrix} \right\}, \begin{pmatrix} \alpha_2 \\ \beta_2 \end{pmatrix} = \begin{pmatrix} \epsilon \\ b \end{pmatrix}.$$

The redux of M is $(\alpha_0\alpha_1\alpha_2, \beta_0\beta_1\beta_2) = (bab, abb)$. The set M has two singleton reduxes,

$$M_1 = \begin{pmatrix} \alpha_0 \\ \beta_0 \end{pmatrix} G_1^* \begin{pmatrix} \alpha_1\alpha_2 \\ \beta_1\beta_2 \end{pmatrix} = \begin{pmatrix} b \\ a \end{pmatrix} \begin{pmatrix} ac \\ ca \end{pmatrix}^* \begin{pmatrix} ab \\ bb \end{pmatrix}$$

and,

$$M_2 = \begin{pmatrix} \alpha_0\alpha_1 \\ \beta_0\beta_1 \end{pmatrix} G_2^* \begin{pmatrix} \alpha_2 \\ \beta_2 \end{pmatrix} = \begin{pmatrix} bab \\ ab \end{pmatrix} \begin{pmatrix} bab \\ bab \end{pmatrix}^* \begin{pmatrix} \epsilon \\ b \end{pmatrix}.$$

The set $G_1 \cup \{(\alpha_1\alpha_2\alpha_0, \beta_1\beta_2\beta_0)\} = \{(ac, ca)\} \cup \{(abb, bba)\} = \{(ac, ca), (abb, bba)\}$ has a unique common inner witness, say $z_1 = a = (ac)^*a \cap (abb)^*a$ and no common outer witness since $(ca)^*c \cap (bba)^*bb = \emptyset$. By Theorem 8 (a), the unique common inner witness of the singleton redux M_1 of M is $[\alpha_0z_1, \beta_0]_R = [ba, a]_R = b$.

The set $G_2 \cup \{(\alpha_2\alpha_0\alpha_1, \beta_2\beta_0\beta_1)\} = \{(bab, bab)\}$ has infinitely many common witnesses. Thus the singleton redux M_2 is a subset of powers of the primitive root of the redux using Theorem 8 (c), i.e., $M_2 = (bab, abb)^+$. Thus M_2 have infinitely many common inner witnesses $(bab)^*b$ and common outer witnesses $(abb)^*ab$.

By Theorem 9, M has a unique common inner witness $b \cap (bab)^*b = b$, that equals to the intersection of the common inner witness of its singleton reduxes M_1 and M_2 .

The proof of the common witness theorems of singleton redux as well as the monoid closure are given in Section 5. Finally, using those results we prove the conjugacy theorem for monoid closure (Theorem 7) in Section 6

4. Existence of Common Witness for Kleene Closure

In this section, we prove the direction (1) $\Rightarrow$ (4) of Theorem 6 recalled in the following lemma.

Lemma 4. *For a set of pairs G , if G^* is conjugate then $R(G)$ has a common witness.*

We prove the lemma when G is finite by case analysis and then extend it for a countably infinite set of pairs using a compactness argument. Towards proving this, we need the following auxiliary result proven using Equal Length Lemma. A corollary of the proposition below is that if G^* is conjugate and $R(G)$ only consists of pairs of identical length, then $R(G)$ has a common witness.

Proposition 14. *Let $G = \{(u_i, v_i) \mid i \in \{1, \dots, k\}\}$ be a set of $k \geq 2$ conjugate pairs such that all pairs in $R(G)$ are of equal length. If $(u_1, v_1)^*(u_2, v_2)^* \dots (u_k, v_k)^*$ is conjugate then either $x_1 = x_2 = \dots = x_k$ or $y_1 = y_2 = \dots = y_k$ where (x_i, y_i) is a cut of the primitive pair (ρ_{u_i}, ρ_{v_i}) for $i \in [k]$.*

Proof. Proof is by induction on the number of pairs in $R(G)$.

Base Case: When $R(G)$ has only 2 pairs, i.e., let $R(G) = \{(\rho_{u_1}, \rho_{v_1}), (\rho_{u_2}, \rho_{v_2})\}$. There exist ℓ_1, ℓ_2 such that $\ell_2 > \ell_1 + 2$, $\ell_1 > 2$ and $(\rho_{u_1}, \rho_{v_1})^{\ell_1} (\rho_{u_2}, \rho_{v_2})^{\ell_2} \in (u_1, v_1)^* (u_2, v_2)^*$ and hence it is conjugate. From Equal Length Lemma we get either $x_1 = x_2$ or $y_1 = y_2$.

We also consider the case when $R(G)$ has exactly 3 pairs, i.e., let $R(G) = \{(\rho_{u_1}, \rho_{v_1}), (\rho_{u_2}, \rho_{v_2}), (\rho_{u_3}, \rho_{v_3})\}$. For $i, j \in [3]$, there exists set of pairs $\{(\rho_{u_i}, \rho_{v_i})^{\ell_i} (\rho_{u_j}, \rho_{v_j})^{\ell_j} \mid \ell_j > \ell_i + 2, \ell_i > 2\} \subset (u_1, v_1)^* (u_2, v_2)^* (u_3, v_3)^*$ that is conjugate. Thus, the pairs (ρ_{u_i}, ρ_{v_i}) and (ρ_{u_j}, ρ_{v_j}) satisfy either $x_i = x_j$ or $y_i = y_j$ by Equal Length Lemma where (x_i, y_i) and (x_j, y_j) are cuts of (ρ_{u_i}, ρ_{v_i}) (ρ_{u_j}, ρ_{v_j}) respectively. There are 3 cases.

1. All pairs in $R(G)$ are identical. Then there exists cuts (x_i, y_i) for each pair $(\rho_{u_i}, \rho_{v_i}) \in R(G)$ for $i \in [3]$ such that $x_1 = x_2 = x_3 = \varepsilon$. Hence, the proposition holds.
2. Exactly one pair in $R(G)$ is identical. Let (ρ_{u_1}, ρ_{v_1}) and (ρ_{u_2}, ρ_{v_2}) are distinct pairs with unique nonempty cuts (x_1, y_1) and (x_2, y_2) respectively. By Lemma 3, either $x_1 = x_2$ or $y_1 = y_2$. Assume that $x_1 = x_2$. Also, since (ρ_{u_3}, ρ_{v_3}) is identical pair, either $\rho_{u_3} = x_i$ or $\rho_{u_3} = y_i$ for $i \in [1, 2]$. If there exists an $i \in [1, 2]$ such that $\rho_{u_3} = x_i$, then $x_1 = x_2 = \rho_{u_3} = x_3$ where $(x_3, y_3) = (\rho_{u_3}, \varepsilon)$ is a cut of (ρ_{u_3}, ρ_{v_3}) . Suppose for all $i \in [1, 2]$, $\rho_{u_3} = y_i$, then $y_1 = y_2 = \rho_{u_3} = y_3$ where $(x_3, y_3) = (\varepsilon, \rho_{u_3})$ is a cut of (ρ_{u_3}, ρ_{v_3}) . Hence, the proposition holds.
3. Exactly one pair in $R(G)$ is non-identical. WLOG assume that (ρ_{u_1}, ρ_{v_1}) and (ρ_{u_3}, ρ_{v_3}) are the identical pairs, (ρ_{u_2}, ρ_{v_2}) is non-identical with unique nonempty cut (x_2, y_2) .

Claim 1. $\rho_{u_1} = \rho_{u_3}$.

Proof. Consider the pair $(u, v) = (u_1, v_1)^j (u_2, v_2)^l (u_3, v_3)^j$ where j and l are set such that $|u_2^l| > 4 \cdot |u_1^j| = 4 \cdot |u_3^j|$. Substituting with their primitive roots, we get

$$\begin{aligned} u &= \overbrace{u_1 \dots u_1}^{B_1} \overbrace{u_2 \dots u_2}^{B_2} \overbrace{u_3 \dots u_3}^{B_3} = \rho_{u_1} \dots \rho_{u_1} x_2 y_2 \dots x_2 y_2 \dots x_2 y_2 \rho_{u_3} \dots \rho_{u_3} \\ v &= v_1 \dots v_1 v_2 \dots v_2 v_3 \dots v_3 = \rho_{u_1} \dots \rho_{u_1} y_2 x_2 \dots y_2 x_2 \dots y_2 x_2 \rho_{u_3} \dots \rho_{u_3} \end{aligned}$$

Let B_1, B_2, B_3 denote the block of u_1 's, u_2 's and u_3 's. Similarly, B'_1, B'_2, B'_3 denote the block of v_1 's, v_2 's and v_3 's. Since (u, v) is conjugate, it has a cut (p, q) . We do a case analysis on p .

1. p ends within $B_1(\rho_{u_1})^{-1}$. In this case, $p \in \rho_{u_1}^*$ by Case II. of Cut Lemma since there exist atleast a factor ρ_{u_1} of q in u to match with the prefix ρ_{u_1} of q in v . On further matching q in u and v , since $|\rho_{u_1}| = |x_2 y_2| = |y_2 x_2| = |\rho_{u_3}|$, we get $\rho_{u_1} = x_2 y_2 = y_2 x_2 = \rho_{u_3}$.
2. p ends after block $B_2 \rho_{u_3}$. This case is symmetric to the above case, where we get $q \in \rho_{u_3}^*$ by Case II. of Cut Lemma. On further matching p in u and v , since $|\rho_{u_3}| = |x_2 y_2| = |y_2 x_2| = |\rho_{u_1}|$, we get $\rho_{u_3} = x_2 y_2 = y_2 x_2 = \rho_{u_1}$.
3. p ends within $\rho_{u_1} B_2 \rho_{u_3}$. WLOG assume that p ends within the first half of $\rho_{u_1} B_2 \rho_{u_3}$. Since $|\rho_{u_1} B_2 \rho_{u_3}|/2 > |B_2|/2 \geq 2 \cdot |u_1^j|$, on matching q in u and v , the prefix u_1^j of v matches entirely with block B_2 and there exist at least one factor of $x_2 y_2$ following that. Using Cut Lemma, the prefix $p(u_1^j)$ in u ends in x_2 . Since $|\rho_{u_1}| = |x_2 y_2| = |y_2 x_2|$, we get $\rho_{u_1} = y_2 x_2 = x_2 y_2$, and this contradicts that (u_2, v_2) is a non-identical pair. Similarly, we get a contradiction when p ends withing second half of $\rho_{u_1} B_2 \rho_{u_3}$. $\square$

Thus, we get $\rho_{u_1} = \rho_{u_3}$ when exactly one pair in $R(G)$ is non-identical. Since (x_2, y_2) is the unique nonempty cut of (u_2, v_2) , by Lemma 3, either $x_2 = \rho_{u_1} = \rho_{u_3}$ or $y_2 = \rho_{u_1} = \rho_{u_3}$. In either case the proposition holds.

Inductive Case: Assume that the statement holds for k pairs in $R(G)$, i.e., let

$$R(G) = \{(\rho_{u_1}, \rho_{v_1}), \dots, (\rho_{u_k}, \rho_{v_k})\}.$$

By the induction hypothesis, if $(u_1, v_1)^* \dots (u_k, v_k)^*$ is conjugate, then either $x_1 = \dots = x_k$ or $y_1 = \dots = y_k$. Without loss of generality assume that $x_1 = \dots = x_k$. We prove that the statement also holds for $k+1$ pairs.

Let $G' \supseteq G$ such that $R(G') = R(G) \cup \{(\rho_{u_{k+1}}, \rho_{v_{k+1}})\}$, where $(\rho_{u_{k+1}}, \rho_{v_{k+1}})$ is the conjugate primitive root of $(u_{k+1}, v_{k+1}) \in G'$ and has the same length as the pairs in $R(G)$. Assume that $(u_1, v_1)^* \dots (u_k, v_k)^* (u_{k+1}, v_{k+1})^*$ is conjugate. Then for every $i \in [k]$ there exists a set of pairs

$$\{(\rho_{u_i}, \rho_{v_i})^{\ell_i} (\rho_{u_{k+1}}, \rho_{v_{k+1}})^{\ell_{k+1}} \mid \ell_{k+1} > \ell_i + 2, \ell_i > 2\} \subseteq (u_1, v_1)^* \dots (u_k, v_k)^* (u_{k+1}, v_{k+1})^*$$

that is conjugate. Hence, by Equal Length Lemma, for every $i \in [k]$ the pairs (ρ_{u_i}, ρ_{v_i}) and $(\rho_{u_{k+1}}, \rho_{v_{k+1}})$ satisfy either

$$x_i = x_{k+1} \quad \text{or} \quad y_i = y_{k+1},$$

where (x_i, y_i) and (x_{k+1}, y_{k+1}) are cuts of (ρ_{u_i}, ρ_{v_i}) and $(\rho_{u_{k+1}}, \rho_{v_{k+1}})$ respectively.

We partition $R(G)$ into two sets: $I = \{(\rho, \rho') \in R(G) \mid \rho = \rho'\}$ containing all identical pairs, and $J = R(G) \setminus I$ containing all non-identical pairs. We distinguish the following cases.

1. If $J = \emptyset$, then all pairs in $R(G)$ are identical. Hence there exist cuts (x_i, y_i) such that

$$x_1 = \dots = x_k = \varepsilon.$$

If $(\rho_{u_{k+1}}, \rho_{v_{k+1}})$ is also identical, we can take $x_{k+1} = \varepsilon$ and the statement follows.

If $(\rho_{u_{k+1}}, \rho_{v_{k+1}})$ is non-identical, it has a unique nonempty cut (x_{k+1}, y_{k+1}) . By Lemma 3 together with Claim 1, for all $i, j \in [k]$ either

$$x_{k+1} = \rho_{u_i} = \rho_{u_j} \quad \text{or} \quad y_{k+1} = \rho_{u_i} = \rho_{u_j}.$$

Thus

$$\rho_{u_1} = \dots = \rho_{u_k} = x_{k+1} \quad \text{or} \quad \rho_{u_1} = \dots = \rho_{u_k} = y_{k+1},$$

and the proposition holds.

2. Suppose $J \neq \emptyset$.

- (a) If $I = \emptyset$, then every pair in $R(G)$ is non-identical and has a unique nonempty cut (x_i, y_i) for $i \in [k]$. If $(\rho_{u_{k+1}}, \rho_{v_{k+1}})$ is also non-identical with unique cut (x_{k+1}, y_{k+1}) , then by Lemma 3 either $x_i = x_{k+1}$ for some $i \in [k]$ or $y_i = y_{k+1}$ for all $i \in [k]$. In the former case we obtain $x_1 = \dots = x_k = x_{k+1}$ by the induction hypothesis, and in the latter case $y_1 = \dots = y_k = y_{k+1}$.

If $(\rho_{u_{k+1}}, \rho_{v_{k+1}})$ is identical, then Lemma 3 implies that either $x_i = \rho_{u_{k+1}}$ for some $i \in [k]$ or $y_i = \rho_{u_{k+1}}$ for all $i \in [k]$. Using the induction hypothesis we again obtain the desired equality.

- (b) Suppose both I and J are nonempty. Without loss of generality let

$$I = \{(\rho_{u_1}, \rho_{v_1}), \dots, (\rho_{u_{j-1}}, \rho_{v_{j-1}})\} \text{ and}$$

$$J = \{(\rho_{u_j}, \rho_{v_j}), \dots, (\rho_{u_k}, \rho_{v_k})\}$$

for some $1 < j \leq k$. Each pair in J is non-identical and hence has a unique nonempty cut.

By the induction hypothesis we obtain

$$\rho_{u_1} = \dots = \rho_{u_{j-1}} = x_j = \dots = x_k. \tag{8}$$

If $(\rho_{u_{k+1}}, \rho_{v_{k+1}})$ is non-identical with cut (x_{k+1}, y_{k+1}) , then applying the base case for three pairs yields either

$$\rho_{u_1} = x_i = x_{k+1} \quad \text{for some } i \in \{j, \dots, k\}, \text{ or}$$

$$\rho_{u_1} = y_i = y_{k+1} \quad \text{for all } i \in \{j, \dots, k\}.$$

Using (8) we obtain the desired equality of all x 's or all y 's.

If $(\rho_{u_{k+1}}, \rho_{v_{k+1}})$ is identical, then Lemma 3 implies that either $x_i = \rho_{u_{k+1}}$ for some $i \in \{j, \dots, k\}$ or $y_i = \rho_{u_{k+1}}$ for all such i . Combining this with (8) and Claim 1 again yields the required equality.

□

4.1. For a Finite Set of Pairs

The common witness theorem for a finite set is a straightforward corollary of the below lemma.

Lemma 5. *Let $G = \{(u_i, v_i) \mid 1 \leq i \leq k\}$ be a finite set of $k \in \mathbb{N}$ pairs. If the set $(u_1, v_1)^* (u_2, v_2)^* \cdots (u_k, v_k)^*$ is conjugate then $R(G)$ as well G has a common witness.*

Proof. When $k = 1$, G has only one pair (u, v) and by assumption it is conjugate. By Theorem 1, (u, v) has a witness. From Proposition 9, we obtain that the witnesses of $R(G) = \{(\rho_u, \rho_v)\}$ are same as that of (u, v) .

Next we assume that $k > 1$. Let $\approx$ be the equivalence relation on G whereby $(u, v) \approx (u', v')$ if $\rho_u \sim \rho_{u'}$, i.e., the primitive roots of the pairs are conjugates. Assume that $\approx$ has d equivalence classes. Clearly $1 \leq d \leq k$. We do a cases analysis on whether $d = 1$ or otherwise.

If $\approx$ has only one equivalence class, then the primitive roots of all the pairs in G are conjugates. Consequently, their lengths are identical. Since $(u_1, v_1)^* \cdots (u_k, v_k)^*$ is conjugate and all the pairs in $R(G)$ have identical lengths, by Proposition 14, $R(G)$ has a common witness.

Now we assume that $d > 1$. Choose d pairs $(u_1, v_1), (u_2, v_2), \dots, (u_d, v_d)$ from each equivalence class. We construct a pair $(u, v) \in (u_1, v_1)^* (u_2, v_2)^* \cdots (u_d, v_d)^* \subseteq (u_1, v_1)^* \cdots (u_k, v_k)^*$ and show that (u, v) is conjugate only if $R(G)$ has a common witness.

Let m be the least common multiple of $|u_1|, \dots, |u_d|$. Let $\ell_{ij} = |u_i| + |u_j| - \gcd(|u_i|, |u_j|) > 0$ for $1 \leq i, j \leq d$ and $i \neq j$. Let $\ell = \max \{\ell_{ij} \mid 1 \leq i, j \leq d, i \neq j\}$. Let N be a multiple of m that is $> 2\ell$.

Let $(u, v) = (u_1, v_1)^{j_1} (u_2, v_2)^{j_2} \cdots (u_d, v_d)^{j_d}$ such that $j_1, \dots, j_d > 2$ and $|u_i^{j_i}| = N$ for each $1 \leq i \leq d$.

$$\begin{aligned} u &= \overbrace{u_1 \cdots u_1}^N \overbrace{u_2 \cdots u_2}^N \cdots \overbrace{u_d \cdots u_d}^N \\ v &= v_1 \cdots v_1 v_2 \cdots v_2 \cdots v_d \cdots v_d \end{aligned}$$

Since (u, v) is conjugate, it has a cut, say (p, q) . Substituting each pair in (u, v) with their primitive roots, we get

$$\begin{aligned} u &= \rho_{u_1} \cdots \rho_{u_1} \rho_{u_2} \cdots \rho_{u_2} \cdots \rho_{u_d} \cdots \rho_{u_d} = pq \\ v &= \rho_{v_1} \cdots \rho_{v_1} \rho_{v_2} \cdots \rho_{v_2} \cdots \rho_{v_d} \cdots \rho_{v_d} = qp \end{aligned}$$

Let (x_i, y_i) be a cut of (ρ_{u_i}, ρ_{v_i}) for $1 \leq i \leq d$.

Let $B_1, B_2, \dots, B_d$ represent the blocks in u , and let $B'_1, B'_2, \dots, B'_d$ represent the blocks in v .

The cut in u can be either within the first block B_1 , or the last block B_d , or anywhere between the first and the last blocks in u . We do a case analysis on all the possible cuts (p, q) of (u, v) and show that there exists a common witness of $R(G)$ in each of the cases. We can assume that both p and q are nonempty. When either p or q is empty, then all pairs (u_i, v_i) in (u, v) are identical, and hence either $x_1 = \cdots = x_d = \varepsilon$ or $y_1 = \cdots = y_d = \varepsilon$.

Case 1: When the cut in u is in the first block B_1 . We make a further case analysis depending upon if the cut is within the first half or the second half of the first block.

Suppose the cut in u is within the first half of the block, i.e., p is of length at most $N/2$. In this case, since the length of each block are equal, the cut in v is within the second half of the last block B'_d , i.e., p is a suffix of v of length at most $N/2$.

$$\begin{aligned} u &= \overbrace{\rho_{u_1} \cdots \cdots \rho_{u_1} \rho_{u_2} \cdots \rho_{u_2} \cdots \rho_{u_d} \cdots \rho_{u_d}}^p \\ v &= \underbrace{\rho_{v_1} \cdots \cdots \rho_{v_1} \rho_{v_2} \cdots \rho_{v_2} \cdots \rho_{v_d} \cdots \rho_{v_d}}_q \underbrace{\cdots \cdots \rho_{v_d}}_p \end{aligned}$$

We obtain $q = p^{-1}(B_1 B_2 \cdots B_d) = (B'_1 B'_2 \cdots B'_d) p^{-1}$.

Claim 2. *The following holds for each $1 \leq i \leq d$.*

1. B_i is of the form $p q_i$ and B'_i is of the form $q_i p$, and
2. p is of the form $(x_i y_i)^{m_i} x_i$ and q_i is of the form $(y_i x_i)^{m_i} y_i$ for some $m_i \geq 0$.

Proof. Proof is by induction on i .

1. *Base Case:* when $i = 1$. We compare the prefixes of q in u and v . Since $|p| \leq N/2$, the prefix of q in u must begin within the first block B_1 . Also, there must be at least one occurrence of the factor $\rho_{u_1} = x_1 y_1$ following the cut. Since q in v starts with $\rho_{v_1} = y_1 x_1$, the cut should be at the end of x_1 by Case I. of Cut Lemma. Note that by Case II. of Cut Lemma, the cut can end after y_1 but then $\rho_{u_1} = x_1 y_1 = y_1 x_1 = \rho_{v_1}$ and hence either $x_1 = \varepsilon$ or $y_1 = \varepsilon$. Hence, in both case, we get $p = (x_1 y_1)^{m_1} x_1$ for some integer $m_1 \geq 0$. Consequently, the prefix of q in the block B_1 , denoted as q_1 , is of the form $y_1 (x_1 y_1)^{n_1}$, for some $n_1 > 0$. After matching q_1 in v , we observe that a factor equal to p appears in the suffix of the block B'_1 , as shown below.

$$\begin{aligned} u &= \overbrace{(x_1 y_1)^{m_1} x_1}^p \overbrace{y_1 (x_1 y_1)^{n_1}}^{q_1} \overbrace{\rho_{u_2} \cdots \rho_{u_2} \cdots \rho_{u_d} \cdots \rho_{u_d}}^{q_1^{-1} q} \\ v &= \underbrace{y_1 (x_1 y_1)^{n_1}}_{q_1} \underbrace{(x_1 y_1)^{m_1} x_1}_{=p} \rho_{v_2} \cdots \rho_{v_2} \cdots \rho_{v_d} \cdots \rho_{v_d} = qp \end{aligned}$$

2. *Inductive Case:* Assume the claim is true for first i blocks where $1 \leq i < k$.

$$\begin{aligned} u &= \overbrace{(x_1 y_1)^{m_1} x_1}^p \overbrace{y_1 (x_1 y_1)^{n_1}}^{q_1} \cdots \overbrace{(x_i y_i)^{m_i} x_i}^p \overbrace{y_i (x_i y_i)^{n_i}}^{q_i} \overbrace{\rho_{u_{i+1}} \cdots \rho_{u_{i+1}} \cdots \rho_{u_d} \cdots \rho_{u_d}}^{q'' = (q_1 p \cdots q_{i-1} p q_i)^{-1} q} = pq \\ v &= \underbrace{y_1 (x_1 y_1)^{n_1}}_{q_1} \underbrace{(x_1 y_1)^{m_1} x_1}_p \cdots \underbrace{y_i (x_i y_i)^{n_i}}_{q_i} \underbrace{(x_i y_i)^{m_i} x_i}_p \rho_{v_{i+1}} \cdots \rho_{v_{i+1}} \cdots \rho_{v_d} \cdots \rho_{v_d} = qp \end{aligned}$$

Let q'' denote the remaining suffix of q in u after the i -th block B_i . We obtain $q'' = (q_1 p \cdots q_{i-1} p q_i)^{-1} q$. By comparing the prefixes of q'' in u and v , we get that the suffix of the block B'_i in v , that is equal to p , matches within the first half of the block B_{i+1} in u since $|p| < N/2$. Moreover, the matching should end at x_{i+1} by Case I. of Cut Lemma. Note that by Case II. of Cut Lemma, the matching can end after y_{i+1} but then $\rho_{v_{i+1}} = y_{i+1} x_{i+1} = x_{i+1} y_{i+1} = \rho_{u_{i+1}}$ and hence either $x_{i+1} = \varepsilon$ or $y_{i+1} = \varepsilon$. Hence, $p = (x_{i+1} y_{i+1})^{m_{i+1}} x_{i+1}$ for some integer $m_{i+1} \geq 0$. Consequently, the remaining suffix of the block B_{i+1} , denoted by q_{i+1} , is of the form $(y_{i+1} x_{i+1})^{n_{i+1}} y_{i+1}$ for some integer $n_{i+1} > 0$. After matching q_{i+1} in B'_{i+1} , we observe that a factor equal to p appears in the suffix of the block B'_{i+1} . Hence, $B_{i+1} = p q_{i+1}$ and $B'_{i+1} = q_{i+1} p$.

□

From the above claim, it follows that $q = q_1 p q_2 p \cdots p q_d$ and

$$p = (x_1 y_1)^{m_1} x_1 = (x_2 y_2)^{m_2} x_2 = \cdots = (x_d y_d)^{m_d} x_d$$

for $m_1, \dots, m_d \geq 0$.

From Claim 3, the above equation holds for any two pairs between the equivalence classes. By Proposition 11, we obtain p is a common inner witness of $R(G)$.

Next we assume the cut in u is in the second half of B_1 . We compare the prefixes of q in u and v and deduce that there exist a common factor of length at least $N/2 > \ell > \ell_{12}$ between block B'_1 and block B_2 . From Theorem 5, ρ_{v_1} is conjugate to ρ_{u_2} , that is in turn is conjugate to ρ_{v_2} . From transitivity of conjugacy, ρ_{v_1} is conjugate to ρ_{v_2} , which contradicts the fact that (u_1, v_1) and (u_2, v_2) belong to different equivalence classes. Hence cut in second half of B_1 is not possible.

Case 2: When the cut in u is in the last block B_d . We make a further case analysis depending upon if the cut in u is in the first half or second half of the last block B_d . The proof is symmetric to that of the previous case. In particular, when the cut is in the second half of the last block,

$$q = (y_1 x_1)^{m_1} y_1 = (y_2 x_2)^{m_2} y_2 = \dots = (y_d x_d)^{m_d} y_d.$$

By Claim 3 and Proposition 11, q is a common outer witness of $R(G)$.

Case 3: When the cut in u is within the block B_j for $1 < j < d$. WLOG, assume that the cut in u is within the first half of the block B_j . In this case the cut in v will be within the second half of the block B_{d-j+1} .

$$\begin{array}{l} u = \overbrace{\rho_{u_1} \dots \rho_{u_1} \dots \rho_{u_j}}^p \dots \overbrace{\rho_{u_j} \dots \rho_{u_d}}^q \\ v = \underbrace{\rho_{v_1} \dots \rho_{v_1} \dots \rho_{v_{d-j+1}}}_q \dots \underbrace{\rho_{v_{d-j+1}} \dots \rho_{u_d}}_p \end{array}$$

By matching q in u and v , we get ρ_{v_1} and ρ_{u_j} shares a common factor of length at least $N/2 > \ell > \ell_{1j}$ and hence they are conjugates to each other by Theorem 5. Since ρ_{u_j} is conjugate to ρ_{v_j} , by transitivity of conjugacy we obtain ρ_{v_1} and ρ_{v_j} are conjugates. This contradicts the fact that (u_1, v_1) and (u_j, v_j) belongs to different equivalence classes. Hence cut in B_j where $1 < j < d$ is not possible.

Claim 3. Let (u_i, v_i) and (u_j, v_j) be two pairs in different equivalence class where $i, j \in [d]$. Let (x_i, y_i) and (x_j, y_j) be a cut of the primitive root of (u_i, v_i) and (u_j, v_j) respectively. The following holds.

1. If $(x_i y_i)^{m_i} x_i = (x_j y_j)^{m_j} x_j$, then for any pair $(u_{j'}, v_{j'})$ that belongs to the same equivalence class as (u_j, v_j) , $(x_i y_i)^{m_i} x_i = (x_{j'} y_{j'})^{m_j} x_{j'}$ with $(x_{j'}, y_{j'})$ being a cut of the primitive root of $(u_{j'}, v_{j'})$.
2. If $(y_i x_i)^{m_i} y_i = (y_j x_j)^{m_j} y_j$, then for any pair $(u_{j'}, v_{j'})$ that belongs to the same equivalence class as (u_j, v_j) , $(y_i x_i)^{m_i} y_i = (y_{j'} x_{j'})^{m_j} y_{j'}$ with $(x_{j'}, y_{j'})$ being a cut of the primitive root of $(u_{j'}, v_{j'})$.

Proof. We prove Item 1. The proof of Item 2 is symmetric.

For contradiction, assume that $(x_i y_i)^{m_i} x_i = (x_j y_j)^{m_j} x_j \neq (x_{j'} y_{j'})^{m_j} x_{j'}$. Construct a pair $(\bar{u}, \bar{v})$ similar to (u, v) by keeping only the pairs (u_i, v_i) , (u_j, v_j) and $(u_{j'}, v_{j'})$ such that the total length of each pairs are equal to N . WLOG, assume, $i < j < j'$

$$\begin{array}{l} \bar{u} = \overbrace{u_i \dots u_i}^N \overbrace{u_j \dots u_j}^N \overbrace{u_{j'} \dots u_{j'}}^N \\ \bar{v} = v_i \dots v_i v_j \dots v_j v_{j'} \dots v_{j'} \end{array}$$

Since $(\bar{u}, \bar{v}) \subseteq (u_1, v_1)^* \dots (u_k, v_k)^*$, it is conjugate with a cut. After doing a case analysis on the cut, we get three possible cases.

1. There exist integers $n_i, n_j, n_{j'} \geq 0$ such that $(x_i y_i)^{m_i} x_i = (x_j y_j)^{n_j} x_j = (x_{j'} y_{j'})^{n_{j'}} x_{j'}$.

Both $n_i = m_i$ and $n_j = m_j$. Otherwise if either $n_i \neq m_i$ or $n_j \neq m_j$, then the set of pairs $\{(u_i, v_i), (u_j, v_j)\}$ has more than one common witness. By Proposition 13, they have infinitely many common witnessess and their primitive roots are the same. This contradicts that (u_i, v_i) and (u_j, v_j) are in different equivalence class. Since (u_j, v_j) and $(u_{j'}, v_{j'})$ belongs to the same equivalence class, $|x_j y_j| = |x_{j'} y_{j'}|$. Hence, we get $n_{j'} = n_j = m_j$. Thus, $(x_i y_i)^{m_i} x_i = (x_j y_j)^{m_j} x_j = (x_{j'} y_{j'})^{m_j} x_{j'}$

2. There exist integers $n_i, n_j, n_{j'} \geq 0$ such that $(y_i x_i)^{n_i} y_i = (y_j x_j)^{n_j} y_j = (y_{j'} x_{j'})^{n_{j'}} y_{j'}$

Since $(x_i y_i)^{m_i} x_i = (x_j y_j)^{m_j} x_j$, we get there is more than one common witness for the set of pairs $\{(u_i, v_i), (u_j, v_j)\}$ and from Proposition 13, we get they have infinitely many common witnesses. Thus the primitive roots are the same, and contradicts that (u_i, v_i) and (u_j, v_j) are in different equivalence class.

3. The block of u_i 's, u_j 's and $u_{j'}$'s share a common factor of length atleast at least $N/2 \geq \ell$ and via Theorem 5, we get their primitive roots are conjugates to each other. This contradicts that (u_i, v_i) and (u_j, v_j) are in different equivalence class. $\square$

Hence, for a finite set of pairs G , $(u_1, v_1)^* \cdots (u_k, v_k)^*$ is conjugate only if $R(G)$ has a common witness. By Proposition 12, we also conclude that G has a common witness. $\square$

Hence, we proved Lemma 4 for the finite case.

4.2. For an Infinite Set of Pairs

We now extend Lemma 4 from a finite set to an infinite set of pairs.

Lemma 6 (Compactness Theorem). *Let G be an infinite set of pairs. If every finite subset of G has a common witness, then G has a common witness.*

Proof. From Proposition 13, if a set has a witness, it has exactly one common witness or infinitely many common witnesses. Given that every finite subset of G has a common witness, there are two possible cases: a finite subset of G with a unique witness exists, or every finite subset of G has infinitely many witnesses.

1. Assume that there exists a finite subset G_f of G with exactly one common witness, say z . We claim that z is a common witness of G as well. By assumption, the finite set $G_f \cup \{(u, v)\}$ has a common witness, for any pair $(u, v) \in G$. Moreover, the witness for this set must be z ; otherwise, it contradicts the uniqueness of the witness of G_f . This implies that z is a witness for any pair in G . Hence z is a common witness of G .
2. Next we assume that every finite subset of G has infinitely many common witnesses. Take any pair (u_i, v_i) and (u_j, v_j) from G . The set $\{(u_i, v_i), (u_j, v_j)\}$ is a finite set with infinitely many witnesses by assumption. Therefore, from Proposition 13, both (u_i, v_i) and (u_j, v_j) have the same primitive root. Since primitive roots are unique, the primitive root of every pair in G is the same. From Proposition 9, the witnesses of the primitive root is same as that of the witnesses of each pair in G . Hence, G has a common witness. $\square$

The proof of Lemma 4 is a straightforward corollary of the Compactness Theorem. If G^* is conjugate, then the closure of every finite subset of G is also conjugate. Using Lemma 5, every finite subset of G has a common witness. Using Compactness Theorem, G has a common witness. From Proposition 12, we conclude that $R(G)$ has a common witness.

This concludes the proof of Common Witness Theorem (Theorem 6).

5. Existence of Common Witness for Monoid Closure

In this section, we prove the equivalence between conjugacy and the presence of a common witness in sumfree sets. We begin by proving Theorem 8 for sumfree sets that contain only one Kleene star. Subsequently, we establish Theorem 9 that extends the result to general sumfree sets.

5.1. Common Witness of a Singleton Redux

We prove Theorem 8 by showing $(1) \Rightarrow (2) \Rightarrow (3) \Rightarrow (1)$. In this subsection, we assume that the sumfree set has a nonempty redux as stated in Theorem 8. It is trivial that $(3) \Rightarrow (1)$.

Now we proceed to prove $(1) \Rightarrow (2)$, namely, if a sumfree set $M = (\alpha_0, \beta_0)G^*(\alpha_1, \beta_1)$ is conjugate, then there exists a common witness of $G \cup \{(\alpha_1\alpha_0, \beta_1\beta_0)\}$. We first prove this direction when G is just a singleton set and later generalise it to any arbitrary set of pairs G .

Proposition 15. *If a set $M = (\alpha_0, \beta_0)(u, v)^*(\alpha_1, \beta_1)$ is conjugate then the set $\{(u, v), (\alpha_1\alpha_0, \beta_1\beta_0)\}$ has a common witness.*

Proof. Since M is conjugate, (u, v) is also conjugate by Lemma 1. Consider the pair $(u', v') = (\alpha_0, \beta_0)(u, v)^{4n}(\alpha_1, \beta_1)$ where $n|u| \geq 2 \cdot (\text{length of the redux of } M)$. Let (x, y) denote a cut of the primitive root of the conjugate pair (u, v) .

We now examine the possible cuts, say (p, q) , of (u', v') and show that in each case, a common witness of $\{(u, v), (\alpha_1\alpha_0, \beta_1\beta_0)\}$ exists.

Case I: When the cut in u' is within α_0 . Then $p = \alpha'_0$ is a prefix of α_0 and $\alpha_0 = \alpha'_0\alpha''_0$ for some word α''_0 . Substituting (u, v) with powers of (xy, yx) ,

$$\begin{aligned} u' &= \overbrace{\alpha'_0}^p \overbrace{\alpha''_0 xy \cdots xy \alpha_1}^q \\ v' &= \beta_0 yx \cdots yx \beta_1 \end{aligned}$$

Comparing prefixes of q in u' and v' , we obtain three possible cases for β_0 .

- (a) β_0 is a proper prefix of α''_0 : After matching β_0 with the prefix of q in u' , we find that the remaining suffix of α''_0 matches with the prefix of the block $yx \cdots yx$ in v' . Since the total length of the block $yx \cdots yx$ is greater than $4|\alpha_0|$, it follows that α''_0 must end within the first half of the block. Furthermore, using Cases I. (a), I. (b), I. (e) and II. of Cut Lemma, it should end after a y since there exists an xy after α''_0 in u' . Thus, we can express α''_0 as

$$\alpha''_0 = \beta_0(yx)^m y \quad (9)$$

for some integer $m \geq 0$. Continuing to match q in u' and v' , we obtain

$$u' = \alpha'_0 \alpha''_0 \overbrace{xy \cdots x}^=(yx)^m y \alpha_1 = pq \quad (10)$$

$$v' = \underbrace{\beta_0(yx)^m y}_{\alpha''_0} \underbrace{xy \cdots x}_{=} \beta_1 = qp = \alpha''_0 \overbrace{xy \cdots x}^=(yx)^m y \alpha_1 \alpha'_0 \quad (11)$$

By equating the sets for v' on both sides of Equation (11), we get

$$\beta_1 = (yx)^m y \alpha_1 \alpha'_0. \quad (12)$$

Concatenating Equation (9) and Equation (12), we obtain

$$(yx)^m y \alpha_1 \alpha_0 = \beta_1 \beta_0 (yx)^m y.$$

From Theorem 1, we get $(yx)^m y$ is a outer witness for $(\alpha_1\alpha_0, \beta_1\beta_0)$, and it is also a outer witness of (u, v) using Proposition 9. Therefore, $(yx)^m y$ is a common outer witness of $\{(u, v), (\alpha_1\alpha_0, \beta_1\beta_0)\}$.

- (b) The case when $\beta_0 = \alpha''_0$:

$$u' = \alpha'_0 \alpha''_0 \overbrace{xy \cdots xy}^= \alpha_1 = pq \quad (13)$$

$$v' = \underbrace{\beta_0}_{\alpha''_0} \underbrace{yx \cdots yx}_{=} \beta_1 = qp = \alpha''_0 \overbrace{xy \cdots xy}^= \alpha_1 \alpha'_0 \quad (14)$$

Equating v' on both sides of the Equation (14), we get that $xy = yx$ and

$$\beta_1 = \alpha_1 \alpha'_0. \quad (15)$$

Appending the equation $\beta_0 = \alpha''_0$ to Equation (15), we get $\alpha_1 \alpha_0 = \beta_1 \beta_0$. Hence $(\alpha_1 \alpha_0, \beta_1 \beta_0)$ is conjugate with ε as a witness. Since $xy = yx$, we can also deduce that (u, v) is an identical pair with ε as a witness. Therefore, ε is a common witness of $\{(u, v), (\alpha_1 \alpha_0, \beta_1 \beta_0)\}$.

- (c) α''_0 is a proper prefix of β_0 : Since the total length of block $xy \cdots xy$ is at least $4|\beta_0|$, it follows that β_0 must end within the first half of the block of xy . Moreover, it should end after an x by Cases I. (c), I. (d), I. (e) and II. of Cut Lemma since there is at least one yx after β_0 in v' . Therefore,

$$\beta_0 = \alpha''_0 (xy)^m x \quad (16)$$

for some integer $m \geq 0$. Continuing with the analysis, we have:

$$u' = \alpha'_0 \overbrace{\alpha''_0 (xy)^m x}^{\beta_0} \overbrace{yx \cdots y}^{\beta_0} \alpha_1 = pq \quad (17)$$

$$v' = \beta_0 \underbrace{yx \cdots y}_{=} (xy)^m x \beta_1 = qp = \overbrace{\alpha''_0 (xy)^m x}^{\beta_0} \overbrace{yx \cdots y}^{\beta_0} \alpha_1 \alpha'_0 \quad (18)$$

By equating the sets for v' on both sides of Equation (18), we get

$$(xy)^m x \beta_1 = \alpha_1 \alpha'_0. \quad (19)$$

Concatenating Equation (16) and Equation (19), we obtain

$$\alpha_1 \alpha_0 (xy)^m x = (xy)^m x \beta_1 \beta_0.$$

From Theorem 1, we get $(xy)^m x$ is an inner witness of $(\alpha_1 \alpha_0, \beta_1 \beta_0)$, and it is also an inner witness for (u, v) using Proposition 9. Thus, $(xy)^m x$ is a common inner witness of $\{(u, v), (\alpha_1 \alpha_0, \beta_1 \beta_0)\}$.

Case 2: When the cut in u' is within the block of $(u, v) \cdots (u, v)$. Then p ends within the block of (u, v) 's. There are two cases based on whether the cut in u' is within the first half or the second half of the block of $(u, v) \cdots (u, v)$.

- (a) When p ends within the first half of the block of (u, v) 's:

$$\begin{aligned} u' &= \alpha_0 \overbrace{xy \cdots xy}^{\text{cut region}} \overbrace{xy \cdots xy}^{\geq 2n \text{ times}} \alpha_1 = pq \\ v' &= \beta_0 yx \cdots yxyx \cdots yx \beta_1 = qp \end{aligned}$$

We compare the prefixes of q in u' and v' . Since the length of the remaining half of the block of xy 's is still greater than $2n|u| > 2|\beta_0|$, it follows that β_0 in v' matches within the block of xy 's in u' and there is at least one xy occurring after it. Moreover, it ends after an x by Cases I. (c), I. (d), I. (e) and II. of Cut Lemma, as there is at least one yx in v' after β_0 . Therefore,

$$p\beta_0 = \alpha_0 (xy)^m x \quad (20)$$

for some integer $m \geq 0$.

$$u' = \overbrace{\alpha_0 (xy)^m x}^{p\beta_0} \overbrace{yx \cdots y}^{\beta_0} \alpha_1 = pq \quad (21)$$

$$v' = \beta_0 \underbrace{yx \cdots y}_{=} (xy)^m x \beta_1 = qp = \beta_0 \overbrace{yx \cdots y}^{\beta_0} \alpha_1 p \quad (22)$$

By equating the sets for v' on both sides of Equation (22), we get

$$\begin{aligned} (xy)^m x \beta_1 = \alpha_1 p &\Rightarrow (xy)^m x \beta_1 \beta_0 = \alpha_1 p \beta_0 && \text{(Appending } \beta_0) \\ &\Rightarrow (xy)^m x \beta_1 \beta_0 = \alpha_1 \alpha_0 (xy)^m x && \text{(Substituting Equation (20))} \end{aligned}$$

Therefore we obtain,

$$\alpha_1 \alpha_0 (xy)^m x = (xy)^m x \beta_1 \beta_0 .$$

From Theorem 1, $(xy)^m x$ is an inner witness of $(\alpha_1 \alpha_0, \beta_1 \beta_0)$, and it is also an inner witness for (u, v) using Proposition 9. Therefore, $(xy)^m x$ is a common inner witness of $\{(u, v), (\alpha_1 \alpha_0, \beta_1 \beta_0)\}$.

(b) When p ends within the second half of the block of (u, v) 's:

$$\begin{aligned} u' &= \alpha_0 \overbrace{xy \cdots xy}^{\geq 2n \text{ times}} \overbrace{xy \cdots xy}^{\text{cut region}} \alpha_1 = pq \\ v' &= \beta_0 yx \cdots yx yx \cdots yx \beta_1 = qp \end{aligned}$$

We compare the suffixes of p in u' and v' . Since the suffix of p within the block xy is still greater than $2n|u| > 2|\beta_1|$, it follows that the suffix β_1 in v' matches within the block of xy 's in u' and there is at least one xy occurring before it. Moreover, it starts with a y by Cases I. (c), I. (d), I. (e) and II. of Cut Lemma since there is at least one yx before β_1 . Hence,

$$\beta_1 q = (yx)^m y \alpha_1 \tag{23}$$

for some integer $m \geq 0$.

$$u' = \alpha_0 \overbrace{xy \cdots x}^{\beta_1 q} \overbrace{(yx)^m y \alpha_1}^{\beta_1 q} = pq \tag{24}$$

$$v' = \beta_0 (yx)^m y \underbrace{xy \cdots x}_{=} \beta_1 = qp = q \alpha_0 \overbrace{xy \cdots x}^{\beta_1 q} \beta_1 \tag{25}$$

By equating v' on both sides of the Equation (25), we get

$$\begin{aligned} \beta_0 (yx)^m y = q \alpha_0 &\Rightarrow \beta_1 \beta_0 (yx)^m y = \beta_1 q \alpha_0 && \text{(Concatenating } \beta_1 \text{ on the left side)} \\ &\Rightarrow \beta_1 \beta_0 (yx)^m y = (yx)^m y \alpha_1 \alpha_0 && \text{(Substituting Equation (23))} \end{aligned}$$

Therefore, we obtain

$$(yx)^m y \alpha_1 \alpha_0 = \beta_1 \beta_0 (yx)^m y .$$

From Theorem 1, we get $(yx)^m y$ is an outer witness of $(\alpha_1 \alpha_0, \beta_1 \beta_0)$, and it is also an outer witness for (u, v) using Proposition 9. Therefore, $(yx)^m y$ is a common outer witness of $\{(u, v), (\alpha_1 \alpha_0, \beta_1 \beta_0)\}$.

Case 3: When the cut in u' is within α_1 . Then $q = \alpha_1''$ is a suffix of α_1 and $\alpha_1 = \alpha_1' \alpha_1''$ for some word α_1' .

$$\begin{aligned} u' &= \overbrace{\alpha_0 xy \cdots xy}^p \overbrace{\alpha_1'}^q \alpha_1'' \\ v' &= \beta_0 yx \cdots yx \beta_1 \end{aligned}$$

This case is symmetric to *Case 1*, where the cut in u' is within α_0 .

Thus, the pair $(u', v') \in M$ is conjugate only if there exist a common witness for the set $\{(u, v), (\alpha_1 \alpha_0, \beta_1 \beta_0)\}$. $\square$

Now we extend the above proposition to an arbitrary set G .

Lemma 7. *If a set $M = (\alpha_0, \beta_0)G^*(\alpha_1, \beta_1)$ is conjugate then one of the following is true:*

1. $|\mathcal{W}(G)| = \infty$ and $R(G) \cup \{(\alpha_1\alpha_0, \beta_1\beta_0)\}$ has a common witness.
2. $|\mathcal{W}(G)| = 1$ and there exists a pair $(u, v) \in G$ such that $W(G) = W(G \cup \{(\alpha_1\alpha_0, \beta_1\beta_0)\}) = W(\{(u, v), (\alpha_1\alpha_0, \beta_1\beta_0)\})$.

Proof. Since M is conjugate, G^* is conjugate by Lemma 1. Further, G^* is conjugate iff G has a common witness by Theorem 6. The set G has two possibilities: it either has a unique common witness or infinitely many common witnesses using Proposition 13.

Assume that G has infinitely many common witnesses. From Proposition 13, each pair in G have the same primitive root, say (ρ, ρ') . The common witnesses of G are the same as that of the witnesses of (ρ, ρ') using Proposition 12. Since M is conjugate, $(\alpha_0, \beta_0)(\rho^n, \rho'^n)^*(\alpha_1, \beta_1)$ is conjugate for some $n \geq 1$. From Proposition 15, there exists a common witness of $\{(\rho^n, \rho'^n), (\alpha_1\alpha_0, \beta_1\beta_0)\}$. Furthermore, the witness of (ρ, ρ') is the same as the witness of (ρ^n, ρ'^n) by Proposition 9. Therefore, we can conclude that there exists a common witness of $\{(\rho, \rho'), (\alpha_1\alpha_0, \beta_1\beta_0)\}$, and thus of $G \cup \{(\alpha_1\alpha_0, \beta_1\beta_0)\}$.

Next we assume that G has a unique common witness. WLOG, assume that G has a unique common inner witness z . It suffices to show that z is an inner witness of $(\alpha_1\alpha_0, \beta_1\beta_0)$. Every pair in G^* has a common witness with $(\alpha_1\alpha_0, \beta_1\beta_0)$ using Proposition 15. If there exists a pair (u, v) in G^* such that $\{(u, v), (\alpha_1\alpha_0, \beta_1\beta_0)\}$ has infinitely many common witnesses, then both (u, v) and $(\alpha_1\alpha_0, \beta_1\beta_0)$ have the same primitive root by Proposition 13. Since z is an inner witness of (u, v) , z is an inner witness of its primitive root, and thus also an inner witness of $(\alpha_1\alpha_0, \beta_1\beta_0)$ using Proposition 9. Suppose instead that every pair in G^* has a unique common witness with $(\alpha_1\alpha_0, \beta_1\beta_0)$. Since G has a unique common inner witness z , there exists two distinct pairs (u, v) and (u', v') of length greater than z such that z is the unique common inner witness for the set $\{(u, v), (u', v')\}$. These pairs can be expressed as $(u, v) = (zr, rz)$ and $(u', v') = (zr', r'z)$, where $r \neq r'$. Consider the following three pairs formed using (u, v) and (u', v') :

$$\begin{aligned} (u_1, v_1) &= (u'uu, v'vv) = (zr'zrzr, r'zrzr), \\ (u_2, v_2) &= (uu'u, vv'v) = (zrzr'zr, rzr'zrz), \\ (u_3, v_3) &= (uuu', vv'v') = (zrzr'zr', rzr'zr'z). \end{aligned}$$

For $1 \leq i, j \leq 3$ with $i \neq j$, the following properties hold:

1. $|(u_i, v_i)| = |(u_j, v_j)|$.
2. $|\rho_{u_i}, \rho_{v_i}| = |\rho_{u_j}, \rho_{v_j}|$. Since u_i and u_j (resp. v_i and v_j) are conjugates, their primitive roots are conjugates, and hence are of equal length, i.e., $|\rho_{u_i}| = |\rho_{u_j}|$ (resp. $|\rho_{v_i}| = |\rho_{v_j}|$).
3. $(u_i, v_i) \neq (u_j, v_j)$. Otherwise, if $u_i = u_j$ (or $v_i = v_j$), it would follow that $r'zr = rzr'$ (or $r'zrzr = rzr'zr'$), implying that r' is a outer witness of $(zr, rz) = (u, v)$. Since r' is also an outer witness of $(u', v') = (zr', r'z)$, it contradicts the assumption that z is a unique common witness of the set $\{(u, v), (u', v')\}$. Hence, $u_i \neq u_j$ and $v_i \neq v_j$.
4. $(\rho_{u_i}, \rho_{v_i}) \neq (\rho_{u_j}, \rho_{v_j})$. Since $u_i \neq u_j$ (resp. $v_i \neq v_j$) and $|\rho_{u_i}| = |\rho_{u_j}|$ (resp. $|\rho_{v_i}| = |\rho_{v_j}|$), we get that $\rho_{u_i} \neq \rho_{u_j}$ (resp. $\rho_{v_i} \neq \rho_{v_j}$).

Let (x_i, y_i) be the cut of (ρ_{u_i}, ρ_{v_i}) for $1 \leq i \leq 3$. The set $\{(u_1, v_1), (u_2, v_2), (u_3, v_3)\}$ has a unique common witness, otherwise their primitive roots will be same by Proposition 13 which is a contradiction. Since z is an inner witness of each (u_i, v_i) , z is the unique common inner witness of $\{(u_1, v_1), (u_2, v_2), (u_3, v_3)\}$, and thus of $\{(\rho_{u_1}, \rho_{v_1}), (\rho_{u_2}, \rho_{v_2}), (\rho_{u_3}, \rho_{v_3})\}$ by Proposition 12. Since $(u_1, v_1)^*(u_2, v_2)^*(u_3, v_3)^* \subset G^*$ is conjugate and $|\rho_{u_1}, \rho_{v_1}| = |\rho_{u_2}, \rho_{v_2}| = |\rho_{u_3}, \rho_{v_3}|$, $z = x_1 = x_2 = x_3$ using Proposition 14.

By assumption, each pair (u_i, v_i) has a unique common witness with $(\alpha_1\alpha_0, \beta_0\beta_1)$. Thus, at least two of these three pairs, say (u_i, v_i) and (u_j, v_j) , either both have unique common inner witnesses with $(\alpha_1\alpha_0, \beta_1\beta_0)$, or both have unique common outer witnesses with $(\alpha_1\alpha_0, \beta_1\beta_0)$. Assume that the set $\{(u_i, v_i), (\alpha_1\alpha_0, \beta_1\beta_0)\}$ and $\{(u_j, v_j), (\alpha_1\alpha_0, \beta_1\beta_0)\}$ both has a unique common inner witness, say z_i and z_j respectively. Let (α, β) be the primitive root of $(\alpha_1\alpha_0, \beta_1\beta_0)$. There exists integers $m_i, m_j, n_i, n_j \in \mathbb{N}$ such that

$$z_i = (x_i y_i)^{m_i} x_i = (\alpha \beta)^{n_i} \alpha \quad (26)$$

$$z_j = (x_j y_j)^{m_j} x_j = (\alpha\beta)^{n_j} \alpha \quad (27)$$

Suppose both $m_i, m_j \geq 1$, then $x_i y_i$ and $x_j y_j$ are both equal length prefixes of $(\alpha\beta)^* \alpha$, and hence $x_i y_i = x_j y_j$, which contradicts that $\rho_{u_i} \neq \rho_{u_j}$. Hence, we get that either $m_i = 0$ or $m_j = 0$. WLOG, assume that $m_i = 0$, i.e., $x_i = (\alpha\beta)^{n_i} \alpha$ and since $z = x_i = x_j$, we get $z \in (\alpha\beta)^* \alpha$, and hence is an inner witness of $(\alpha_1 \alpha_0, \beta_1 \beta_0)$. Infact, z is the unique common inner witness of (u_i, v_i) and $(\alpha_1 \alpha_0, \beta_1 \beta_0)$. Since z (the unique common inner witness of G) is also an inner witness of $(\alpha_1 \alpha_0, \beta_1 \beta_0)$, we get that z is a common inner witness of $G \cup \{(\alpha_1 \alpha_0, \beta_1 \beta_0)\}$.

The case where the set $\{(u_i, v_i), (\alpha_1 \alpha_0, \beta_1 \beta_0)\}$ and $\{(u_j, v_j), (\alpha_1 \alpha_0, \beta_1 \beta_0)\}$ both has a unique common outer witness is invalid and leads to a contradiction, since we get y_i and y_j are equal length prefixes of $(\beta\alpha)^* \beta$, and hence $y_i = y_j$. This contradicts $\rho_{u_i} \neq \rho_{u_j}$ since $x_i = x_j$. $\square$

The remaining direction to prove is (2) $\Rightarrow$ (3) in Theorem 8. This direction states that if there exists a common witness for both G and $(\alpha_1 \alpha_0, \beta_1 \beta_0)$ for a given set $M = (\alpha_0, \beta_0) G^* (\alpha_1, \beta_1)$, then M has a common witness. It is a straightforward corollary of the below lemma.

Lemma 8. *Let $M = (\alpha_0, \beta_0) G^* (\alpha_1, \beta_1)$ be a sumfree set with nonempty redux. If there exists a common witness z' for $G \cup \{(\alpha_1 \alpha_0, \beta_1 \beta_0)\}$, then one of the following cases is true:*

- (a) *If z' is a unique common inner witness, then M has a unique common witness $z = [\alpha_0 z', \beta_0]_R = [\alpha_1, z' \beta_1]_L$. Moreover, if $|\alpha_0 z'| \geq |\beta_0|$ or equivalently $|\alpha_1| \leq |z' \beta_1|$, then z is a common inner witness, otherwise it is a common outer witness.*
- (b) *If z' is a unique common outer witness, then M has a unique common witness $z = [\alpha_0, \beta_0 z']_R = [z' \alpha_1, \beta_1]_L$. Moreover, if $|z' \alpha_1| \geq |\beta_1|$ or equivalently $|\alpha_0| \leq |\beta_0 z'|$, then z is a common outer witness, otherwise it is a common inner witness.*
- (c) *If $G \cup \{(\alpha_1 \alpha_0, \beta_1 \beta_0)\}$ have infinitely many common witnesses, then M is a subset of powers of the primitive root of its redux. Thus, M has infinitely many witnesses.*

Proof. There are two cases to consider depending upon if $G \cup \{(\alpha_1 \alpha_0, \beta_1 \beta_0)\}$ has a common inner or outer witness.

Case (a): When z' is a common inner witness of $G \cup \{(\alpha_1 \alpha_0, \beta_1 \beta_0)\}$. The following equations hold:

$$\alpha_1 \alpha_0 z' = z' \beta_1 \beta_0 \quad (28)$$

$$u z' = z' v \text{ for any pair } (u, v) \in G^* \quad (29)$$

We claim that $z = [\alpha_0 z', \beta_0]_R = [\alpha_1, z' \beta_1]_L$ is a common witness for M . There are two cases depending upon whether β_0 is a suffix of $\alpha_0 z'$ or vice-versa in Equation (28).

- (a) When β_0 is a suffix of $\alpha_0 z'$ or equivalently, when α_1 is a prefix of $z' \beta_1$. We get $z = \alpha_0 z' \beta_0^{-1} = \alpha_1^{-1} z' \beta_1$, or equivalently $\alpha_0 z' = z \beta_0$ and $\alpha_1 z = z' \beta_1$. We show that z is a common inner witness for M .

For any $(u, v) \in G^*$,

$$\begin{aligned} \alpha_0 u \alpha_1 z &= \alpha_0 u z' \beta_1 && \text{(Substituting } \alpha_1 z = z' \beta_1) \\ &= \alpha_0 z' v \beta_1 && (z' \text{ is an inner witness of } (u, v)) \\ &= z \beta_0 v \beta_1 && \text{(Substituting } \alpha_0 z' = z \beta_0) \end{aligned}$$

- (b) When $\alpha_0 z'$ is a suffix of β_0 or equivalently $z' \beta_1$ is a prefix of α_1 . We get $z = \beta_0 (\alpha_0 z')^{-1} = (z' \beta_1)^{-1} \alpha_1$, or equivalently $\alpha_1 = z' \beta_1 z$ and $z \alpha_0 z' = \beta_0$. We show that z is a common outer witness for M .

For any $(u, v) \in G^*$,

$$\begin{aligned} z \alpha_0 u \alpha_1 &= z \alpha_0 u z' \beta_1 z && \text{(Substituting } \alpha_1 = z' \beta_1 z) \\ &= z \alpha_0 z' v \beta_1 z && (z' \text{ is an inner witness of } (u, v)) \\ &= \beta_0 v \beta_1 z && \text{(Substituting } z \alpha_0 z' = \beta_0) \end{aligned}$$

Case (b): When z' is a common outer witness of G and $(\alpha_1\alpha_0, \beta_1\beta_0)$. Therefore, the following equations hold:

$$z'\alpha_1\alpha_0 = \beta_1\beta_0z' \quad (30)$$

$$z'u = vz' \text{ for any pair } (u, v) \in G^* \quad (31)$$

We claim that $z = [\alpha_0, \beta_0z']_R = [z'\alpha_1, \beta_1]_L$ is a witness for M . There are two cases depending upon if α_0 is a suffix of β_0z' or vice-versa in Equation (30).

- (a) When α_0 is a suffix of β_0z' or equivalently, β_1 is a prefix of $z'\alpha_1$. We get $z = \beta_0z'\alpha_0^{-1} = \beta_1^{-1}z'\alpha_1$, or equivalently $z\alpha_0 = \beta_0z'$ and $z'\alpha_1 = \beta_1z$. We show that z is a common outer witness for M .

For any $(u, v) \in G^*$,

$$\begin{aligned} z\alpha_0u\alpha_1 &= \beta_0z'u\alpha_1 && \text{(Substituting } z\alpha_0 = \beta_0z') \\ &= \beta_0vz'\alpha_1 && (z' \text{ is an outer witness of } (u, v)) \\ &= \beta_0v\beta_1z && \text{(Substituting } z'\alpha_1 = \beta_1z) \end{aligned}$$

- (b) If β_0z' is a suffix of α_0 or equivalently, $z'\alpha_1$ is a prefix of β_1 . Therefore, $z = \alpha_0(\beta_0z')^{-1} = (z'\alpha_1)^{-1}\beta_1$, or equivalently $\alpha_0 = z\beta_0z'$ and $z'\alpha_1z = \beta_1$. We show that z is a common inner witness for M .

For any $(u, v) \in G^*$,

$$\begin{aligned} \alpha_0u\alpha_1z &= z\beta_0z'u\alpha_1z && \text{(Substituting } \alpha_0 = z\beta_0z') \\ &= z\beta_0vz'\alpha_1z && (z' \text{ is an outer witness of } (u, v)) \\ &= z\beta_0v\beta_1 && \text{(Substituting } z'\alpha_1z = \beta_1) \end{aligned}$$

Therefore, if there exists a common witness z' for G and $(\alpha_1\alpha_0, \beta_1\beta_0)$, then there also exists a common witness z for M . Furthermore, since the redux remains constant in the word equation that relates the common witness of M and those of $G \cup \{(\alpha_1\alpha_0, \beta_1\beta_0)\}$, each distinct common witness of M corresponds to a distinct common witness of $G \cup \{(\alpha_1\alpha_0, \beta_1\beta_0)\}$, and vice-versa. Consequently, if z' is the unique common witness for $G \cup \{(\alpha_1\alpha_0, \beta_1\beta_0)\}$ then z is the unique common witness of M , and conversely.

Suppose the set $G \cup \{(\alpha_1\alpha_0, \beta_1\beta_0)\}$ has infinitely many witnesses. According to Proposition 13, all the pairs in that set have the same primitive root, let us say (ρ, ρ') . Therefore, $G^*(\alpha_1, \beta_1)(\alpha_0, \beta_0)$ is a subset of powers of (ρ, ρ') and is conjugate. Since M is a cyclic shift of $G^*(\alpha_1, \beta_1)(\alpha_0, \beta_0)$ and is also conjugate, it is also a subset of powers of a primitive root, let us say (ρ_m, ρ'_m) , that is a cyclic shift of (ρ, ρ') . Moreover, α_1 (resp. β_1) is an inner (resp. outer) witness of (ρ, ρ_m) (resp. (ρ', ρ'_m)). We observe that (ρ_m, ρ'_m) is the primitive root of the redux of M . Hence, M is a subset of powers of the primitive root of its redux. $\square$

5.2. Common Witness of a Sumfree Set

We prove Theorem 9 in this subsection.

Lemma 9. Given a sumfree set $M = (\alpha_0, \beta_0)G_1^*(\alpha_1, \beta_1) \cdots (\alpha_{k-1}, \beta_{k-1})G_k^*(\alpha_k, \beta_k)$. The following are equivalent.

1. z is a common inner (resp. outer) witness of M .
2. z is a common inner (resp. outer) witness of each of its singleton redux.

Proof. Let $M = (\alpha_0, \beta_0)G_1^*(\alpha_1, \beta_1) \cdots (\alpha_{k-1}, \beta_{k-1})G_k^*(\alpha_k, \beta_k)$ be a sumfree set. For $i \in \{1, \dots, k\}$, let M_i denote the singleton redux of M keeping only the Kleene star G_i^* , i.e.,

$$M_i = (\alpha_0 \cdots \alpha_{i-1}, \beta_0 \cdots \beta_{i-1})G_i^*(\alpha_i \cdots \alpha_k, \beta_i \cdots \beta_k)$$

The proof of (1) $\Rightarrow$ (2) is trivial, i.e., if z is a common witness of M , then z is also a common witness of $M_i \subseteq M$.

We prove (2) $\Rightarrow$ (1). WLOG, assume that z is a common *inner* witness of each M_i 's. If the redux is an empty pair $(\varepsilon, \varepsilon)$, then $M = G_1^* G_2^* \dots G_k^*$. It is easy to prove that z is a common inner witness of M by showing that for any arbitrary pair $(u_i, v_i) \in G_i^*$, $u_1 u_2 \dots u_k z = z v_1 v_2 \dots v_k$. Proof by induction on k .

$$\begin{aligned} u_1 \dots u_{k-1} u_k z &= u_1 \dots u_{k-1} z v_k && \text{(Since } u_k z = z v_k \text{)} \\ &= z v_1 \dots v_{k-1} v_k && \text{(By Induction Hypothesis)} \end{aligned}$$

Now, assume that the redux is nonempty. Since each M_i has a common witness, there exist a common witness, say z_i , for $G_i \cup \{(\alpha_i \dots \alpha_k \alpha_0 \dots \alpha_{i-1}, \beta_i \dots \beta_k \beta_0 \dots \beta_{i-1})\}$ by Theorem 8. There are 3 possible cases for z_i .

(a) z_i is a unique common inner witness. Therefore, for any pair $(u_i, v_i) \in G_i^*$,

$$u_i z_i = z_i v_i \quad (32)$$

$$\alpha_i \dots \alpha_k \alpha_0 \dots \alpha_{i-1} z_i = z_i \beta_i \dots \beta_k \beta_0 \dots \beta_{i-1} \quad (33)$$

$$z = \alpha_0 \dots \alpha_{i-1} z_i (\beta_0 \dots \beta_{i-1})^{-1} = (\alpha_i \dots \alpha_k)^{-1} z_i \beta_i \dots \beta_k \quad \text{(By Lemma 8 (a))} \quad (34)$$

(b) z_i is a unique common outer witness. Therefore, for any pair $(u_i, v_i) \in G_i^*$,

$$z_i u_i = v_i z_i \quad (35)$$

$$z_i \alpha_i \dots \alpha_k \alpha_0 \dots \alpha_{i-1} = \beta_i \dots \beta_k \beta_0 \dots \beta_{i-1} z_i \quad (36)$$

$$z = \alpha_0 \dots \alpha_{i-1} (\beta_0 \dots \beta_{i-1} z_i)^{-1} = (z_i \alpha_i \dots \alpha_k)^{-1} \beta_i \dots \beta_k \quad \text{(By Lemma 8 (b))} \quad (37)$$

(c) When $G_i \cup \{(\alpha_i \dots \alpha_k \alpha_0 \dots \alpha_{i-1}, \beta_i \dots \beta_k \beta_0 \dots \beta_{i-1})\}$ has infinitely many witnesses, by Proposition 13, it is a set of powers of the same primitive root say (ρ_i, ρ'_i) . Therefore z_i belongs to witnesses of (ρ_i, ρ'_i) . From Lemma 8 (c), the set M_i reduces to a set of powers of the primitive root of the redux $(\alpha_0 \dots \alpha_k, \beta_0 \dots \beta_k)$, say (ρ, ρ') . Note that $\alpha_i \dots \alpha_k$ (*resp.* $\alpha_0 \dots \alpha_{i-1}$) is an inner (*resp.* outer) witness of (ρ_i, ρ) . Similarly $\beta_i \dots \beta_k$ (*resp.* $\beta_0 \dots \beta_{i-1}$) is an inner (*resp.* outer) witness of (ρ'_i, ρ') .

We show that z is a common inner witness of M , i.e., for any arbitrary pair $(u_i, v_i) \in G_i^*$ (possibly empty), we prove $\alpha_0 u_1 \alpha_1 u_2 \alpha_2 \dots \alpha_{k-1} u_k \alpha_k z = z \beta_0 v_1 \beta_1 v_2 \beta_2 \dots \beta_{k-1} v_k \beta_k$. The proof is by induction on the number of singleton reduxes, $0 \leq i \leq k$.

Base Case. When $i = 0$, it is vacuously true since z is a witness of the redux.

Inductive Case. Assume for induction that it is true for the first $i - 1$ singleton reduxes, i.e.,

$$\alpha_0 u_1 \alpha_1 u_2 \alpha_2 \dots u_{i-1} \alpha_{i-1} \dots \alpha_k z = z \beta_0 v_1 \beta_1 v_2 \beta_2 \dots v_{i-1} \beta_{i-1} \dots \beta_k.$$

We prove it for the first i singleton reduxes, i.e., we show

$$\alpha_0 u_1 \alpha_1 u_2 \alpha_2 \dots u_{i-1} \alpha_{i-1} u_i \alpha_i \dots \alpha_k z = z \beta_0 v_1 \beta_1 v_2 \beta_2 \dots v_{i-1} \beta_{i-1} v_i \beta_i \dots \beta_k.$$

There are 3 possible cases for the common witness z_i of $G_i \cup \{(\alpha_i \dots \alpha_k \alpha_0 \dots \alpha_{i-1}, \beta_i \dots \beta_k \beta_0 \dots \beta_{i-1})\}$.

1. When z_i is a unique common inner witness. From Equation (34), $z = (\alpha_i \dots \alpha_k)^{-1} z_i \beta_i \dots \beta_k$.

$$\begin{aligned} &\alpha_0 u_1 \alpha_1 \dots u_{i-1} \alpha_{i-1} u_i \alpha_i \alpha_{i+1} \dots \alpha_k z \\ &= \alpha_0 u_1 \alpha_1 \dots u_{i-1} \alpha_{i-1} u_i z_i \beta_i \dots \beta_k && \text{(Subs. } z \text{)} \\ &= \alpha_0 u_1 \alpha_1 \dots u_{i-1} \alpha_{i-1} z_i v_i \beta_i \dots \beta_k && (u_i z_i = z_i v_i) \\ &= \alpha_0 u_1 \alpha_1 \dots u_{i-1} \alpha_{i-1} \alpha_i \dots \alpha_k z (\beta_i \dots \beta_k)^{-1} v_i \beta_i \dots \beta_k && \text{(Subs. } z_i \text{)} \\ &= z \beta_0 v_1 \beta_1 \dots v_{i-1} \beta_{i-1} \beta_i \dots \beta_k (\beta_i \dots \beta_k)^{-1} v_i \beta_i \dots \beta_k && \text{(Inductive Hypothesis)} \\ &= z \beta_0 v_1 \beta_1 \dots v_{i-1} \beta_{i-1} v_i \beta_i \dots \beta_k && \text{(Simplifying)} \end{aligned}$$

2. When z_i is a unique outer witness. By Equation (37), $z = (z_i \alpha_i \cdots \alpha_k)^{-1} \beta_i \cdots \beta_k$.

$$\begin{aligned}
& \alpha_0 u_1 \alpha_1 \cdots u_{i-1} \alpha_{i-1} u_i \alpha_i \alpha_{i+1} \cdots \alpha_k z \\
&= \alpha_0 u_1 \alpha_1 \cdots u_{i-1} \alpha_{i-1} u_i \alpha_i \alpha_{i+1} \cdots \alpha_k (z_i \alpha_i \cdots \alpha_k)^{-1} \beta_i \cdots \beta_k & (\text{Subs. } z) \\
&= \alpha_0 u_1 \alpha_1 \cdots u_{i-1} \alpha_{i-1} u_i z_i^{-1} \beta_i \cdots \beta_k & (\text{Simplifying}) \\
&= \alpha_0 u_1 \alpha_1 \cdots u_{i-1} \alpha_{i-1} z_i^{-1} v_i \beta_i \cdots \beta_k & (u_i = z_i^{-1} v_i z_i) \\
&= \alpha_0 u_1 \alpha_1 \cdots u_{i-1} \alpha_{i-1} \alpha_i \cdots \alpha_k z (\beta_i \cdots \beta_k)^{-1} v_i \beta_i \cdots \beta_k & (\text{Subs. } z_i^{-1}) \\
&= z \beta_0 v_1 \beta_1 \cdots v_{i-1} \beta_{i-1} \beta_i \cdots \beta_k (\beta_i \cdots \beta_k)^{-1} v_i \beta_i \cdots \beta_k & (\text{Inductive Hypothesis}) \\
&= z \beta_0 v_1 \beta_1 \cdots v_{i-1} \beta_{i-1} v_i \beta_i \cdots \beta_k & (\text{Simplifying})
\end{aligned}$$

3. When z_i is a witness of the primitive root (ρ_i, ρ'_i) of $G_i \cup \{(\alpha_i \cdots \alpha_k \alpha_0 \cdots \alpha_{i-1}, \beta_i \cdots \beta_k \beta_0 \cdots \beta_{i-1})\}$ (The case where $G_i \cup \{(\alpha_i \cdots \alpha_k \alpha_0 \cdots \alpha_{i-1}, \beta_i \cdots \beta_k \beta_0 \cdots \beta_{i-1})\}$ have infinitely many witnesses). Here (u_i, v_i) is some m^{th} power of (ρ_i, ρ'_i) . Since z is a witness of a singleton redux, it is also a witness of the redux $(\alpha_0 \cdots \alpha_k, \beta_0 \cdots \beta_k)$, and hence a witness of its primitive root (ρ, ρ') . We also know that $\alpha_i \cdots \alpha_k$ is an inner witness of (ρ_i, ρ) and $\beta_i \cdots \beta_k$ is an inner witness of (ρ'_i, ρ') .

$$\begin{aligned}
& \alpha_0 u_1 \alpha_1 \cdots u_{i-1} \alpha_{i-1} u_i \alpha_i \alpha_{i+1} \cdots \alpha_k z \\
&= \alpha_0 u_1 \alpha_1 \cdots u_{i-1} \alpha_{i-1} (\rho_i)^m \alpha_i \alpha_{i+1} \cdots \alpha_k z & (\text{Subs. } u_i) \\
&= \alpha_0 u_1 \alpha_1 \cdots u_{i-1} \alpha_{i-1} \alpha_i \alpha_{i+1} \cdots \alpha_k (\rho)^m z & (\alpha_i \cdots \alpha_k \text{ is an i.w. of } (\rho_i, \rho)) \\
&= \alpha_0 u_1 \alpha_1 \cdots u_{i-1} \alpha_{i-1} \alpha_i \alpha_{i+1} \cdots \alpha_k z (\rho')^m & (z \text{ is a witness of } (\rho, \rho')) \\
&= \alpha_0 u_1 \alpha_1 \cdots u_{i-1} \alpha_{i-1} \alpha_i \alpha_{i+1} \cdots \alpha_k z (\beta_i \cdots \beta_k)^{-1} (\rho'_i)^m \beta_i \cdots \beta_k & (\beta_i \cdots \beta_k \text{ is an i.w. of } (\rho'_i, \rho')) \\
&= \alpha_0 u_1 \alpha_1 \cdots u_{i-1} \alpha_{i-1} \alpha_i \alpha_{i+1} \cdots \alpha_k z (\beta_i \cdots \beta_k)^{-1} v_i \beta_i \cdots \beta_k & (\text{Subs. } v_i = (\rho'_i)^m) \\
&= z \beta_0 v_1 \beta_1 \cdots v_{i-1} \beta_{i-1} \beta_i \cdots \beta_k (\beta_i \cdots \beta_k)^{-1} v_i \beta_i \cdots \beta_k & (\text{Inductive Hypothesis}) \\
&= z \beta_0 v_1 \beta_1 \cdots v_{i-1} \beta_{i-1} v_i \beta_i \cdots \beta_k & (\text{Simplifying})
\end{aligned}$$

Thus z is a common inner witness of M . □

6. Conjugacy of Monoid Closures

The entire section is devoted to the proof of Theorem 7.

The proof of ($\Leftarrow$) direction of Theorem 7 is straightforward. For Item (1), if M has a common witness, then there exist a witness for every pair of words produced by M and hence M is conjugate. For Item (2), since each G_i^* is conjugate for $i \in [1, k]$ and $\alpha_0 \cdots \alpha_k \sim \beta_0 \cdots \beta_k$, we get that M produces pairs of words of equal length. Moreover, each pair of words in M is a cyclic shift of $v\rho^i$ for some $i \geq 0$, and hence it is conjugate.

We prove the ($\Rightarrow$) direction in Theorem 7 in the rest of the section. The case when M is of the form $(\alpha_0, \beta_0)G_1^*(\alpha_1, \beta_1)$ holds already from Theorem 8. We now assume that $k \geq 2$.

Since M is conjugate, each G_i^* for $i \in [1, k]$ is conjugate by Corollary 1. Using Theorem 6, we get that each G_i has a common witness. From Proposition 13, either $|\mathcal{W}(G_i^*)| = \infty$ or $|\mathcal{W}(G_i^*)| = 1$.

First, we show that when the redux of M is empty, M has a common witness. If the redux of M is empty, then each singleton redux of M is of the form $G_i^*, i \in [1, k]$ and $M = G_1^* G_2^* \cdots G_k^*$. We define $P = \bigcup_{i=1}^k P_i$ where

- $P_i = \{(\rho_i^{n_i}, \rho_i^{m_i})\}$ if $|\mathcal{W}(G_i^*)| = \infty$, where $\{(\rho_i, \rho'_i)\} = R(G_i^*)$ and $n_i > 0$ is such that $(\rho_i^{n_i}, \rho_i^{m_i}) \in G_i^*$, and
- $P_i = \{(u_1, v_1), (u_2, v_2)\}$ if $|\mathcal{W}(G_i^*)| = 1$, where $(u_1, v_1), (u_2, v_2) \in G_i^*$ are such that $\mathcal{W}(\{(u_1, v_1), (u_2, v_2)\}) = \mathcal{W}(G_i^*)$.

Observe that

$$\prod_{i=1}^k \left(\prod_{p \in P_i} p^* \right) \subseteq M \tag{38}$$

is conjugate. By Lemma 5, the set P has a common witness z . Therefore z is a common witness for each of the G_i^* by construction and Proposition 9. Therefore, z is a common witness for each of the singleton redux namely G_i^* for $i \in [1, k]$. By virtue of Lemma 9, M has a common witness. Thus if redux of M is empty M has a common witness.

Henceforth we assume that the redux of M is nonempty. Let

$$C = |\alpha_0 \alpha_1 \cdots \alpha_k| = |\beta_0 \beta_1 \cdots \beta_k|. \quad (39)$$

Definition 8 (Canonical Pair). A pair $(u', v') \in M$ is canonical if it is of the form:

$$(u, v) = (\alpha_0, \beta_0)(u_1, v_1)^{j_1}(\alpha_1, \beta_1)(u_2, v_2)^{j_2} \cdots (u_k, v_k)^{j_k}(\alpha_k, \beta_k) \quad (40)$$

where $(u_i, v_i) \in G_i^*$ and $j_i \in \mathbb{N}$, for $i \in [1, k]$, satisfies the following conditions.

1. Let $i \in [1, k]$. Then,

- (a) $(u_i, v_i) = (\rho_i^n, \rho_i'^n)$ if $|W(G_i^*)| = \infty$, where $R(G_i) = \{(\rho_i, \rho_i')\}$ and $n \geq 1$ is such that $(\rho_i^n, \rho_i'^n) \in G_i^*$, and
- (b) $(u_i, v_i) = (u, v) \in G_i^*$ if $|W(G_i^*)| = 1$, where $(u, v) \in G_i^*$ is such that

$$W(\{(u, v), (\alpha_i \cdots \alpha_k \alpha_1 \cdots \alpha_{i-1}, \beta_i \cdots \beta_k \beta_1 \cdots \beta_{i-1})\}) = W(G_i^*) \quad (41)$$

guaranteed by Lemma 7.

2. If $|W(G_i^*)| = 1$ for some $i \in [1, k]$, then $\rho_{u_j} \not\sim \rho_{u_l}$ for some $j, l \in [1, k]$.

We next establish the existence of canonical pairs.

Lemma 10. Let ρ be a primitive word. If $|W(G_i^*)| = 1$, then there exists a pair $(u_i, v_i) \in G_i^*$ such that $\rho_{u_i} \not\sim \rho$ and $W(\{(u_i, v_i), (\alpha_i \cdots \alpha_k \alpha_0 \cdots \alpha_{i-1}, \beta_i \cdots \beta_k \beta_0 \cdots \beta_{i-1})\}) = W(G_i^*)$.

Proof. Assume that G_i^* has a unique common witness and let ρ be an arbitrary primitive word.

Let $p = (\alpha_i \cdots \alpha_k \alpha_0 \cdots \alpha_{i-1}, \beta_i \cdots \beta_k \beta_0 \cdots \beta_{i-1})$. Since $|W(G_i^*)| = 1$, there exist at least two pairs $(u_1, v_1), (u_2, v_2) \in G_i^*$ such that $\rho_{u_1} \neq \rho_{u_2}$. Hence, by Proposition 3, we can construct three pairs in G_i^* that have pairwise distinct primitive roots. Therefore, there exist two pairs $(u_1, v_1), (u_2, v_2) \in G_i^*$ that have primitive roots different from that of p . Therefore any two pairs from the set $\{(u_1, v_1), (u_2, v_2), p\}$ has a unique common witness by Corollary 2. Moreover, by Lemma 7,

$$W(\{(u_1, v_1), p\}) = W(\{(u_2, v_2), p\}) = W(G_i^*). \quad (42)$$

If either $\rho_{u_1} \not\sim \rho$ or $\rho_{u_2} \not\sim \rho$, then we choose the corresponding pair as (u_i, v_i) , and hence the claim follows. Otherwise, assume that $\rho_{u_1} \sim \rho \sim \rho_{u_2}$.

Next consider the words $(u_3, v_3) = (u_1 u_2, v_1 v_2)$ and $(u_4, v_4) = (u_3 u_1, v_3 v_1)$. By Proposition 4, we deduce that $\rho_{u_3} \not\sim \rho$ and $\rho_{u_4} \not\sim \rho$. If ρ_{u_3} differs from the primitive root of $\text{fst}(p)$, then $W(\{(u_3, v_3), p\}) = W(G_i^*)$ and $\rho_{u_3} \not\sim \rho$. Hence the word $(u_i, v_i) = (u_3, v_3)$ satisfies the claim. Otherwise, ρ_{u_3} is the primitive root of $\text{fst}(p)$. By Proposition 3, $\rho_{u_3} \notin \{\rho_{u_1}, \rho_{u_2}\}$ since $\rho_{u_1} \neq \rho_{u_2}$. By applying Proposition 3 again, we get $\rho_{u_4} \neq \rho_{u_3}$. Hence, the primitive roots of u_4 and $\text{fst}(p)$ differ. Thus, $(u_i, v_i) = (u_4, v_4)$ satisfies the claim. $\square$

Corollary 3. If M is conjugate then there exists a canonical pair in M .

Proof. We construct a canonical pair (u, v) in M by choosing, for each $i \in [1, k]$, a pair $(u_i, v_i) \in G_i^*$ satisfying Item 1 of Definition 8. Item 1.(a) holds trivially, while Item 1.(b) follows from Lemma 7. It remains to verify Item 2 in Definition 8. If $|W(G_i^*)| = \infty$ for every $i \in [1, k]$, then Item 2 is vacuously satisfied. Otherwise, suppose that there exists an index $i \in [1, k]$ such that $|W(G_i^*)| = 1$. If there exists $j \neq i \in [1, k]$ such that $\rho_{u_i} \not\sim \rho_{u_j}$, then we are done. Otherwise, using Lemma 10 we choose a pair $(u'_i, v'_i) \in G_i^*$ that satisfies $\rho_{u'_i} \not\sim \rho_{u_i}$ and Item 1.(b). $\square$

Definition 9 (Differentially-Pumped Pair). A Differentially-Pumped Pair is a canonical pair of the form

$$(u, v) = (\alpha_0, \beta_0)(u_1, v_1)^{j_1}(\alpha_1, \beta_1)(u_2, v_2)^{j_2} \cdots (u_k, v_k)^{j_k}(\alpha_k, \beta_k) \quad (43)$$

where $j_1 > 2k \cdot \max(C, \ell)$ where $\ell = 2 \cdot \max\{|u_i| \mid i \in [1, k]\}$, and $j_2, \dots, j_k$ obeys

$$2 \cdot (|u_1^{j_1}| + |u_2^{j_2}| + \cdots + |u_{i-1}^{j_{i-1}}| + C) < |u_i^{j_i}|. \quad (44)$$

Observe that $\ell > |u_i| + |u_j| - \gcd(|u_i|, |u_j|) > 0$ (Fine & Wilf index between u_i and u_j) for $i, j \in [1, k]$.

Since M is conjugate, a differentially-pumped pair $(u, v) \in M$ is conjugate and has a cut, say (p, q) . Towards proving Theorem 7, we do a case analysis on the cut depending whether the cut (p, q) is short or long (defined below).

Definition 10 (Short & Long Cuts). Let $(u, v) \in M$ be the differentially pumped pair from Definition 9. Let (p, q) be a cut of (u, v) . We say p is short if the prefix $p\beta_0$ of u is a prefix of $\alpha_0 u_1^{j_1-1}$ as shown below.

$$\begin{aligned} u &= \overbrace{\alpha_0 u_1 \cdots u_1}^{p\beta_0} \cdots \overbrace{u_1 \alpha_1 u_2 \cdots u_2 \alpha_2 \cdots u_k \cdots u_k \alpha_k}^{\beta_0^{-1}q} \\ v &= \underbrace{\beta_0 v_1 \cdots v_1 \beta_1 v_2 \cdots v_2 \beta_2 \cdots v_k \cdots v_k \beta_k}_{q} \end{aligned}$$

Similarly, q is short if the prefix $q\alpha_0$ of v is a prefix of $\beta_0 v_1^{j_1-1}$. The cut (p, q) is short if either p or q is short. When p, q or the cut (p, q) is not short, they are long.

Definition 11 (Blocks of a Canonical Pair). Given a differentially-pumped pair from Definition, $B_1, B_2, \dots, B_k$ represent the factors of $u_1^{j_1}, u_2^{j_2}, \dots, u_k^{j_k}$ respectively in u . Likewise $B'_1, B'_2, \dots, B'_k$ represent the factors $v_1^{j_1}, v_2^{j_2}, \dots, v_k^{j_k}$ respectively in v .

$$\begin{aligned} u &= \alpha_0 \overbrace{u_1 \cdots u_1}^{j_1 \text{ times}} \alpha_1 \overbrace{u_2 \cdots u_2}^{j_2 \text{ times}} \alpha_2 \cdots \overbrace{u_k \cdots u_k}^{j_k \text{ times}} \alpha_k &= \alpha_0 \overbrace{x_1 y_1 \cdots x_1 y_1}^{B_1} \alpha_1 \overbrace{x_2 y_2 \cdots x_2 y_2}^{B_2} \alpha_2 \cdots \overbrace{x_k y_k \cdots x_k y_k}^{B_k} \alpha_k \\ v &= \beta_0 v_1 \cdots v_1 \beta_1 v_2 \cdots v_2 \beta_2 \cdots v_k \cdots v_k \beta_k &= \beta_0 \overbrace{y_1 x_1 \cdots y_1 x_1}^{B'_1} \beta_1 \overbrace{y_2 x_2 \cdots y_2 x_2}^{B'_2} \beta_2 \cdots \overbrace{y_k x_k \cdots y_k x_k}^{B'_k} \beta_k \end{aligned}$$

Also, let (x_i, y_i) denote a cut of (ρ_{u_i}, ρ_{v_i}) for each $i \in [1, k]$.

In the next two subsections, we prove that whenever a differentially-pumped pair has a short cut, M has a common witness (See Proposition 16). Otherwise, if it has a long cut then we show that there exist words v, ρ such that $\text{fst}(G_i^*) \subseteq \rho^*$ for all $i \in [1, k]$, and $\text{fst}(M) \sim \text{snd}(M) \subseteq v\rho^*$ where either $v = \epsilon$ or $v \sim \alpha_0 \cdots \alpha_k$ (See Proposition 17). As a consequence, Theorem 7 follows.

6.1. Short cut for Differentially-pumped pair

In this subsection, we prove that whenever a differentially-pumped pair has a short cut, then M has a common witness.

Proposition 16. Assume that M is conjugate and let $(u, v) \in M$ be a differentially-pumped pair from Definition 9. If (u, v) has a short cut then M has a common witness.

First, we prove some preparatory lemmas.

Lemma 11. Assume that (u, v) from Definition 9 has a short cut (p, q) such that p is short. Then the following conditions hold.

1. Let $w_1 = [\alpha_0, p\beta_0]_L$. If α_0 is a prefix of $p\beta_0$, then w_1 is an inner witness of (u_1, v_1) , and $B_1 = w_1 q_1$ and $B'_1 = q_1 w_1$ for some word q_1 . Otherwise, $p\beta_0$ is a prefix of α_0 and w_1 is an outer witness of (u_1, v_1) , and $B_1 = q_1 w_1$ and $B'_1 = w_1 q_1$ for some word q_1 .

2. For $i \in [2, k]$,

- (a) if w_{i-1} is an inner witness of (u_{i-1}, v_{i-1}) , then $w_i = [\alpha_{i-1}, w_{i-1}\beta_{i-1}]_L$ satisfies the following: If α_{i-1} is a prefix of $w_{i-1}\beta_{i-1}$, then w_i is an inner witness of (u_i, v_i) , and $B_i = w_i q_i$ and $B'_i = q_i w_i$ for some word q_i . Otherwise, $w_{i-1}\beta_{i-1}$ is a prefix of α_{i-1} and w_i is an outer witness of (u_i, v_i) , and $B_i = q_i w_i$ and $B'_i = w_i q_i$ for some word q_i .
- (b) If w_{i-1} is an outer witness of (u_{i-1}, v_{i-1}) , then $w_i = [w_{i-1}\alpha_{i-1}, \beta_{i-1}]_L$ satisfies the following: If $w_{i-1}\alpha_{i-1}$ is a prefix of β_{i-1} , then w_i is an inner witness of (u_i, v_i) , and $B_i = w_i q_i$ and $B'_i = q_i w_i$ for some word q_i . Otherwise, β_{i-1} is a prefix of $w_{i-1}\alpha_{i-1}$ and w_i is an outer witness of (u_i, v_i) , and $B_i = q_i w_i$ and $B'_i = w_i q_i$ for some word q_i .

3. If w_k is an inner witness of (u_k, v_k) , then $w_1 = [\alpha_k \alpha_0, w_k \beta_k \beta_0]_L$. Moreover, if w_1 is an inner witness (u_1, v_1) , then $w_1 = (\alpha_k \alpha_0)^{-1} w_k \beta_k \beta_0$, and $w_1 = (w_k \beta_k \beta_0)^{-1} \alpha_k \alpha_0$ otherwise. If w_k is an outer witness of (u_k, v_k) , then $w_1 = [w_k \alpha_k \alpha_0, \beta_k \beta_0]_L$. Moreover, if w_1 is an inner witness (u_1, v_1) , then $w_1 = (w_k \alpha_k \alpha_0)^{-1} \beta_k \beta_0$, and $w_1 = (\beta_k \beta_0)^{-1} w_k \alpha_k \alpha_0$ otherwise.

Proof. 1. Let $w_1 = [\alpha_0, p\beta_0]_L$. There are two cases: either α_0 is a prefix of $p\beta_0$, or $p\beta_0$ is a prefix of α_0 . In either case, we show that w_1 is a witness of (u_1, v_1) .

- α_0 is a prefix of $p\beta_0$: We show that w_1 is an inner witness of (u_1, v_1) , and there exists a word q_1 such that $B_1 = w_1 q_1$ and $B'_1 = q_1 w_1$.

Since α_0 is a prefix of $p\beta_0$ and p is short, $w_1 = \alpha_0^{-1} p\beta_0$ is a prefix of $B_1(u_1)^{-1}$ in u . We match $(p\beta_0)^{-1}u$ with $\beta_0^{-1}vp^{-1}$. Since $\beta_0^{-1}vp^{-1}$ starts with y_1x_1 and since there is at least one factor x_1y_1 following $p\beta_0$, either $w_1 \in (x_1y_1)^*x_1$ (by Case I. of Cut Lemma) or $w_1 \in (x_1y_1)^*$ and $x_1y_1 = y_1x_1$ (by II. of Cut Lemma). In the former case, w_1 is an inner witness of $(x_1y_1, y_1x_1) = (\rho_{u_1}, \rho_{v_1})$ and hence is also an inner witness of (u_1, v_1) by Proposition 9. Similarly, when $w_1 \in (x_1y_1)^*$ and $x_1y_1 = y_1x_1$, we get w_1 is an inner witness of (x_1y_1, y_1x_1) as well as (u_1, v_1) .

Let $q_1 = w_1^{-1}B_1$. Clearly, q_1 is of the form $(y_1x_1)^*y_1$ or $(x_1y_1 = y_1x_1)^*$. In both cases, after matching q_1 in block B'_1 of v , we observe that a factor equal to w_1 appears in the suffix of the block of B'_1 , i.e., $B'_1 = q_1 w_1$ (as shown below).

$$\begin{array}{l} u = \alpha_0 \overbrace{x_1y_1 \cdots x_1}^{w_1} \overbrace{\cdots x_1y_1}^{q_1} \alpha_1 x_2y_2 \cdots x_2y_2 \alpha_2 \cdots x_ky_k \cdots x_ky_k \alpha_k \\ v = \beta_0 \underbrace{y_1x_1 \cdots y_1}_{q_1} \underbrace{x_1 \cdots y_1x_1}_{w_1} \beta_1 y_2x_2 \cdots y_2x_2 \beta_2 \cdots y_kx_k \cdots y_kx_k \beta_k \end{array}$$

- When $p\beta_0$ is a prefix of α_0 : We show that w_1 is an outer witness of (u_1, v_1) , and there exists a word q_1 such that $B_1 = q_1 w_1$ and $B'_1 = w_1 q_1$.

Since $p\beta_0$ is a prefix of α_0 and $|\alpha_0| \leq C < j_1$, $|w_1| = |(p\beta_0)^{-1}\alpha_0| < |v_1^{j_1}|$. Therefore w_1 is a prefix of $B'_1(v_1)^{-1}$ in v . On matching $(p\beta_0)^{-1}u$ and $\beta_0^{-1}vp^{-1}$, since the block B_1 in u starts with x_1y_1 and since there is at least one factor y_1x_1 in B'_1 after matching w_1 , either $w_1 \in (y_1x_1)^*y_1$ (by Case I. of Cut Lemma) or $w_1 \in (y_1x_1)^*$ and $x_1y_1 = y_1x_1$ (by II. of Cut Lemma). In the former case, w_1 is an outer witness of $(x_1y_1, y_1x_1) = (\rho_{u_1}, \rho_{v_1})$ and hence is also an outer witness of (u_1, v_1) by Proposition 9. When $w_1 \in (y_1x_1)^*$, we get $x_1y_1 = y_1x_1$ and hence w_1 is an outer witness of (x_1y_1, y_1x_1) as well as (u_1, v_1) .

Let $q_1 = (w_1)^{-1}B'_1$. As in the case above, $q_1 \in (x_1y_1)^*x_1$ or $q_1 \in (x_1y_1 = y_1x_1)^*$. In both cases, after matching q_1 in block B_1 of u , a factor equal to w_1 appears in the suffix of the block of B_1 , i.e., $B_1 = q_1 w_1$ (as shown below).

$$\begin{aligned}
u &= \overbrace{\alpha_0}^{p\beta_0w_1} \overbrace{x_1y_1 \cdots x_1}^{q_1} \overbrace{y_1 \cdots x_1y_1}^{w_1} \alpha_1 x_2y_2 \cdots x_2y_2 \alpha_2 \cdots x_ky_k \cdots x_ky_k \alpha_k \\
v &= \beta_0 \underbrace{y_1x_1 \cdots y_1}_{w_1} \underbrace{x_1 \cdots y_1x_1}_{q_1} \beta_1 y_2x_2 \cdots y_2x_2 \beta_2 \cdots y_kx_k \cdots y_kx_k \beta_k
\end{aligned}$$

2. Proof by induction on $i \in [2, k]$. The base case and the inductive step are identical. Hence we treat only the inductive step.

WLOG assume that w_{i-1} is an inner witness of (u_{i-1}, v_{i-1}) . The case where w_{i-1} is an outer witness of (u_{i-1}, v_{i-1}) is symmetric. In this case, $B_{i-1} = w_{i-1}q_{i-1}$ and $B'_{i-1} = q_{i-1}w_{i-1}$ (as shown below). Moreover, $|w_{i-1}| \leq |u_{i-1}^{j_{i-1}}|$.

$$\begin{aligned}
u &= \cdots \alpha_{i-2} \overbrace{x_{i-1}y_{i-1} \cdots x_{i-1}}^{w_{i-1}} \overbrace{\cdots x_{i-1}y_{i-1}}^{q_{i-1}} \alpha_{i-1} x_iy_i \cdots x_iy_i \cdots \\
v &= \cdots \beta_{i-2} \underbrace{y_{i-1}x_{i-1} \cdots x_{i-1}}_{q_{i-1}} \underbrace{\cdots y_{i-1}x_{i-1}}_{w_{i-1}} \beta_{i-1} y_ix_i \cdots y_ix_i \cdots
\end{aligned}$$

On matching further, there are two cases for $w_i = [\alpha_{i-1}, w_{i-1}\beta_{i-1}]_L$ as follows.

- *When α_{i-1} is a prefix of $w_{i-1}\beta_{i-1}$:* Here, $w_i = (\alpha_{i-1})^{-1}w_{i-1}\beta_{i-1}$ begins at the block B_i in u as depicted in Figure 4. Since $|w_i| \leq |w_{i-1}\beta_{i-1}| \leq |u_{i-1}^{j_{i-1}}| + C < |u_i^{j_i}|/2$, w_i ends within block B_i and there exists at least one x_iy_i in B_i immediately following w_i . Let $B_i = w_iq_i$ for some word q_i . On further matching q_i in u and v , since the block B'_i in v starts with y_ix_i and since q_i consists at least one factor x_1y_1 , either $w_i \in (x_iy_i)^*x_i$ (by Case I. of Cut Lemma) or $w_i \in (x_iy_i)^*$ and $x_iy_i = y_ix_i$ (by II. of Cut Lemma). In the former case, w_i is an inner witness of $(x_iy_i, y_ix_i) = (\rho_{u_i}, \rho_{v_i})$ and hence also an inner witness of (u_i, v_i) by Proposition 9. When $w_i \in (x_iy_i)^*$, we get $x_iy_i = y_ix_i$ and hence w_i is an inner witness of (x_iy_i, y_ix_i) as well as (u_i, v_i) .

Since $B_i = w_iq_i$, after matching q_i in block B'_i of v , we observe that a factor equal to w_i appears in the suffix of the block of B'_i , i.e., $B'_i = q_iw_i$ (as shown below).

$$\begin{aligned}
u &= \cdots \alpha_{i-2} \overbrace{x_{i-1}y_{i-1} \cdots x_{i-1}}^{w_{i-1}} \overbrace{\cdots x_{i-1}y_{i-1}}^{q_{i-1}} \alpha_{i-1} \overbrace{x_iy_i \cdots x_iy_i}^{w_i} \overbrace{\cdots x_iy_i}^{q_i} \alpha_i \cdots \\
v &= \cdots \beta_{i-2} \underbrace{y_{i-1}x_{i-1} \cdots x_{i-1}}_{q_{i-1}} \underbrace{\cdots y_{i-1}x_{i-1}}_{w_{i-1}} \beta_{i-1} \underbrace{y_ix_i \cdots y_ix_i}_{q_i} \underbrace{\cdots y_ix_i}_{w_i} \beta_i \cdots
\end{aligned}$$

Figure 4: w_i when w_{i-1} is an inner witness and α_{i-1} is a prefix of $w_{i-1}\beta_{i-1}$

- *When $w_{i-1}\beta_{i-1}$ is a prefix of α_{i-1} :* Here, $w_i = (w_{i-1}\beta_{i-1})^{-1}\alpha_{i-1}$ begins at the block B'_i in v . Since $|w_i| \leq |\alpha_{i-1}| \leq C < |u_i^{j_i}|/2$, w_i ends within block B'_i and there exists at least one y_ix_i in B'_i immediately following w_i . Let $B'_i = w_iq_i$ for some word q_i . On further matching q_i in u and v , since the block B_i in u starts with x_iy_i and since q_i contains at least one y_ix_i , either $w_i \in (y_ix_i)^*y_i$ (by Case I. of Cut Lemma) or $w_i \in (y_ix_i)^*$ and $x_iy_i = y_ix_i$ (by II. of Cut Lemma). In the former case, w_i is an outer witness of $(x_iy_i, y_ix_i) = (\rho_{u_i}, \rho_{v_i})$ and hence also an outer witness of (u_i, v_i) by Proposition 9. When $w_i \in (y_ix_i)^*$, we get $x_iy_i = y_ix_i$ and hence w_i is an outer witness of (x_iy_i, y_ix_i) as well as (u_i, v_i) .

Since $B'_i = w_iq_i$, after matching q_i in block B_i of u , a factor equal to w_i appears in the suffix of the block of B_i , i.e., $B_i = q_iw_i$ (as shown below).

Thus, the claim holds for $i \in \{2, \dots, k\}$.

$$\begin{aligned}
u &= \cdots \alpha_{i-2} \overbrace{x_{i-1}y_{i-1} \cdots x_{i-1}}^{w_{i-1}} \overbrace{\cdots x_{i-1}y_{i-1}}^{q_{i-1}} \alpha_{i-1} \overbrace{x_i y_i \cdots}^{q_i} \overbrace{\cdots x_i y_i}^{w_i} \alpha_i \cdots \\
v &= \cdots \beta_{i-2} \overbrace{y_{i-1}x_{i-1} \cdots}^{q_{i-1}} \overbrace{x_{i-1} \cdots y_{i-1}x_{i-1}}^{w_{i-1}} \beta_{i-1} \overbrace{y_i x_i \cdots}^{w_i} \overbrace{\cdots y_i x_i}^{q_i} \beta_i \cdots
\end{aligned}$$

Figure 5: w_i when w_{i-1} is an inner witness and $w_{i-1}\beta_{i-1}$ is a prefix of α_{i-1}

3. We prove the case when both w_1 and w_k are inner witnesses of (u_1, v_1) and (u_k, v_k) respectively (as shown below). All other cases are symmetric.

$$\begin{aligned}
u &= \alpha_0 \overbrace{x_1 y_1 \cdots x_1}^{w_1} \overbrace{\cdots x_1 y_1}^{q_1} \alpha_1 x_2 y_2 \cdots x_2 y_2 \alpha_2 \cdots \overbrace{x_k y_k \cdots x_k}^{w_k} \overbrace{\cdots x_k y_k}^{q_k} \alpha_k \\
v &= \beta_0 \overbrace{y_1 x_1 \cdots y_1}^{q_1} \overbrace{x_1 \cdots y_1 x_1}^{w_1} \beta_1 y_2 x_2 \cdots y_2 x_2 \beta_2 \cdots \overbrace{y_k x_k \cdots}^{q_k} \overbrace{x_k \cdots y_k x_k}^{w_k} \beta_k
\end{aligned}$$

We get that

$$\begin{aligned}
w_k \beta_k &= \alpha_k p && \text{(By matching } u \text{ and } v) \\
w_k \beta_k \beta_0 &= \alpha_k p \beta_0 && \text{(Appending } \beta_0) \\
&= \alpha_k \alpha_0 w_1. && \text{(Subs. } p\beta_0 = \alpha_0 w_1) \quad \square
\end{aligned}$$

Lemma 12. For each $i \in [1, k]$, let w_i be the witness of (u_i, v_i) given by Lemma 11. Then for each $j \in [1, k]$ such that $j > i$, the following holds.

1. If w_i is an inner witness of (u_i, v_i) , then $w_j = [\alpha_i \cdots \alpha_{j-1}, w_i \beta_i \cdots \beta_{j-1}]_L$. More precisely, if w_j is an inner witness of (u_j, v_j) , then $w_j = (\alpha_i \cdots \alpha_{j-1})^{-1} w_i \beta_i \cdots \beta_{j-1}$, otherwise $w_j = (w_i \beta_i \cdots \beta_{j-1})^{-1} \alpha_i \cdots \alpha_{j-1}$.
2. If w_i is an outer witness of (u_i, v_i) , then $w_j = [w_i \alpha_i \cdots \alpha_{j-1}, \beta_i \cdots \beta_{j-1}]_L$. More precisely, if w_j is an inner witness of (u_j, v_j) , then $w_j = (w_i \alpha_i \cdots \alpha_{j-1})^{-1} \beta_i \cdots \beta_{j-1}$, otherwise $w_j = (\beta_i \cdots \beta_{j-1})^{-1} w_i \alpha_i \cdots \alpha_{j-1}$.

Proof. Proof by induction on $j - i$. Base case is when $j = i + 1$ and the claim holds by Lemma 11. WLOG assume w_i is an inner witness of (u_i, v_i) and the case 1. holds for j with $i < j < k$, and we show that it also holds for $j + 1$. There are two cases:

- When w_j is an inner witness of (u_j, v_j) : $w_j = (\alpha_i \cdots \alpha_{j-1})^{-1} w_i \beta_i \cdots \beta_{j-1}$ (by inductive hypothesis). From Lemma 11, we get that w_{j+1} is a witness of (u_{j+1}, v_{j+1}) and

$$\begin{aligned}
w_{j+1} &= [\alpha_j, w_j \beta_j]_L && \text{(By Lemma 11)} \\
&= [\alpha_j, (\alpha_i \cdots \alpha_{j-1})^{-1} w_i \beta_i \cdots \beta_{j-1} \beta_j]_L && \text{(Subs. } w_j) \\
&= [\alpha_i \cdots \alpha_{j-1} \alpha_j, w_i \beta_i \cdots \beta_{j-1} \beta_j]_L && \text{(Since } [u, v]_L = [wu, wv]_L \text{ for all } w)
\end{aligned}$$

If w_{j+1} is an inner witness of (u_{j+1}, v_{j+1}) , then

$$\begin{aligned}
w_{j+1} &= (\alpha_j)^{-1} w_j \beta_j \\
&= (\alpha_j)^{-1} (\alpha_i \cdots \alpha_{j-1})^{-1} w_i \beta_i \cdots \beta_{j-1} \beta_j && \text{(Subs. } w_j \text{ by I.H.)} \\
&= (\alpha_i \cdots \alpha_{j-1} \alpha_j)^{-1} w_i \beta_i \cdots \beta_{j-1} \beta_j
\end{aligned}$$

Similarly, if w_{j+1} is an outer witness of (u_{j+1}, v_{j+1}) , then

$$\begin{aligned}
w_{j+1} &= (w_j \beta_j)^{-1} \alpha_j \\
&= ((\alpha_i \cdots \alpha_{j-1})^{-1} w_i \beta_i \cdots \beta_{j-1} \beta_j)^{-1} \alpha_j && \text{(Subs. } w_j \text{ by I.H.)} \\
&= (w_i \beta_i \cdots \beta_{j-1} \beta_j)^{-1} \alpha_i \cdots \alpha_{j-1} \alpha_j
\end{aligned}$$

- When w_j is an outer witness: $w_j = (w_i\beta_i \cdots \beta_{j-1})^{-1}\alpha_i \cdots \alpha_{j-1}$ (by inductive hypothesis). From Lemma 11, we get that w_{j+1} is a witness of (u_{j+1}, v_{j+1}) and

$$\begin{aligned} w_{j+1} &= [w_j\alpha_j, \beta_j]_L && \text{(By Lemma 11)} \\ &= [(w_i\beta_i \cdots \beta_{j-1})^{-1}(\alpha_i \cdots \alpha_{j-1})\alpha_j, \beta_j]_L && \text{(Subs. } w_j) \\ &= [\alpha_i \cdots \alpha_{j-1}\alpha_j, w_i\beta_i \cdots \beta_{j-1}\beta_j]_L && ([u, v]_L = [wu, wv]_L \text{ for all } w) \end{aligned}$$

If w_{j+1} is an inner witness of (u_{j+1}, v_{j+1}) , then

$$\begin{aligned} w_{j+1} &= (w_j\alpha_j)^{-1}\beta_j \\ &= ((w_i\beta_i \cdots \beta_{j-1})^{-1}\alpha_i \cdots \alpha_{j-1}\alpha_j)^{-1}\beta_j && \text{(Subs. } w_j \text{ by I.H.)} \\ &= (\alpha_i \cdots \alpha_{j-1}\alpha_j)^{-1}w_i\beta_i \cdots \beta_{j-1}\beta_j \end{aligned}$$

Similarly, if w_{j+1} is an outer witness of (u_{j+1}, v_{j+1}) , we can show that $w_{j+1} = (w_i\beta_i \cdots \beta_{j-1}\beta_j)^{-1}\alpha_i \cdots \alpha_{j-1}\alpha_j$.

The case where w_i is an outer witness of (u_i, v_i) is symmetric. $\square$

Lemma 13. For each $i \in [1, k]$, let w_i be the witness of (u_i, v_i) given by Lemma 11. For each $i \in [1, k]$, if w_i is an inner (resp. outer) witness of (u_i, v_i) , then w_i is also an inner (resp. outer) witness of $(\alpha_i \cdots \alpha_k \alpha_0 \cdots \alpha_{i-1}, \beta_i \cdots \beta_k \beta_0 \cdots \beta_{i-1})$.

Proof. WLOG assume that w_i is an inner witness of (u_i, v_i) . The other case is symmetric. Using Lemma 12,

$$\begin{aligned} w_k &= [\alpha_i \cdots \alpha_{k-1}, w_i\beta_i \cdots \beta_{k-1}]_L, \\ w_i &= (\alpha_1 \cdots \alpha_{i-1})^{-1}w_1\beta_1 \cdots \beta_{i-1} \end{aligned} \tag{45}$$

We perform a case analysis on whether w_1 and w_k are inner/outer witnesses. Assume that w_1 is an inner witness of (u_1, v_1) and w_k is an outer witness of (u_k, v_k) . Hence,

$$w_k = (w_i\beta_i \cdots \beta_{k-1})^{-1}\alpha_i \cdots \alpha_{k-1} \tag{46}$$

Using Item 3 of Lemma 11, we get that

$$w_1 = (w_k\alpha_k\alpha_0)^{-1}\beta_k\beta_0. \tag{47}$$

Therefore,

$$\begin{aligned} w_i &= (\alpha_1 \cdots \alpha_{i-1})^{-1}w_1\beta_1 \cdots \beta_{i-1} && \text{(Using Equation (45))} \\ &= (\alpha_1 \cdots \alpha_{i-1})^{-1}(w_k\alpha_k\alpha_0)^{-1}\beta_k\beta_0\beta_1 \cdots \beta_{i-1} && \text{(Subs. Equation (47))} \\ &= (w_k\alpha_k\alpha_0\alpha_1 \cdots \alpha_{i-1})^{-1}\beta_k\beta_0\beta_1 \cdots \beta_{i-1} \\ &= ((w_i\beta_i \cdots \beta_{k-1})^{-1}\alpha_i \cdots \alpha_{k-1}\alpha_k\alpha_0\alpha_1 \cdots \alpha_{i-1})^{-1}\beta_k\beta_0\beta_1 \cdots \beta_{i-1} && \text{(Subs. Equation (46))} \\ &= (\alpha_i \cdots \alpha_{k-1}\alpha_k\alpha_0\alpha_1 \cdots \alpha_{i-1})^{-1}w_i\beta_i \cdots \beta_{k-1}\beta_k\beta_0\beta_1 \cdots \beta_{i-1} \end{aligned}$$

Hence w_i is also an inner witness of $(\alpha_i \cdots \alpha_k \alpha_0 \cdots \alpha_{i-1}, \beta_i \cdots \beta_k \beta_0 \cdots \beta_{i-1})$. All other cases for w_1 and w_k are analogous. $\square$

Lemma 14. Assume that M is conjugate and a differentially-pumped pair in M has a short cut. If any two singleton redxes of M have unique common witnesses, then they are identical. Moreover, either both are inner, or both are outer witnesses.

Proof. Let

$$M_i = (\alpha_0 \cdots \alpha_{i-1}, \beta_0 \cdots \beta_{i-1})G_i^*(\alpha_i \cdots \alpha_k, \beta_i \cdots \beta_k) \tag{48}$$

$$M_j = (\alpha_0 \cdots \alpha_{j-1}, \beta_0 \cdots \beta_{j-1})G_j^*(\alpha_j \cdots \alpha_k, \beta_j \cdots \beta_k) \tag{49}$$

be two singleton reduxes with unique common witnesses c_i and c_j respectively. Using Theorem 8, we get that the set $G_i \cup \{(\alpha_i \cdots \alpha_k \alpha_0 \cdots \alpha_{i-1}, \beta_i \cdots \beta_k \alpha_0 \cdots \beta_{i-1})\}$ has a unique common witness, say z_i . Moreover, by construction, z_i is also the unique common witness of the pairs (u_i, v_i) used in the canonical pair from Definition 9 and $(\alpha_i \cdots \alpha_k \alpha_0 \cdots \alpha_{i-1}, \beta_i \cdots \beta_k \beta_0 \cdots \beta_{i-1})$. Analogously, let z_j be the unique common witness of the set $G_j \cup \{(\alpha_j \cdots \alpha_k \alpha_0 \cdots \alpha_{j-1}, \beta_j \cdots \beta_k \alpha_0 \cdots \beta_{j-1})\}$ and $\{(u_j, v_j), (\alpha_j \cdots \alpha_k \alpha_0 \cdots \alpha_{j-1}, \beta_j \cdots \beta_k \alpha_0 \cdots \beta_{j-1})\}$ where (u_j, v_j) is from Definition 9. From Lemma 12, $z_i = w_i$ and $z_j = w_j$. WLOG assume that $i < j$. There are four cases depending on whether w_i and w_j is an inner or outer witness.

- *When w_i and w_j are inner witnesses:* By Lemma 12, $w_j = (\alpha_i \cdots \alpha_{j-1})^{-1} w_i \beta_i \cdots \beta_{j-1}$. By Lemma 8, $c_i = [\alpha_0 \cdots \alpha_{i-1} w_i, \beta_0 \cdots \beta_{i-1}]_R$ and $c_j = [\alpha_0 \cdots \alpha_{j-1} w_j, \beta_0 \cdots \beta_{j-1}]_R$. Assume that c_j is a common inner witness. We show that $c_i = c_j$ and it is also an inner witness.

$$\begin{aligned}
c_j &= [\alpha_0 \cdots \alpha_{j-1} w_j, \beta_0 \cdots \beta_{j-1}]_R \\
&= \alpha_0 \cdots \alpha_{j-1} w_j (\beta_0 \cdots \beta_{j-1})^{-1} && \text{(By Lemma 8)} \\
&= \alpha_0 \cdots \alpha_{j-1} (\alpha_i \cdots \alpha_{j-1})^{-1} w_i \beta_i \cdots \beta_{j-1} (\beta_0 \cdots \beta_{j-1})^{-1} && \text{(Subs. } w_j) \\
&= \alpha_0 \cdots \alpha_{i-1} w_i (\beta_0 \cdots \beta_{i-1})^{-1} && \text{(Simplifying)} \\
&= [\alpha_0 \cdots \alpha_{i-1} w_i, \beta_0 \cdots \beta_{i-1}]_R \\
&= c_i
\end{aligned}$$

We also deduce that c_i is also a common inner witness by Lemma 8. Similarly, we can show that when c_j is a common outer witness of M_j , then $c_i = c_j$ is also a common outer witness of M_i .

- *When w_i and w_j is a unique common outer witness:* the case the similar to above.
- *When w_i is a unique common inner witness and w_j is a unique common outer witness:* By Lemma 12, $w_j = (w_i \beta_i \cdots \beta_{j-1})^{-1} \alpha_i \cdots \alpha_{j-1}$. By Lemma 8, $c_i = [\alpha_0 \cdots \alpha_{i-1} w_i, \beta_0 \cdots \beta_{i-1}]_R$ and $c_j = [\alpha_0 \cdots \alpha_{j-1}, \beta_0 \cdots \beta_{j-1} w_j]_R$. Assume that c_j is a common inner witness. We show that $c_i = c_j$ and it is also an inner witness.

$$\begin{aligned}
c_j &= [\alpha_0 \cdots \alpha_{j-1}, \beta_0 \cdots \beta_{j-1} w_j]_R \\
&= \alpha_0 \cdots \alpha_{j-1} (\beta_0 \cdots \beta_{j-1} w_j)^{-1} && \text{(By Lemma 8)} \\
&= \alpha_0 \cdots \alpha_{j-1} (\beta_0 \cdots \beta_{j-1} (w_i \beta_i \cdots \beta_{j-1})^{-1} \alpha_i \cdots \alpha_{j-1})^{-1} && \text{(Subs. } w_j) \\
&= \alpha_0 \cdots \alpha_{j-1} (\beta_0 \cdots \beta_{i-1} (w_i)^{-1} \alpha_i \cdots \alpha_{j-1})^{-1} && \text{(Simplifying)} \\
&= \alpha_0 \cdots \alpha_{i-1} (\beta_0 \cdots \beta_{i-1} (w_i)^{-1})^{-1} \\
&= \alpha_0 \cdots \alpha_{i-1} w_i (\beta_0 \cdots \beta_{i-1})^{-1} \\
&= [\alpha_0 \cdots \alpha_{i-1} w_i, \beta_0 \cdots \beta_{i-1}]_R \\
&= c_i
\end{aligned}$$

We also deduce that c_i is also a common inner witness by Lemma 8. Similarly, we can show that when c_j is a common outer witness of M_j , $c_i = c_j$ is also a common outer witness of M_i .

- *When w_i is a unique common outer witness and w_j is a unique common inner witness:* the case is similar to above. $\square$

Proof of Proposition 16. Now we prove that M has a common witness. Let M_i for $i \in [1, k]$ denote the singleton redux of M obtained from M by substituting all but G_i^* by $(\varepsilon, \varepsilon)$. Since M is conjugate, each of its singleton redux, M_i for $i \in [1, k]$ is also conjugate. By Theorem 8, each M_i has a common witness.

Since the redux of M is nonempty, if a singleton redux has infinitely many common witnesses, then it is a subset of powers of the primitive root of the redux by Theorem 8(c). Hence, any witness for the primitive root of the redux is a witness for that singleton redux of M . If all singleton reduxes have infinitely many witnesses, they are all subsets of powers of the primitive root. Hence, all of them share a common witness, and by Lemma 9 M has a common witness.

Otherwise assume that there exists a singleton redux with a unique common witness. By Lemma 14, if any two singleton reduxes have unique witnesses, then they are the same. Thus, all singleton reduxes with unique witnesses share a unique witness; Let it be z . Since z is also a witness for the redux. By Proposition 9, it is a witness for any power of the primitive root of the redux. Therefore, z is also a witness for all singleton reduxes with infinitely many witnesses by Theorem 8(c). Hence, z is a common witness of each singleton reduxes of M . By virtue of Lemma 9, M has a common witness when p is short.

□

6.2. Long cut for differentially-pumped pair

In this subsection we prove the following

Proposition 17. *Assume that M is conjugate. Whenever a differentially-pumped pair of M has a long cut, then*

$$M \subseteq (\alpha_0, \beta_0)(\rho_1, \rho'_1)^*(\alpha_1, \beta_1) \cdots (\rho_k, \rho'_k)^*(\alpha_k, \beta_k)$$

such that $\rho_i \sim \rho'_i \sim \rho_j \sim \rho'_j$ for all $i, j \in [1, k]$. Let $\bar{\alpha} = \alpha_1, \dots, \alpha_{k-1}, \alpha_k \alpha_0$, $\bar{\beta} = \beta_1, \dots, \beta_{k-1}, \beta_k \beta_0$. Let $\bar{\varrho}$ and $\bar{\varrho}'$ denote the sequence of pairs

$$\bar{\varrho} = (\rho_1, \rho_2), \dots, (\rho_{k-1}, \rho_k), (\rho_k, \rho_1) \quad (50)$$

$$\bar{\varrho}' = (\rho'_1, \rho'_2), \dots, (\rho'_{k-1}, \rho'_k), (\rho'_k, \rho'_1). \quad (51)$$

Then, one of the following holds.

- For each $i \in [1, k]$, $(\bar{\alpha})_i$ and $(\bar{\beta})_i$ are inner witnesses of $(\bar{\varrho})_i$ and $(\bar{\varrho}')_i$ respectively, and we have

$$\text{fst}(M) \sim \text{snd}(M) \subseteq \rho_i^* \sim \rho_i'^*$$

for each $i \in [1, k]$.

- There is a unique $i \in [1, k]$ such that $(\bar{\alpha})_i$ is not an inner witness of $(\bar{\varrho})_i$ and there is a unique $j \in [1, k]$ such that $(\bar{\beta})_j$ is not an inner witness of $(\bar{\varrho}')_j$, and we have

$$\text{fst}(M) \sim \text{snd}(M) \subseteq \alpha_i \dots \alpha_k \alpha_0 \dots \alpha_{i-1} \rho_i^* \sim \beta_j \dots \beta_k \beta_0 \dots \beta_{j-1} \rho_j'^*.$$

Towards this, we first show that each G_i^* , for $i \in [1, k]$, have infinitely many witnesses and hence $G_i^* \subseteq (\rho_i, \rho'_i)^*$ for some conjugate primitive pair (ρ_i, ρ'_i) for $i \in [1, k]$. Moreover, for all $i, j \in [1, k]$, $\rho_i \sim \rho'_i \sim \rho_j \sim \rho'_j$.

Lemma 15. *Assume that a differentially-pumped pair (u, v) from Definition 9 has a long cut. Then the primitive roots of pairs (u_i, v_i) for $i \in [1, k]$ in (u, v) are conjugate, i.e.,*

$$\rho_{u_1} \sim \rho_{v_1} \sim \rho_{u_2} \sim \rho_{v_2} \cdots \sim \rho_{u_k} \sim \rho_{v_k}. \quad (52)$$

Proof. There are three cases for a long cut (p, q) in (u, v) :

1. The cut p in u ends in the second half of the block B_i for $1 \leq i < k$.
2. The cut p in u ends after the block B_{i-1} and before the second half of the block B_i for $1 < i \leq k$.
3. The cut p in u ends in the second half of the block B_k .

Note that whenever the cut p in u ends in the first half of block B_1 , or ends after the block B_k , either p or q is short. Since p and q are long, $|p\beta_0| > |\alpha_0 u_1^{j_1-1}| \geq |u_1^{j_1-1}| \geq 2k \cdot \max(C, \ell)$ (since $j_1 > 2k \cdot \max(C, \ell)$). Therefore

$$|p| > 2k \cdot \max(C, \ell) - |\beta_0|. \quad (53)$$

For $i \in [1, k]$, let B_i^f and B_i^s represent the first and second halves of the block B_i , respectively. Similarly, let $B_i'^f$ and $B_i'^s$ represent the first and second halves of the block B_i' , respectively. Clearly, $|B_i^f| = |B_i'^f|$ and $|B_i^s| = |B_i'^s|$. Since $|B_1| > 2k \cdot \max(C, \ell) |B_1^f|$, we have $|B_1^s| \geq k \cdot \max(C, \ell)$. Similarly, from Equation (44), we get that for each $i \in [2, k]$,

$$|B_1| + |B_2| + \dots + |B_{i-1}| + C \leq |B_i^f| = |B_i^s| \quad (54)$$

$$\begin{aligned} u' &= \alpha_0 \underbrace{u_1 \dots}_{B_1^f} \underbrace{\dots u_1}_{B_1^s} \alpha_1 \underbrace{u_2 \dots}_{B_2^f} \underbrace{\dots u_2}_{B_2^s} \alpha_2 \dots \underbrace{u_k \dots}_{B_k^f} \underbrace{\dots u_k}_{B_k^s} \alpha_k \\ v' &= \beta_0 \underbrace{v_1 \dots}_{B_1'^f} \underbrace{\dots v_1}_{B_1'^s} \beta_1 \underbrace{v_2 \dots}_{B_2'^f} \underbrace{\dots v_2}_{B_2'^s} \beta_2 \dots \underbrace{v_k \dots}_{B_k'^f} \underbrace{\dots v_k}_{B_k'^s} \beta_k \end{aligned}$$

We now show that, in all the above cases for long cut, Equation (52) holds.

- When the cut p in u ends in B_i^s for $1 \leq i < k$:

$$\begin{aligned} u &= \alpha_0 \dots \underbrace{B_i^f B_i^s}_{p} \alpha_i \dots B_k \alpha_k \\ v &= \beta_0 \dots B_i'^f B_i'^s \beta_i \dots B_k'^f \underbrace{B_k'^s \beta_k}_{p} \end{aligned}$$

Since $|p| < |\alpha_0 B_1 \dots \alpha_{i-1} B_i| < |B_1| + \dots + |B_i| + C < |B_k'^s|$, p in v is matched/contained in $B_k'^s \beta_k$. The length of each block B_j for $j \in [1, i]$ in u is at least ℓ . Therefore, by virtue of Theorem 5, $\rho_{u_j} \sim \rho_{v_k}$. Hence, for all $j \in [1, i]$,

$$\rho_{v_j} \sim \rho_{u_j} \sim \rho_{v_k} \sim \rho_{u_k}. \quad (55)$$

Now, on matching the suffixes of q in u and v , we get that a prefix of B_k of length $|p(\beta_k)^{-1}| - |\alpha_k|$ matches with suffix of $B_{k-1}' \beta_{k-1}$. Hence, there exists a common factor between blocks B_k and B_{k-1}' of length atleast

$$|p(\beta_k)^{-1}| - |\alpha_k| - |\beta_{k-1}| = |p| - |\beta_k| - |\alpha_k| - |\beta_{k-1}| \quad (56)$$

$$\geq 2k \cdot \max(C, \ell) - |\beta_0| - |\beta_k| - |\alpha_k| - |\beta_{k-1}| \quad (\text{By Equation (53)}) \quad (57)$$

$$\geq 2k \cdot \max(C, \ell) - 2C \quad (58)$$

$$\geq 2(k-1) \cdot \max(C, \ell) \quad (59)$$

$$> \ell. \quad (60)$$

By Theorem 5, we get that $\rho_{v_k} \sim \rho_{u_k} \sim \rho_{v_{k-1}} \sim \rho_{u_{k-1}}$. Continuing inductively, for all $j \in [i+1, k]$, we get that there exists a common factor between blocks B_j and B_{j-1}' of length atleast

$$|p(\beta_k)^{-1}| - \sum_{n \in [j, \dots, k]} |\alpha_n| - \sum_{n \in [j-1, \dots, k-1]} |\beta_n| = |p| - |\beta_k| - \sum_{n \in [j, \dots, k]} |\alpha_n| - \sum_{n \in [j-1, k-1]} |\beta_n| \quad (61)$$

$$\geq 2k \cdot \max(C, \ell) - 2C \quad (62)$$

$$\geq 2(k-1) \cdot \max(C, \ell) \quad (63)$$

$$> \ell. \quad (64)$$

Hence, for $j \in [i+1, k]$, by Theorem 5, we deduce that

$$\rho_{u_k} \sim \rho_{v_{k-1}} \sim \rho_{u_{k-1}} \dots \sim \rho_{v_{i+1}} \sim \rho_{u_{i+1}}. \quad (65)$$

By combining Equations 55 and 65, we get that Equation (52) holds.

- When the cut p in u ends after the block B_{i-1} and before the block B_i^s for some $1 < i \leq k$:

$$\begin{aligned} u &= \overbrace{\alpha_0 \cdots B_{i-1} \alpha_{i-1} B_i^f}^p \overbrace{B_i^s \alpha_i \cdots \alpha_k}^q \\ v &= \beta_0 \cdots B'_{i-1} \beta_{i-1} B_i'^f B_i'^s \beta_i \cdots \beta_k \end{aligned}$$

In this case, while comparing the prefixes of q in u and v , since $|B_i^f|, |B_i^s| > |B_1| + \cdots |B_{i-1}| + C = |B'_1| + \cdots |B'_{i-1}| + C$, the prefix $\beta_0 B'_1 \cdots B'_{i-1} \beta_{i-1}$ of q in v is matched within the block B_i . Since length of each block B'_j for $j \in [1, i-1]$ is at least ℓ , by virtue of Theorem 5, $\rho_{v_j} \sim \rho_{u_i}$. Therefore, for all $j \in [1, i-1]$

$$\rho_{u_j} \sim \rho_{v_j} \sim \rho_{v_i} \sim \rho_{u_i}. \quad (66)$$

On further matching q in v and u , there exists a suffix of the block B'_i of length at least

$$|\beta_0 B'_1 \cdots B'_{i-1} \beta_{i-1} B'_i| - |\alpha_{i-1} B_i| = |\beta_0 B'_1 \cdots B'_{i-1} \beta_{i-1}| - |\alpha_{i-1}| \quad (67)$$

$$\geq |B'_1| - |\alpha_{i-1}| \quad (\text{Since } i \geq 2) \quad (68)$$

$$\geq 2k \max(C, \ell) - |\alpha_{i-1}| \quad (69)$$

to match with $\alpha_i B_{i+1}$. Hence, B'_i and B_{i+1} share a common factor of length atleast

$$2k \max(C, \ell) - |\alpha_{i-1}| - |\alpha_i| \geq 2k \max(C, \ell) - C \quad (70)$$

$$\geq 2(k-1) \max(C, \ell) \quad (71)$$

$$> \ell. \quad (72)$$

From Theorem 5, we get that $\rho_{u_i} \sim \rho_{v_i} \sim \rho_{u_{i+1}} \sim \rho_{v_{i+1}}$.

Continuing inductively, for all $j \in [i, k-1]$, we get that blocks B'_j and B_{j+1} share a common factor of length atleast

$$2k \max(C, \ell) - \sum_{n \in [i-1, j]} |\alpha_n| - \sum_{n \in [i, j-1]} |\beta_n| \geq 2(k-1) \cdot \max(C, \ell) > \ell.$$

From Theorem 5, we get that $\rho_{v_j} \sim \rho_{u_{j+1}}$ for $j \in [i, k-1]$. By transitivity of conjugacy, we get

$$\rho_{u_i} \sim \rho_{v_i} \sim \rho_{v_{i+1}} \sim \rho_{v_{i+1}} \sim \cdots \sim \rho_{u_k} \sim \rho_{v_k}. \quad (73)$$

Combining Equations 66 and 73, we get that Equation (52) holds.

- The cut p in u ends within B_k^s :

$$\begin{aligned} u &= \alpha_0 \overbrace{B_1^f B_1^s \alpha_1 B_2^f B_2^s \alpha_2 \cdots B_k^f B_k^s}^p \overbrace{\alpha_k}^q \\ v &= \beta_0 \overbrace{B_1'^f B_1'^s \beta_1 B_2'^f B_2'^s \beta_2 \cdots B_k'^f B_k'^s}^p \overbrace{\beta_k}^q \end{aligned}$$

Since q is not short, $|q\alpha_0| > |\beta_0 v_1^{j_1-1}| \geq |v_1^{j_1-1}| \geq 2k \max(C, \ell)$ (since $j_1 > 2k \cdot \max(C, \ell)$). Also, by assumption, $|q| \leq |B_k^s \alpha_k|$. While matching the q in u and v , there exists a maximal $i \in [1, k-1]$, such that $\beta_0 B'_1 \cdots B'_i$ is a proper prefix of q in v . Since $|B_j^f|$ for $j \in [1, i]$ is atleast $\geq k \max(C, \ell) > \ell$, and $|B_j^f|$ is matched within the block B_k , by virtue of Theorem 5, $\rho_{v_j} \sim \rho_{u_k}$. Therefore, for all $j \in [1, i]$

$$\rho_{u_j} \sim \rho_{v_j} \sim \rho_{v_k} \sim \rho_{u_k}. \quad (74)$$

If $i = k - 1$, then we are done. Otherwise, assume that $i \leq k - 2$. Now, matching the prefix of p in u and v , we get that a suffix of block B_{i+1} of length atleast

$$|\alpha_0 \cdots B_i \alpha_i B_{i+1}| - |q^{-1}(\beta_0 B'_1 \cdots B'_{i+1} \beta_{i+1})| = |q| - (|\alpha_0 \cdots \alpha_i| - |\beta_0 \cdots \beta_{i+1}|) \quad (75)$$

$$\geq |q| - C \quad (76)$$

$$\geq 2k \max(C, \ell) \quad (77)$$

$$> \ell \quad (78)$$

matches with block B'_{i+2} , and hence $\rho_{u_{i+1}} \sim \rho_{v_{i+2}}$ by Theorem 5. Continuing inductively, we get for $j \in [(i+1), (k-1)]$, block B_j and B'_{j+1} shares a common factor of length atleast $2(k-1) \max(C, \ell) > \ell$, and by Theorem 5,

$$\rho_{v_j} \sim \rho_{u_j} \sim \rho_{v_{j+1}} \sim \rho_{u_{j+1}}. \quad (79)$$

Combining Equations 74 and 79, we get that Equation (52) holds. $\square$

As a consequence of Lemma 15, we get the following.

Corollary 4. *Assume that M is conjugate. Whenever a differentially-pumped pair in M has a long cut, each Kleene closure G_i^* of M , for $i \in [1, k]$, has infinitely many common witnesses. Moreover,*

$$M \subseteq (\alpha_0, \beta_0)(\rho_1, \rho'_1)^*(\alpha_1, \beta_1) \cdots (\rho_k, \rho'_k)^*(\alpha_k, \beta_k). \quad (80)$$

for some $\rho_1 \sim \rho'_1 \sim \cdots \sim \rho_k \sim \rho'_k$.

Proof. By virtue of Item 3 of Definition 8, if there exists G_i^* , for some $i \in [1, k]$, with a unique common witness, then every differentially pumped pair (u, v) constructed satisfies $\rho_{u_j} \not\sim \rho_{v_\ell}$ for some $j, \ell \in [1, k], j \neq \ell$. Therefore, if a differentially-pumped pair in M has a long cut, each Kleene closure G_i^* has infinitely many witnesses, and therefore $G_i^* \subseteq (\rho_i, \rho'_i)^*$ for some primitive pair (ρ_i, ρ'_i) . Since by Lemma 15, $\rho_i \sim \rho_j$ for all $i, j \in [1, k]$ the claim follows. $\square$

Therefore, in the long cut case,

$$M \subseteq (\alpha_0, \beta_0)(\rho_1, \rho'_1)^* \cdots (\alpha_{k-1}, \beta_{k-1})(\rho_k, \rho'_k)^*(\alpha_k, \beta_k). \quad (81)$$

such that $\rho_1 \sim \rho'_1 \sim \cdots \sim \rho_k \sim \rho'_k$. Hence, $|\rho_1| = \cdots = |\rho_k| = |\rho'_1| = \cdots = |\rho'_k|$, and let d denote this common length where $d = |\rho_1| = \cdots = |\rho_k|$. We use the following shorthand for the rest of the section.

Definition 12. Let $\bar{\alpha} = \alpha_1, \dots, \alpha_{k-1}, \alpha_k \alpha_0$ and $\bar{\beta} = \beta_1, \dots, \beta_{k-1}, \beta_k \beta_0$. Let $\bar{\varrho}$ and $\bar{\varrho}'$ denote the sequence of pairs

$$\bar{\varrho} = (\rho_1, \rho_2), \dots, (\rho_{k-1}, \rho_k), (\rho_k, \rho_1) \quad (82)$$

$$\bar{\varrho}' = (\rho'_1, \rho'_2), \dots, (\rho'_{k-1}, \rho'_k), (\rho'_k, \rho'_1) \quad (83)$$

Definition 13 (d -near). Assume that $(u, v) \in M$ is conjugate with a cut (x, y) . We say $\alpha \in \bar{\alpha}$ and $\beta \in \bar{\beta}$ overlap if some position of α matches (recall Definition 4) with some position of β w.r.t. the cut (x, y) . Note that if α and β do not overlap, then the nearest positions according to the matching are either the first position of α and the last position of β or vice versa. We say α and β are d -near w.r.t. the cut (x, y) for some $d \in \mathbb{N}$ if either they overlap or there are at most d -many positions between their nearest positions.

For a sequence of words $\bar{s} = s_1, s_2, \dots, s_n$, we denote by $(\bar{s})_i$, for $i \in [1, n]$, as the i -th word s_i in the sequence.

Lemma 16. Assume that a differentially-pumped pair (u, v) from Definition 9 has a long cut (p, q) , and let $d = |\rho_1| = \cdots = |\rho_k|$. Then precisely one of the following is true.

- None of the pairs $((\bar{\alpha})_i, (\bar{\beta})_j)$ for $i, j \in [1, k]$ are d -near. In this case, the word $(\bar{\alpha})_i$ is an inner witness of the pair $(\bar{\varrho})_i$, and the word $(\bar{\beta})_i$ is an inner witness of $(\bar{\varrho}')_i$ for each $i \in [1, k]$.

- There exist a unique pair $((\bar{\alpha})_i, (\bar{\beta})_j)$ for some $i \neq j \in [1, k]$ that are d -near w.r.t. the cut (p, q) . Moreover, $(\bar{\alpha})_l$ is an inner witness of the pair $(\bar{\alpha})_i$ for each $l \in [1, k] \setminus \{i\}$, and $(\bar{\beta})_l$ is an inner witness of the pair $(\bar{\beta})_j$ for each $l \in [1, k] \setminus \{j\}$.

Proof. In the definition of (u, v) , recall that $\ell = 2 \cdot \max\{|u_i| \mid i \in [1, k]\} \geq 2 \cdot d$.

Firstly, we prove that in the long cut (p, q) of (u, v) there is at most one pair $(\bar{\alpha})_i$ and $(\bar{\beta})_j$ for some $i, j \in [1, k]$ that are d -near.

Assume there exist $i, i', j, j' \in [1, k]$ such that $(\bar{\alpha})_i$ and $(\bar{\beta})_j$, as well as, $(\bar{\alpha})_{i'}$ and $(\bar{\beta})_{j'}$ are d -near. By definition of the pair (u, v) , we have $i = i'$ if and only if $j = j'$ since a word in $\bar{\alpha}$ can be d -near with at most one word in $\bar{\beta}$ and vice-versa. Hence, assume for the sake of contradiction that $i \neq i'$ and $j \neq j'$.

We claim that $i \neq j$ and $i' \neq j'$. For the sake of contradiction assume that $i = j$, i.e., $(\bar{\alpha})_i$ and $(\bar{\beta})_i$ are d -near. We do a case analysis. When $i = k$, i.e., when $(\bar{\alpha})_k = \alpha_k \alpha_0$ and $(\bar{\beta})_k = \beta_k \beta_0$ are d -near then either p or q is short since $\min(|p|, |q|) \leq \max(|\alpha_k \alpha_0|, |\beta_k \beta_0|) + d$. Therefore,

$$\min(|p\beta_0|, |q\alpha_0|) \leq 2C + d \quad (84)$$

$$< 2C + \ell \quad (85)$$

$$< 2k \cdot \max(C, \ell) \quad (\text{Since } k \geq 2) \quad (86)$$

contradicting the assumption that the cut is long. Next assume that $i \neq k$. In this case, note that $(\bar{\alpha})_i = \alpha_i$ and $(\bar{\beta})_i = \beta_i$. Since α_i and β_i are d -near, matching the suffix $\alpha_i B_{i+1} \alpha_{i+1} \cdots \alpha_{k-1} B_k \alpha_k$ with $\beta_i B'_{i+1} \beta_{i+1} \cdots \beta_{k-1} B'_k \beta_k$, we get a suffix of length $d + \sum_{l=i}^k |\alpha_l| - \sum_{l=i}^k |\beta_l| \leq d + C$ either in u or v to be matched with the prefix of the other. Thus, $\min(|p|, |q|) \leq C + d$, and hence $\min(|p\beta_0|, |q\alpha_0|) \leq 2C + d < 2k \cdot \max(C, \ell)$. Hence, either p or q is short, contradicting the assumption that the cut is long. Therefore, we conclude that $i \neq j$. By a similar argument $i' \neq j'$.

WLOG we can assume that $i < i'$. We have two cases: either $j < j'$ or $j' < j$. Assume $j < j'$. Since $(\bar{\alpha})_i$ and $(\bar{\beta})_j$, as well as, $(\bar{\alpha})_{i'}$ and $(\bar{\beta})_{j'}$ are d -near, by matching the infix of u and v defined by $(\bar{\alpha})_i, (\bar{\alpha})_{i'}$ and $(\bar{\beta})_j, (\bar{\beta})_{j'}$, namely $(\bar{\alpha})_i B_{i+1} \cdots B_{i'} (\bar{\alpha})_{i'}$ and $(\bar{\beta})_j B'_{j+1} \cdots B'_{j'} (\bar{\beta})_{j'}$, we note that

$$\left| |(\bar{\alpha})_i B_{i+1} \cdots B_{i'} (\bar{\alpha})_{i'}| - |(\bar{\beta})_j B'_{j+1} \cdots B'_{j'} (\bar{\beta})_{j'}| \right| \leq C + 2 \cdot d \quad (87)$$

If $i' > j'$, then $|B_{i'}| > 2 \cdot (|B_1| + \cdots + |B_{j'}| + C)$ by definition of pair (u, v) . Hence,

$$\left| |(\bar{\alpha})_i B_{i+1} \cdots B_{i'} (\bar{\alpha})_{i'}| - |(\bar{\beta})_j B'_{j+1} \cdots B'_{j'} (\bar{\beta})_{j'}| \right| \geq 1/2 \cdot |B_{i'}| > k \cdot \max(C, \ell) > C + 2 \cdot d. \quad (88)$$

Therefore, contradicting Equation (87). Similarly, the case when $j' > i'$ also contradicts Equation (87). Thus, the case of $j < j'$ leads to a contradiction.

Next assume that $j' < j$. Observe that $(\bar{\alpha})_i B_{i+1} \cdots B_{i'} (\bar{\alpha})_{i'}$ and $(\bar{\beta})_j B'_{j+1} \cdots B'_k (\bar{\beta})_k B'_1 \cdots B'_{j'} (\bar{\beta})_{j'}$ (or equivalently the factors $(\bar{\alpha})_{i'} B'_{i'+1} \cdots B'_k (\bar{\alpha})_k B_1 \cdots B_i (\bar{\alpha})_i$ and $(\bar{\beta})_{j'} B'_{j'+1} \cdots B'_j (\bar{\beta})_j$) are the factors defined by $(\bar{\alpha})_i, (\bar{\alpha})_{i'}$ in u and $(\bar{\beta})_j, (\bar{\beta})_{j'}$ in v . Let $d' = \left| |(\bar{\alpha})_i B_{i+1} \cdots B_{i'} (\bar{\alpha})_{i'}| - |(\bar{\beta})_j B'_{j+1} \cdots B'_k (\bar{\beta})_k B'_1 \cdots B'_{j'} (\bar{\beta})_{j'}| \right|$. Since $(\bar{\alpha})_i$ and $(\bar{\beta})_j$, as well as, $(\bar{\alpha})_{i'}$ and $(\bar{\beta})_{j'}$ are d -near,

$$d' \leq C + 2 \cdot d. \quad (89)$$

If $i' \neq k$, then $|B'_k| > 2 \cdot (|B_1| + \cdots + |B_{i'}| + C)$. Hence, $d' \geq 1/2 \cdot |B'_k| > C + 2 \cdot d$. Therefore, contradicting Equation (89). The case when $i' = k$ is symmetric to the above case by considering the factors $(\bar{\alpha})_k B_1 \cdots B_i (\bar{\alpha})_i$ and $(\bar{\beta})_{j'} B'_{j'+1} \cdots B'_j (\bar{\beta})_j$ defined by $(\bar{\alpha})_i, (\bar{\alpha})_{i'}$ in u and $(\bar{\beta})_j, (\bar{\beta})_{j'}$ in v .

Therefore, there does not exist $i, i', j, j' \in [1, k]$, $(i, j) \neq (i', j')$, such that $(\bar{\alpha})_i$ and $(\bar{\beta})_j$, as well as, $(\bar{\alpha})_{i'}$ and $(\bar{\beta})_{j'}$ are d -near, i.e., there exists at most one pair (i, j) such that $(\bar{\alpha})_i$ and $(\bar{\beta})_j$ are d -near. Moreover, $i \neq j$.

For some $i \in [1, k]$, assume that $(\bar{\alpha})_i \in \bar{\alpha}$ is not d -near with any word in $\bar{\beta}$. This implies that in the matching of u and v w.r.t. the cut (p, q) , the factor $\rho_i(\bar{\alpha})_{i+1}$ (with $\rho_{i+1} = \rho_1$ when $i = k$) is matched within some block of ρ_j^* for some $j \in [1, k]$. Since $\rho_i \sim \rho_{i+1} \sim \rho'_j$, it follows from Proposition 10 that $(\bar{\alpha})_i$ is an inner witness of $(\bar{\rho})_i = (\rho_i, \rho_{i+1})$. Similarly, for any $i \in [1, k]$, if $(\bar{\beta})_i \in \bar{\beta}$ is not d -near with any word in $\bar{\alpha}$, then $(\bar{\beta})_i$ is an inner witness of $(\bar{\rho}')_i = (\rho'_i, \rho'_{i+1})$ (with $\rho'_{i+1} = \rho'_1$ when $i = k$). Hence, the claim is proved. $\square$

Lemma 17. Let $1 \leq i < j \leq k$. Assume that $(\bar{\alpha})_\ell$ is an inner witness of $(\bar{\rho})_\ell = (\rho_\ell, \rho_{\ell+1})$ (with $\rho_{\ell+1} = \rho_1$ when $\ell = k$), for all $\ell \in [i, j]$. Then, $(\bar{\alpha})_i \cdots (\bar{\alpha})_j$ is an inner witness of $(fst((\bar{\rho})_i), snd((\bar{\rho})_j))$, and moreover,

$$\rho_i^*(\bar{\alpha})_i \rho_{i+1}^*(\bar{\alpha})_{i+1} \cdots (\bar{\alpha})_j \rho_{j+1}^* = \rho_i^*(\bar{\alpha})_i \cdots (\bar{\alpha})_j \quad (90)$$

$$= (\bar{\alpha})_i \cdots (\bar{\alpha})_j \rho_{j+1}^*. \quad (91)$$

Further, if $1 \leq j < i \leq k$ and $(\bar{\alpha})_\ell$ is an inner witness of $(\bar{\rho})_\ell = (\rho_\ell, \rho_{\ell+1})$ (with $\rho_{\ell+1} = \rho_1$ when $\ell = k$), for all $\ell \in [i, k] \cup [1, j]$, then $(\bar{\alpha})_i \cdots (\bar{\alpha})_k (\bar{\alpha})_1 \cdots (\bar{\alpha})_j$ is an inner witness of $(fst((\bar{\rho})_i), snd((\bar{\rho})_j))$, and moreover,

$$\rho_i^*(\bar{\alpha})_i \rho_{i+1}^* \cdots (\bar{\alpha})_k \rho_1^*(\bar{\alpha})_1 \rho_2^* \cdots (\bar{\alpha})_j \rho_{j+1}^* = \rho_i^*(\bar{\alpha})_i \cdots (\bar{\alpha})_k (\bar{\alpha})_1 \cdots (\bar{\alpha})_j \quad (92)$$

$$= (\bar{\alpha})_i \cdots (\bar{\alpha})_k (\bar{\alpha})_1 \cdots (\bar{\alpha})_j \rho_{j+1}^*. \quad (93)$$

Proof. Since by assumption $\rho_\ell^n(\bar{\alpha})_\ell = (\bar{\alpha})_\ell \rho_{\ell+1}^n$ for all $\ell \in [i, j]$ and $\forall n \geq 0$, we have

$$\rho_i^*(\bar{\alpha})_i \rho_{i+1}^*(\bar{\alpha})_{i+1} \cdots (\bar{\alpha})_{j-1} \rho_j^*(\bar{\alpha})_j \rho_{j+1}^* = \rho_i^*(\bar{\alpha})_i \rho_{i+1}^*(\bar{\alpha})_{i+1} \cdots (\bar{\alpha})_{j-1} \rho_j^*(\bar{\alpha})_j \quad (94)$$

$$= \rho_i^*(\bar{\alpha})_i \rho_{i+1}^*(\bar{\alpha})_{i+1} \cdots \rho_{j-1}^*(\bar{\alpha})_{j-1} (\bar{\alpha})_j \quad (95)$$

$$= \rho_i^*(\bar{\alpha})_i \cdots (\bar{\alpha})_j \quad (\text{By induction on the interval } [i, j]) \quad (96)$$

By an analogous argument,

$$\rho_i^*(\bar{\alpha})_i \rho_{i+1}^*(\bar{\alpha})_{i+1} \cdots (\bar{\alpha})_{j-1} \rho_j^*(\bar{\alpha})_j \rho_{j+1}^* = (\bar{\alpha})_i \cdots (\bar{\alpha})_j \rho_{j+1}^*. \quad (97)$$

From the above, we infer that $(\bar{\alpha})_i \cdots (\bar{\alpha})_j$ is an inner witness of the pair $(\rho_i, \rho_{j+1}) = (fst((\bar{\rho})_i), snd((\bar{\rho})_j))$.

Next assume that $1 \leq j \leq i \leq k$ and $(\bar{\alpha})_\ell$ is an inner witness of $(\bar{\rho})_\ell = (\rho_\ell, \rho_{\ell+1})$ (with $\rho_{\ell+1} = \rho_1$ when $\ell = k$), for all $\ell \in [i, k] \cup [1, j]$. By the above observation,

$$\rho_i^*(\bar{\alpha})_i \rho_{i+1}^* \cdots (\bar{\alpha})_k \rho_1^* = (\bar{\alpha})_i \cdots (\bar{\alpha})_k \rho_1^* \quad (98)$$

$$= \rho_i^*(\bar{\alpha})_i \cdots (\bar{\alpha})_k \quad (99)$$

$$\rho_1^*(\bar{\alpha})_1 \rho_2^* \cdots (\bar{\alpha})_j \rho_{j+1}^* = (\bar{\alpha})_1 \cdots (\bar{\alpha})_j \rho_{j+1}^* \quad (100)$$

$$= \rho_1^*(\bar{\alpha})_1 \cdots (\bar{\alpha})_j \quad (101)$$

From the above equations, we get that

$$\rho_i^*(\bar{\alpha})_i \rho_{i+1}^* \cdots (\bar{\alpha})_k \rho_1^*(\bar{\alpha})_1 \rho_2^* \cdots (\bar{\alpha})_j \rho_{j+1}^* = \rho_i^*(\bar{\alpha})_i \rho_{i+1}^* \cdots (\bar{\alpha})_k \rho_1^*(\bar{\alpha})_1 \cdots (\bar{\alpha})_j \quad (\text{Using Equation (101)}) \quad (102)$$

$$= \rho_i^*(\bar{\alpha})_i \cdots (\bar{\alpha})_k (\bar{\alpha})_1 \cdots (\bar{\alpha})_j \quad (\text{Using Equation (99)}) \quad (103)$$

$$= (\bar{\alpha})_i \cdots (\bar{\alpha})_k (\bar{\alpha})_1 \cdots (\bar{\alpha})_j \rho_{j+1}^* \quad (\text{Similarly}) \quad (104)$$

From the above, we infer that $(\bar{\alpha})_i \cdots (\bar{\alpha})_k (\bar{\alpha})_1 \cdots (\bar{\alpha})_j$ is an inner-witness of the pair $(\rho_i, \rho_{j+1}) = (fst((\bar{\rho})_i), snd((\bar{\rho})_j))$. $\square$

Proof of Proposition 17. From Corollary 4, we get that $M \subseteq (\alpha_0, \beta_0)(\rho_1, \rho'_1)^*(\alpha_1, \beta_1) \cdots (\rho_k, \rho'_k)^*(\alpha_k, \beta_k)$ for some $\rho_1 \sim \rho'_1 \sim \cdots \sim \rho_k \sim \rho'_k$. By virtue of Lemma 16, there are two cases.

Case 1: When the word $(\bar{\alpha})_i$ is an inner witness of $(\bar{\rho})_i$ for each $i \in [1, k]$. Notice that by Lemma 17, $(\bar{\alpha})_1 \cdots (\bar{\alpha})_k$ is an inner witness of (ρ_1, ρ_1) , and hence $(\bar{\alpha})_1 \cdots (\bar{\alpha})_k \in \rho_1^*$. We show that $fst(M)$ is a cyclic shift of ρ_1^* as follows.

$$fst(M) \subseteq \alpha_0 \rho_1^* \alpha_1 \rho_2^* \alpha_2 \cdots \rho_k^* \alpha_k \quad (105)$$

$$\sim \rho_1^* \alpha_1 \rho_2^* \alpha_2 \cdots \rho_k^* \alpha_k \alpha_0 \quad (106)$$

$$= \rho_1^*(\bar{\alpha})_1 \rho_2^*(\bar{\alpha})_2 \cdots \rho_k^*(\bar{\alpha})_k \quad (107)$$

$$= \rho_1^*(\bar{\alpha})_1 (\bar{\alpha})_2 \cdots (\bar{\alpha})_k \quad (\text{By Lemma 17}) \quad (108)$$

$$\subseteq \rho_1^*. \quad (\text{Since } (\bar{\alpha})_1 \cdots (\bar{\alpha})_k \text{ is an i.w of } (\rho_1, \rho_1)) \quad (109)$$

By a similar argument, we show that $\text{snd}(M) \subseteq \rho_1'^*$. Therefore, since M is conjugate, $M \subseteq (\rho_1, \rho_1')^*$ and hence has a common witness. Moreover, since $\rho_1' \sim \rho_1$, $\text{fst}(M)$ and $\text{snd}(M)$ are cyclic shifts of ρ_1^* .

Case 2: When there exists a unique pair (i, j) , $i \neq j \in [1, k]$, such that for all $l \in [1, k] \setminus \{i\}$, $(\bar{\alpha})_l$ is an inner witness of the pair $(\bar{\varrho})_l$, and for all $l \in [1, k] \setminus \{j\}$, $(\bar{\beta})_l$ is an inner witness of the pair $(\bar{\varrho}')_l$.

$$\text{fst}(M) \subseteq \alpha_0 \rho_1^* \alpha_1 \rho_2^* \alpha_2 \cdots \rho_k^* \alpha_k \quad (110)$$

$$\sim \rho_1^* \alpha_1 \rho_2^* \alpha_2 \cdots \rho_k^* \alpha_k \alpha_0 \quad (111)$$

$$= \rho_1^* (\bar{\alpha})_1 \rho_2^* (\bar{\alpha})_2 \cdots \rho_k^* (\bar{\alpha})_k \quad (112)$$

$$\sim (\bar{\alpha})_i \rho_{i+1}^* \cdots (\bar{\alpha})_k \rho_1^* (\bar{\alpha})_1 \cdots (\bar{\alpha})_{i-1} \rho_i^* \quad (113)$$

$$= (\bar{\alpha})_i \cdots (\bar{\alpha})_k (\bar{\alpha})_1 \cdots (\bar{\alpha})_{i-1} \rho_i^* \quad (\text{By Lemma 17}) \quad (114)$$

Therefore, we have $\text{fst}(M) \sim \text{snd}(M) \subseteq (\bar{\alpha})_i \cdots (\bar{\alpha})_k (\bar{\alpha})_1 \cdots (\bar{\alpha})_{i-1} (\rho_i)^*$. By an analogous argument, $\text{fst}(M) \sim \text{snd}(M) \subseteq (\bar{\beta})_j \cdots (\bar{\beta})_k (\bar{\beta})_1 \cdots (\bar{\beta})_{j-1} (\rho_j')^*$, and the claim follows. $\square$

7. Computing Witness of a Sumfree Expression

In this section, we give a decision procedure to compute the common witness of a sumfree expression, if it exists. A sumfree expression can have no common witness, a unique common witness, or infinitely many common witnesses. Thus, the set of common witnesses (abbreviated as the *witness set*) is either empty, singleton, or infinite. Whenever there are infinitely many common witnesses for an expression, the witnesses are the same as those of its primitive root (Proposition 13). In that case we compute the primitive root as their finite representation.

The following proposition shows that there is a bound to the size of the unique common witness of two conjugate pairs if it exists, which aids in computing the common witness of two pairs in linear time.

Proposition 18. *If two conjugate primitive pairs (u_1, v_1) and (u_2, v_2) have a unique common witness z , then $|z| \leq 2 \cdot \max(|u_1|, |u_2|)$.*

Proof. Let z be a common inner witness. Therefore, $z = (x_1 y_1)^{n_1} x_1 = (x_2 y_2)^{n_2} x_2$ for some $n_1, n_2 \geq 0$ where (x_i, y_i) is a cut of the primitive pair (u_i, v_i) for $i \in \{1, 2\}$. We claim either n_1 or n_2 is less than 2. Suppose not, i.e., $n_1 \geq 2$ and $n_2 \geq 2$. Thus u_1^ω and u_2^ω share a common prefix of length at least $|u_1| + |u_2|$. From Theorem 4, they have the same primitive root. It implies that $x_1 y_1 = x_2 y_2$ since u_1 and u_2 are primitive words. Since $(x_1 y_1)^{n_1} x_1 = (x_2 y_2)^{n_2} x_2$, $x_1 y_1 = x_2 y_2$ and $|x_1|, |x_2| < |x_1 y_1|$, we obtain $n_1 = n_2$, and hence, $x_1 = x_2$. This implies $y_1 = y_2$ and hence, $y_1 x_1 = y_2 x_2$. Both (u_1, v_1) and (u_2, v_2) are the same word; thus, they have infinitely many common witnesses, that is a contradiction. Hence $|z| \leq 2 \cdot \max\{|u_1|, |u_2|\}$.

The case for common outer witness is symmetric. $\square$

The above proposition holds true for any two conjugate pairs (not necessarily primitive) by Proposition 12.

Proposition 19. *The witness set of two conjugate pairs can be computed in linear time.*

Proof. Since the common witness for a set and its primitive root are the same (Proposition 12), we compute the witness set for the primitive roots of the given two conjugate pairs. We can compute the primitive roots in linear time by Proposition 2. Let $G = \{(u_1, v_1), (u_2, v_2)\}$ be the set of primitive roots of the given conjugate pairs and let (x_i, y_i) be the cut of (u_i, v_i) for $i \in \{1, 2\}$. The cut of the primitive pair (u_i, v_i) can be computed in linear time by finding the first nontrivial occurrence of v_i in $u_i u_i$.

One of the following possibilities holds true for G : it has no common witness, a unique common witness, or infinitely many common witnesses. The following algorithm outlines the computation of the witness set of G :

1. Check if the primitive pairs are identical, i.e., verify if $u_1 = u_2$ and $v_1 = v_2$. If yes, then G has infinitely many common witnesses by Proposition 13. The witness is finitely represented by the primitive pair (u_1, v_1) . This step takes linear time w.r.t. the length of the primitive pairs.

2. If the pairs are not identical, then check if G has a unique common witness using Proposition 18 as follows: WLOG assume that $|u_1| > |u_2|$. According to Proposition 18, if a unique common witness exists for G , its length is at most $2 \cdot \max(|u_1|, |u_2|) = 2 \cdot |u_1|$. Thus, it suffices to check whether $(x_1y_1)^\omega$ and $(x_2y_2)^\omega$ share equal prefixes of length $|x_1|$ or $|x_1y_1x_1|$, that also end in x_2 . If it is satisfied, then G has a unique common witness. This step can be performed in linear time w.r.t. the length of the primitive pairs.
3. If none of the above holds, then G has no common witness.

The overall complexity of the algorithm is linear w.r.t the length of the primitive roots of the given pairs. $\square$

The witness set of a sumfree expression is equal to the intersection of witness sets of each of its singleton reduxes. So first, we show how to compute the witness set of a sumfree expression with only one Kleene star, in effect the witness set of a singleton redux. Using this procedure, we show how to compute the witness set of a general sumfree expression.

Lemma 18. *Let $M = (\alpha_0, \beta_0)E^*(\alpha_1, \beta_1)$ be a sumfree expression. Given the witness set of E , we can compute the witness set of M in time $O(m + n)$ where m is the size of the expression M , and n is the size of the witness of E .*

Proof. If redux of M is empty, then the witness set of M is same as the witness set of E by Theorem 6. Now assume that M has a nonempty redux. From Theorem 8, M has a common witness iff $E \cup \{(\alpha_1\alpha_0, \beta_1\beta_0)\}$ has a common witness. The common witness of M is computed from the common witness of E and $(\alpha_1\alpha_0, \beta_1\beta_0)$.

The idea is to check if a common witness exists for $E \cup \{(\alpha_1\alpha_0, \beta_1\beta_0)\}$ using Lemma 7. If it exists, using that we compute the common witness for M . There are two possibilities for common witnesses of E :

1. E has a unique common inner (resp. outer) witness z . By Item 2 of Lemma 7, it suffices to check if z is a common inner (resp. outer) witness of $(\alpha_1\alpha_0, \beta_1\beta_0)$. This can be checked in $O(m + n)$ time using Theorem 1. If so, z is the common witness of $E \cup \{(\alpha_1\alpha_0, \beta_1\beta_0)\}$. Now compute the common witness of M using Theorem 8(a) (resp. Theorem 8(b)). This can be computed in $O(m + n)$ time. Otherwise, $E \cup \{(\alpha_1\alpha_0, \beta_1\beta_0)\}$ has no common witness and hence, M has no common witness by Theorem 8.
2. E has infinitely many common witnesses. In this case, the witnesses of E are the same as that of its primitive root, say (ρ, ρ') . From Item 1 of Lemma 7, it suffices to check if (ρ, ρ') and $(\alpha_1\alpha_0, \beta_1\beta_0)$ has a common witness. For this, first check if the primitive root of $(\alpha_1\alpha_0, \beta_1\beta_0)$ is equal to (ρ, ρ') . This step takes time $O(m + n)$. We have two cases:
 - (a) If $(\alpha_1\alpha_0, \beta_1\beta_0)$ have same primitive root as that of E , then E and $(\alpha_1\alpha_0, \beta_1\beta_0)$ have infinitely many common witnesses by Proposition 13. In this case, M is a set of powers of the primitive root of the redux by Theorem 8(c). Thus M has infinitely many witnesses. Compute the primitive root of its redux using Proposition 2. This step takes $O(m)$ time.
 - (b) Otherwise, compute the unique common witness of (ρ, ρ') and $(\alpha_1\alpha_0, \beta_1\beta_0)$ if it exists using Proposition 19. If so, we are back to Case 1; else M has no common witness. This step takes $O(m + n)$ time.

$\square$

Using the above algorithm, we compute the common witness of a general sumfree expression as follows.

Lemma 19. *Let M be a sumfree expression. Given the witness set of each Kleene star in M , we can compute the witness set of M in time $O(m \cdot (m + n))$ where m is the size of the expression and n is the maximum size among the given witnesses.*

Proof. From Lemma 9, the witness set of M is the intersection of the witness sets of its singleton reduxes. Check if each singleton redux of M has a common witness and compute it using Lemma 18. This step takes $O(m \cdot (m + n))$. If there is a singleton redux with no common witness, then M has no common witness by Theorem 9.

Assume that M has a nonempty redux. The algorithm is as follows. Firstly, check if the redux of M is conjugate using Proposition 5 (in time $O(m)$). If yes, then compute the primitive root of the redux, say (ρ_m, ρ'_m) , using Proposition 2 (in time $O(m)$). Otherwise, M has no common witness. There are two cases depending on whether all singleton redux has infinitely many witnesses.

- (a) If all the singleton reduxes have infinitely many witnesses, then M is a set of powers of the primitive root of the redux (ρ_m, ρ'_m) by Theorem 8(c). Thus, M has infinitely many common witnesses.
- (b) If there exists a singleton redux with a unique common witness, say z , then for all other singleton reduxes of M with a unique witness z' , check if $z = z'$ (for all other singleton reduxes z is already a witness by virtue of being a witness of the redux of M). If so, z is the unique common witness of M ; otherwise M has no common witness.

This takes $O(mn + m)$ time. Consider the case where the redux of M is empty. Let M be of the form $E_1^* E_2^* \cdots E_k^*$. We iterate over $i \in \{1, \dots, k-1\}$.

- (a) If both E_i and E_{i+1} has infinitely many witnesses, then the witness set of E_i^* and E_{i+1}^* is finitely represented by their primitive roots. Check if the primitive roots are the same. If yes, continue the procedure for $i = i + 1$. Otherwise, check and compute the common witness, say z , of the primitive roots if exists using Proposition 19 (in time $O(n)$). If not, then M has no common witness. Otherwise, check if z is a common witness of all singleton reduxes $E_{i+2}^*, \dots, E_k^*$ (This can be done in $O(n)$ for each singleton redux — If a singleton redux has a unique witness then check if it is equal to z . Otherwise, if it has infinitely many witness, then it is represented by a primitive root. It suffices to check if z is a witness of that primitive root).
- (b) If either one of E_i and E_{i+1} has a unique witness, say z , then check if z is a common witness of all singleton reduxes $E_{i+2}^*, \dots, E_k^*$.

This step takes $O(mn + n)$ time. Overall, it takes $O(m \cdot (m + n))$. $\square$

Computation of the Witness Set: Given a sumfree expression M , we compute its witness set bottom-up. We start from the innermost Kleene star. It is a pair of words (u, v) . First, we check if (u, v) is conjugate using Proposition 5. If yes, then there are infinitely many common witnesses for $(u, v)^*$, namely the witnesses of its primitive root, otherwise M has no witness. This step can be done in a time linear in the length of (u, v) . Now we recursively use Lemma 19 to compute the common witness of the expression under the Kleene star in each level. If there is no common witness for any level of Kleene star expression, then M is not conjugate.

To find out the complexity of the decision procedure, it suffices to estimate the maximum length of a witness involved in the computation.

Length of the Witness of a Sumfree Expression: We claim that if a sumfree expression M is conjugate, then there exists a witness of length linear in size of M . If M has infinitely many witnesses, from Proposition 13, M is a set of powers of a primitive root. Therefore, there exists a witness of length that is less than that of the length of the primitive root. Next, suppose M has a unique common witness. In that case, there exists a subexpression E_i^* such that E_i^* has a unique common witness, and all Kleene star appearing in E_i has infinitely many witnesses. Thus, all of them have a common witness of length at most $|E_i|$. Therefore, there is a singleton redux M_i of E_i^* that has a unique witness z_i . The size of z_i is linear in M_i and the size of the witnesses of subexpressions of E_i . Both are upper bounded by size of M . Furthermore, the common witnesses for all subsequent levels is unique (if it exists) and its length is bounded by $|M|$.

Complexity of the Algorithm: Since the size of the common witness of M is linear in $|M|$, by Lemma 19, the overall complexity of computing a common witness of a sumfree expression is $O(h \cdot m^2)$ where h is the *star height* of M and m is the length of the expression.

8. Deciding Conjugacy of a Sumfree Expression

Given a sumfree expression $E = (\alpha_0, \beta_0) E_1^* (\alpha_1, \beta_1) \dots E_k^* (\alpha_k, \beta_k)$, $k \geq 1$, the following algorithms decides whether E is conjugate.

1. Check whether E has a common witness using the algorithm described in Section 7. If there exists a common witness, then output YES.

2. Otherwise, verify the following conditions. If any of them fails, output NO.

- (a) $\alpha_0 \dots \alpha_k \sim \beta_0 \dots \beta_k$.
- (b) For $i \in [1, k]$, each E_i^* is conjugate and moreover it has a primitive root (ρ_i, ρ'_i) .
- (c) $\rho_i \sim \rho_j$ for all $i, j \in [1, k]$.
- (d) Let $\bar{\alpha} = (\alpha_1, \dots, \alpha_{k-1}, \alpha_k \alpha_0)$ and $\bar{\rho} = (\rho_1, \rho_2), \dots, (\rho_{k-1}, \rho_k), (\rho_k, \rho_1)$. Then, there is at most one element $i \in [1, k]$ such that $(\bar{\alpha})_i$ is *not* an inner witness of $(\bar{\rho})_i$.
- (e) Let $\bar{\beta} = (\beta_1, \dots, \beta_{k-1}, \beta_k \alpha_0)$ and $\bar{\rho}' = (\rho'_1, \rho'_2), \dots, (\rho'_{k-1}, \rho'_k), (\rho'_k, \rho'_1)$. Then there is at most one element $j \in [1, k]$ such that $(\bar{\beta})_j$ is *not* an inner witness of $(\bar{\rho}')_j$.

3. If i and j do not exist in steps (d) and (e), then output YES.

4. Otherwise, output YES if the below expression is conjugate.

$$\left(\begin{array}{c} \alpha_i \dots \alpha_k \alpha_0 \dots \alpha_{i-1} \\ \beta_j \dots \beta_k \beta_0 \dots \beta_{j-1} \end{array} \right) \left(\begin{array}{c} \rho_i \\ \rho'_j \end{array} \right)^* \quad (115)$$

Note that the above expression is conjugate iff the set $\{(\alpha_i \dots \alpha_k \alpha_0 \dots \alpha_{i-1}, \beta_j \dots \beta_k \beta_0 \dots \beta_{j-1}), (\rho_i, \rho'_j)\}$ has a common witness.

5. In all other cases, Output NO.

The correctness of the above algorithm follows from Theorem 7 and Proposition 17. In the above algorithm, Steps 1, 2.(b), and 4 amounts to computing the common witness of a sumfree expression, and therefore is in polynomial time. Now, Steps 2.(a) and 2.(c) is checking conjugacy of two words, which is in linear time. Steps 2.(d) and 2.(e) are in linear time. Therefore, the algorithm runs in polynomial time.

9. Conclusion

It is shown that the conjugacy problem of a rational relation is decidable. The current decision procedure proceeds through the analysis of rational expressions. In its essence, it is analogous to the boundedness checking of distance automata using factorisation trees [22], though explicit use of factorisation trees are avoided using sumfree rational expressions instead. An obvious question is the existence of an automata-theoretic proof. Factorisation forests remain the primary tool to settle boundedness questions on automata and by that standard the proof approach taken in this paper is natural and quite possibly the most intuitive.

Computing a witness of a given sumfree expression, if one exists, can be done in polynomial time. However, converting a rational expression into a sum of sumfree expressions may result in an exponential blowup. Thus, the algorithm presented in the paper is of exponential time. It remains to find the precise complexity of this problem.

It is natural to look at the conjugacy problem of more general classes, for instance functions definable by a deterministic two-way transducers (regular functions [23]), or by two-way pebble automata (polyregular functions [24]).

Appendix A.

Lemma 20. *Every rational expression E can be converted to one in sumfree normal form E' in exponential time. Moreover, $|E'| \leq 2^{|E|}$.*

Proof. Let E be a rational expression over the monoid $\mathbf{M}$. We assume that the rational expression E is given as a tree e . We take the size of e , denoted as $|e|$, to be the number of nodes in the tree. We inductively define a tree e' that has the same language as the sumfree normal form of the expression E and furthermore, as shown in Figure A.6, it is in the shape of a right-comb with the internal nodes of the spine labelled with $+$'s (and the leaf of the spine is labelled with $\emptyset$) and the pendant left subtrees attached to the spine are sumfree. We call e' as the SNF tree of e .

We obtain an equivalent sumfree normal form expression and its expression tree e' by induction on the structure of E . We prove the following invariant along with the construction of e' .

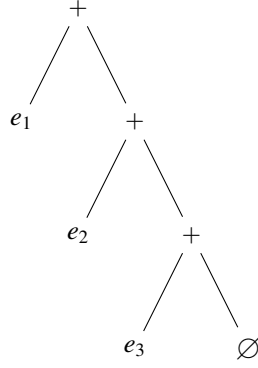


Figure A.6: SNF tree for the SNF expression $E_1 + E_2 + E_3$.

Claim 4. $|e'| \leq 2^{2|e|}$

The following definition is used in the analysis below. Let $N(e')$ denote the number of summands in e' , i.e., $N(e')$ is the number of nodes in the spine of the comb, or equivalently, 1 more than the number of nodes labelled with '+' in e' . Hence $N(e') \leq |e'|$.

Base Case. When E is $\emptyset$ or $m \in M$, then E is already sumfree. The tree e corresponds to a tree with a single node. We take e' to be the tree with 3 nodes in SNF with the left subtree of the root being e . Hence $|e'| = 3$ and the claim holds.

Inductive Case. Assume that G and H are rational expressions with expression trees g and h respectively. Let g' and h' denote their SNF trees. By induction hypothesis, $G \equiv \alpha_1 + \dots + \alpha_k$ and $F \equiv \beta_1 + \dots + \beta_\ell$ such that α_i, β_j , $1 \leq i \leq k$, $1 \leq j \leq \ell$, are sumfree expressions. Also, $|g'| \leq 2^{2|g|}$ and $|h'| \leq 2^{2|h|}$.

1. If $E = G + H$, then by substituting for G and H , we get an equivalent expression of the desired form. This step takes constant time. To obtain e' we replace the leaf of the spine of g' with the root of h' . Clearly,

$$\begin{aligned}
 |e'| &= |g'| + |h'| - 1 \\
 &\leq 2^{2|g|} + 2^{2|h|} - 1 \\
 &\leq 2^{2(|g|+|h|+1)} \\
 &= 2^{2|e|}.
 \end{aligned}$$

2. If $E = G \cdot H$, then by substituting G and H we get $E \equiv (\alpha_1 + \dots + \alpha_k) \cdot (\beta_1 + \dots + \beta_\ell)$. Distributing the monoid operation over the union, we get $E \equiv (\alpha_1\beta_1 + \dots + \alpha_1\beta_\ell) + \dots + (\alpha_k\beta_1 + \dots + \alpha_k\beta_\ell)$, that is in the required form. This step takes time quadratic in the maximum among the length of the SNF expressions G and H .

Assume there are p -many (resp. q -many) pendant subtrees attached to the spine of g' (resp. h'). The tree e' is a right-comb with pq -many pendant subtrees where each subtree is obtained by the pairwise concatenation of pendant subtrees from g' and h' respectively. Clearly, $N(e') = N(g')N(h') - 1$.

$$\begin{aligned}
 |e'| &\leq N(g')N(h') + |g'|N(h') + |h'|N(g') + N(g')N(h') \\
 &\leq 4|g'||h'| \\
 &\leq 4 \cdot 2^{2|g|}2^{2|h|} \\
 &\leq 2^{2(|g|+|h|+1)} \\
 &= 2^{2|e|}.
 \end{aligned}$$

3. Finally, if $E = G^*$, then by repeatedly applying the rational identity $(X + Y)^* = (X^*Y^*)^*$, where X, Y are rational expressions, we get $E = G^* \equiv (\alpha_1 + \alpha_2 + \dots + \alpha_k)^* = (\alpha_1^* \alpha_2^* \dots \alpha_k^*)^*$. This step takes linear time w.r.t. the length of the SNF expression G .

We obtain the tree g' corresponding to g , and construct a new tree h from g' as follows.

- Add an intermediate node labelled with $*$ between each pendant subtree and the spine of g' .
- Replace each $+$ labelled nodes in the spine with concatenation.
- Replace $\emptyset$ in the leaf of spine with epsilon.
- Add a new root node labelled with $*$.

Now, e' is obtained by attaching h as the left subtree of a right-comb in the desired form. Clearly $N(e') = 1$.

$$\begin{aligned}
|e'| &\leq |g'| + N(g') + 3 \\
&\leq |g'| + |g'| + 3 \\
&\leq 2 \cdot 2^{|g|} + 3 \\
&= 2^{2|g|+1} + 3 \\
&\leq 2 \cdot 2^{2|g|+1} \quad (\text{Since } |g| \geq 1, 2^{2|g|+1} \geq 8) \\
&= 2^{2(|g|+1)} \\
&= 2^{2|e|}
\end{aligned}$$

Hence proved that the upper bound on the size of the SNF expression is exponential in the size of the given expression.

Each step of constructing an SNF expression takes polynomial time in the length of its constituent SNF expressions. Therefore, any rational expression can be converted to an equivalent sumfree normal form in exponential time. $\square$

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